

Financial literacy pays: Smarter investing goes beyond AI



30 July 2026

Allianz Research

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Allianz School for Life

Executive Summary



Ludovic Subran
Chief Investment Officer
& Chief Economist
ludovic.subran@allianz.com



Simon Krause
Economist, ESG & Insurance
simon.krause@allianz.com



Michaela Grimm
Senior Economist,
Demography & Social Protection
michaela.grimm@allianz.com



Kathrin Stoffel
Economist, Insurance & Wealth
kathrin.stoffel@allianz.com



Nils Bradtke
Senior Investment Strategist
nils.bradtke@allianz.com



Katharina Utermöhl
Head of Thematic & Policy Research
katharina.uterhoehl@allianz.com

Lucie Trollé
Research Assistant

• **Financial literacy is an important foundation for building, protecting and growing wealth over a lifetime, but it remains the exception.** From children's investment accounts in the US to pension reforms in Germany, responsibility for building wealth and financing retirement is increasingly shifting from institutions to individuals. As financial products become more complex and private provision more important, understanding basic financial concepts such as compound interest, inflation, risk and diversification is more important than ever. Yet financial capability is failing to keep pace. The fourth edition of the Allianz Research financial literacy study, questioning more than 8,000 respondents across eight countries, shows little change since our last survey in 2023¹. Literacy remains worryingly low: only 17% scored as highly financially literate, while one in four scored low. The UK (23%), Austria (22%) and Germany (22%) record the highest share of highly financially literate respondents, Spain (17%), Italy (17%) and Poland (15%) occupy the middle ground, while the US (13%) and France (11%) rank at the bottom. The US also records the largest low-literacy group (33%), suggesting that greater capital-market participation does not automatically translate into greater financial capability.

• **Women, younger respondents and those less educated face the greatest financial challenges, but score lowest on financial literacy – a pattern that is remarkably consistent across countries.** Women need to finance longer retirements and accumulate private wealth, yet are only half as likely as men to achieve high literacy (11% vs. 24%). Younger generations also bear greater responsibility for privately funding their retirement as public pension systems come under growing pressure, yet they record the weakest results²: only 12% of Gen Z score as highly literate, compared with 22% of Baby Boomers. At the intersection of both divides, Gen Z women perform worst (10% high; 40% low), while Boomer men perform best (29% high; 13% low). Education is an even stronger divider: just 5% of respondents with primary education achieve high literacy, compared with 24% of those with tertiary education. Confidence and competence are also misaligned: While 17% of respondents score as highly literate, only 8% consider themselves more knowledgeable than the average investor. Conversely, the number of respondents who judge their financial knowledge as lower than average (48%) is double the number actually scoring as low-literate (26%). This presents the risk of too little confidence leading to inaction, while overconfidence may invite costly mistakes.

¹ For the previous survey, see: Allianz Research (2023), "Playing with a Squared Ball: The Financial Literacy Gender Gap," https://www.allianz.com/en/economic_research/insights/publications/specials_fmo/financial-literacy.html

² See: Allianz Research (2026), "Pension Reform Survey 2026: Everyone knows reform is needed, few expect it to happen", https://www.allianz.com/en/economic_research/insights/publications/specials_fmo/260625-pension-survey.html

- **Smarter investing, not higher saving, is one of the largest untapped sources of household wealth.** The return on savings depends not only on how much households set aside, but also on how effectively they invest. Financial literacy strongly shapes those choices: more than half of low-literacy respondents (53%) cannot identify an investment product they would prioritize, compared with only one in five (21%) of the high-literacy group. A simple simulation shows that shifting half of existing bank deposits in an average household portfolio, in equal parts, into national bonds and equities would have increased annual returns in every country, led by Poland, Germany, and Austria (each +1pp). Such a shift, without considering additional savings, would have lifted per capita financial wealth by 14–21% in real terms over the past two decades in countries with the highest deposit shares. Germany provides a clear example of the scale of the opportunity: the estimated per capita financial surplus over 20 years amounts to EUR11,700 after inflation.
- **AI is lowering the barriers to financial advice without raising capability, making financial literacy more important than ever.** AI makes financial advice more accessible by reducing the cost and increasing availability, but it cannot substitute for the financial literacy needed to evaluate that advice and make better money management decisions. AI adoption is already widespread: almost half of respondents (48%) use AI at least weekly, rising to 77% among Gen Z, while 14% already cite it as a main source of financial advice (26% among Gen Z). Yet greater access does not automatically translate into better decisions. AI users are considerably more confident in their financial knowledge (43% vs. 33% overall), but are no more likely to be highly financially literate (18% vs. 17%). This creates the risk of an overconfidence bias: AI may increase confidence without improving judgment. For financially literate households, AI can complement professional advice. For less financially literate households, however, it may reinforce poor decisions by providing advice they lack the knowledge to assess.
- **Closing financial literacy gaps requires a new policy playbook involving governments, employers and financial institutions.** There is an ever-growing need for credible financial and risk education. While the responsibility for financial security does not sit with households alone, people are being asked to take greater responsibility for their financial future and need the tools and knowledge to make informed decisions. Governments should embed financial education in schools and adult learning, while making the long-term consequences of financial decisions more tangible through pension dashboards and personalized retirement projections. Employers can strengthen financial capability through workplace education, pension guidance and access to trusted advice. Financial institutions should complement these efforts with simpler products, higher advice standards, digital tools and AI-enabled guidance that helps households move from saving to long-term investing. Across all stakeholders, interventions should be broad enough to reach everyone, yet sufficiently targeted to close the persistent literacy gaps among women, younger generations and less-educated adults. At the same time, interventions need to help people calibrate their financial confidence, recognize when professional advice is required and use AI as a complement, rather than a substitute, for trusted financial guidance.



The financial literacy scoreboard: failing to reach the top

In many advanced economies, responsibility for financial security is increasingly shifting from institutions onto individuals, raising the importance of financial literacy. As social security systems come under increasing demographic strain, individuals need to take greater responsibility for securing their retirement through private plans³. At the same time, returns to capital continue to outpace those to labor – a development that could be turbo-charged by the impact of artificial intelligence on employment – making investments rather than wages the key driver of long-term wealth, whether the goal is homeownership or other financial objectives⁴. In this context, financial literacy equips people to make the decisions that build, protect and grow their wealth over a lifetime. The implications extend far beyond personal finances. Financial security underpins health, reduces stress and strengthens people's sense of control over the future⁵. As households assume greater responsibility for retirement planning and wealth accumulation, financial

literacy increasingly determines not only who prospers financially, but who enjoys greater economic resilience and wellbeing. And the challenge is only becoming harder. A rapidly evolving financial landscape – defined by digital assets, increasingly sophisticated financial products and persistent cost-of-living pressures – demands ever greater confidence and competence in managing money.

To understand how well people are equipped to navigate this new financial reality, we surveyed more than 8,000 respondents across eight countries: Austria, France, Germany, Italy, Poland, Spain, the UK and the US⁶. The survey, the fourth edition of our financial literacy study, builds on Allianz Research's financial literacy series, using the same core questionnaire to track developments over time⁷. This year's edition expands the country coverage with the addition of Austria and Poland, with the latter offering a first perspective on financial literacy in Central and Eastern Europe. The

3 Recent pension reform proposals increasingly emphasize funded private retirement savings. In Germany, these include plans for a capital-funded private pension pillar and the introduction of a "Frühstart-Rente", a state-supported investment account for children, designed to encourage long-term capital-market participation from an early age.

4 See Jordà, O., K. Knoll, D. Kuvshinov, M. Schularick and A. Taylor (2019), "The Rate of Return on Everything, 1870–2015," *Quarterly Journal of Economics*, 134(3), 1225–1298. Using long-run historical data for 16 advanced economies, the authors find that the rate of return on wealth has persistently exceeded the rate of economic growth over the past 150 years, with average returns on capital roughly twice as high as economic growth.

5 See: OECD (2023), "International Survey of Adult Financial Literacy," OECD Publishing, Paris. The survey finds that adults with high financial literacy report higher financial well-being and financial resilience than those with low literacy, and that financial literacy helps individuals "develop greater control over personal financial matters."

6 Details on the sample are provided in Appendix A. The complete questionnaire is reproduced in Appendix B.

7 To ensure comparability across countries and survey waves, we measure financial literacy using the same index in each survey wave. For the previous survey, see: Allianz Research (2023), "Playing with a Squared Ball: The Financial Literacy Gender Gap," https://www.allianz.com/en/economic_research/insights/publications/specials_fmo/financial-literacy.html

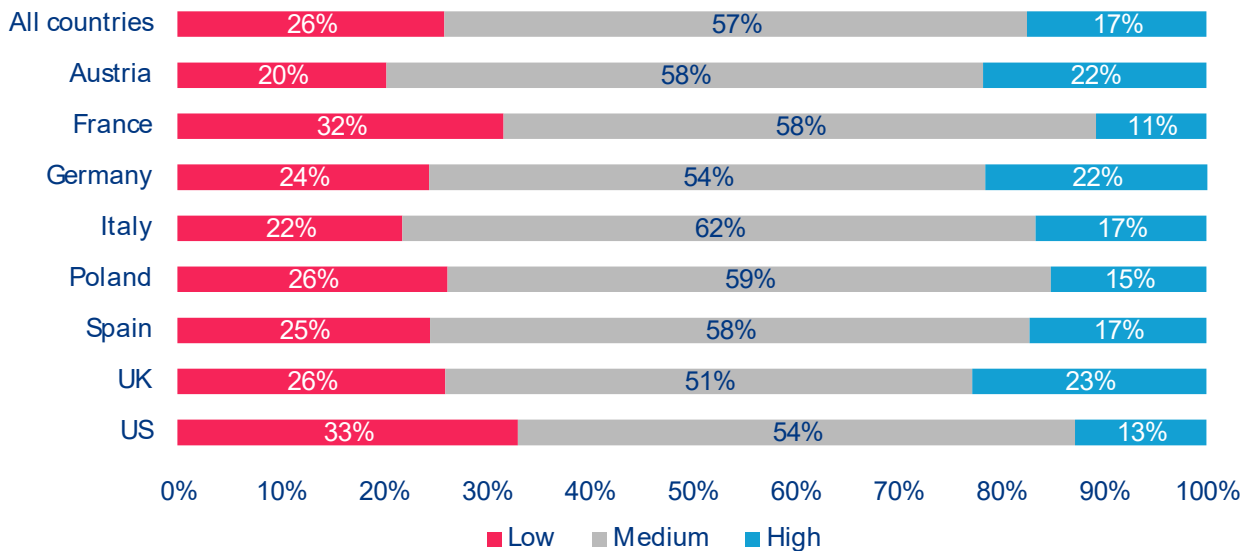
financial literacy score is based on nine questions covering the core concepts underpinning sound financial decision-making – interest, compound interest, inflation, risk diversification and probability – drawing on the seminal work of Professors Annamaria Lusardi and Olivia Mitchell⁸. Respondents are grouped into three levels, or bands, of financial literacy according to the number of correct answers: low (0–2), medium (3–6) and high (7–9). This classification is used throughout the report.

The survey confirms an insufficient level of financial literacy almost everywhere. Only 17.5% of all respondents reach the high-literacy band, while 25.9% fall into the low band. The majority of 56.6% are classified as medium, suggesting a basic grasp of financial concepts, but not the level of knowledge required to navigate increasingly complex financial decisions. Notably, this medium share is strikingly uniform across countries, hovering around 55% almost everywhere, while the differences between countries play out almost entirely in the relative size of their high- and low-literacy bands (Figure 1). Ranked by the shares of respondents with high levels of financial literacy, the UK (22.9% high literacy), Austria (22.0%) and Germany (21.5%) record the strongest results. Spain (17.4%),

Italy (16.9%), and Poland (15.4%) occupy the middle ground, while France (10.7%) and the US (12.9%) rank at the bottom. Among these country differences, the US stands out in particular. Despite having one of the world's most market-based retirement systems, it also records the largest low-literacy group (33.0%). This may be no coincidence. Through 401(k)s, IRAs and default investment options, households can participate in capital markets with little active financial knowledge: contributions are automatic, funds are pre-selected and investing happens largely on autopilot. Being invested, therefore, does not necessarily translate into greater financial capability.

The financial literacy scoreboard has moved little since 2023, although some countries have made limited progress. Among the six countries included in both the 2023 and 2026 surveys, Germany and the UK record the notable improvements, with the share of respondents in the high-literacy band increasing by 5.4 and 7.0pp, respectively. The US also sees a moderate improvement in the share of respondents with high literacy (+2.6pp), while France records only marginal progress (+1.0pp). Spain sees a growing polarization of financial literacy: both the low- (+4.4pp) and high-literacy groups (+1.1pp) grow, while the medium-literacy

Figure 1: Financial literacy levels, share of respondents by country in %



Source: Allianz Research

⁸ The index follows the « Big Three » and extended financial-literacy questions developed in Lusardi, A. and Mitchell, O. S. (2011), «Financial Literacy around the World: An Overview», *Journal of Pension Economics and Finance*, 10(4), 497–508, and Lusardi, A. and Mitchell, O. S. (2014), «The Economic Importance of Financial Literacy: Theory and Evidence», *Journal of Economic Literature*, 52(1), 5–44.

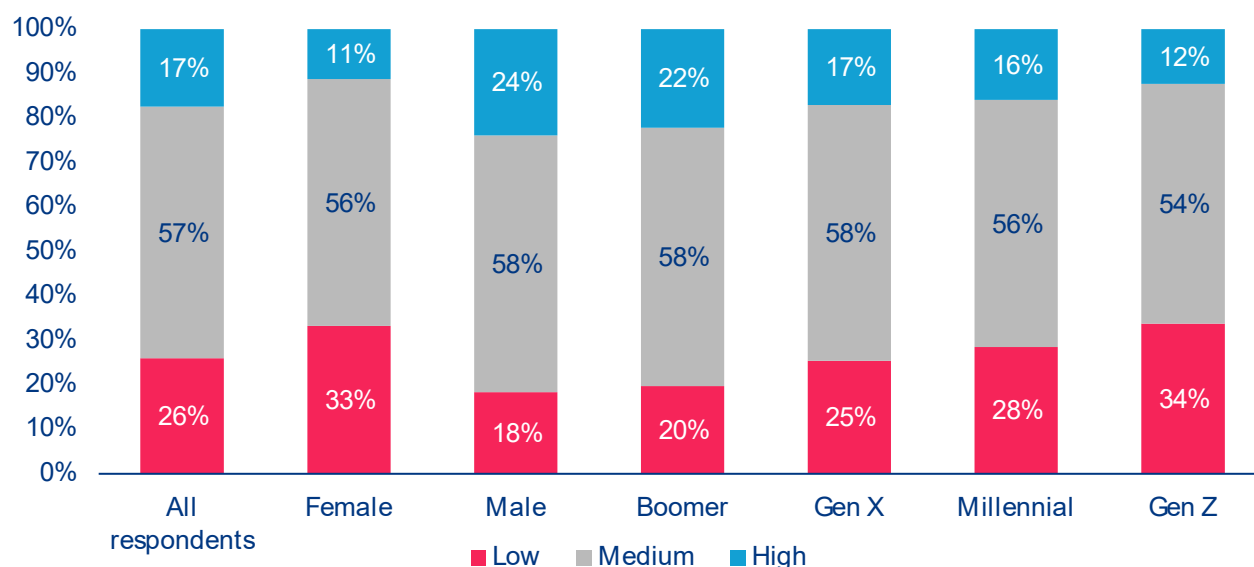
group shrinks (-5.5pp). In contrast, Italy remains broadly unchanged, showing neither major improvement nor deterioration. Overall, the findings suggest that current financial literacy initiatives are failing to deliver broad-based gains, underscoring the need for more sustained and better-targeted interventions.

Male and older players know the game: financial literacy splits sharply by gender and rises steadily with age (Figure 2). First, the gender literacy gap is striking: 23.9% of men reach the high-literacy band, compared with just 11.2% of women. One-third of women (33.3%) fall into the low-literacy band against fewer than one in five men (18.4%). This gap is particularly concerning because women typically live longer, earn less over their lifetime and therefore depend even more on making their savings work effectively. Second, financial literacy increases across the generations⁹: only 12.3% of Gen Z reach the high-literacy band, compared with 22.2% of Baby Boomers, while the low-literacy share falls from 33.8% to 19.7%. This is a worrying mismatch. Younger generations will need to start building private wealth earlier and more consistently than previous cohorts, as they are likely to depend more heavily on defined-contribution

pensions and their own savings. Yet they are the least financially equipped to navigate these responsibilities. Together, these two divides reinforce one another: younger generations and women face the greatest challenge in securing their financial future and record the weakest financial literacy.

A closer look shows that the gender gap is remarkably persistent across countries. In every country, men are around twice as likely as women to reach the high-literacy band (Figure 3). The gap is largest in Austria (30.4% vs 13.5%), Germany (29.5% vs 13.5%) and the UK (30.5% vs 15.6%), meaning that these countries' strong overall performance is driven primarily by men. This matters because women typically accumulate less income and wealth over their lifetime while depending more heavily on it in retirement¹⁰. Therefore, each euro of women's savings and investments matters even more. Yet they are also worse served when they seek financial advice¹¹. Lower financial literacy therefore compounds existing gender disparities rather than offsetting them.

Figure 2: Financial literacy levels, share of respondents by gender and generation in %

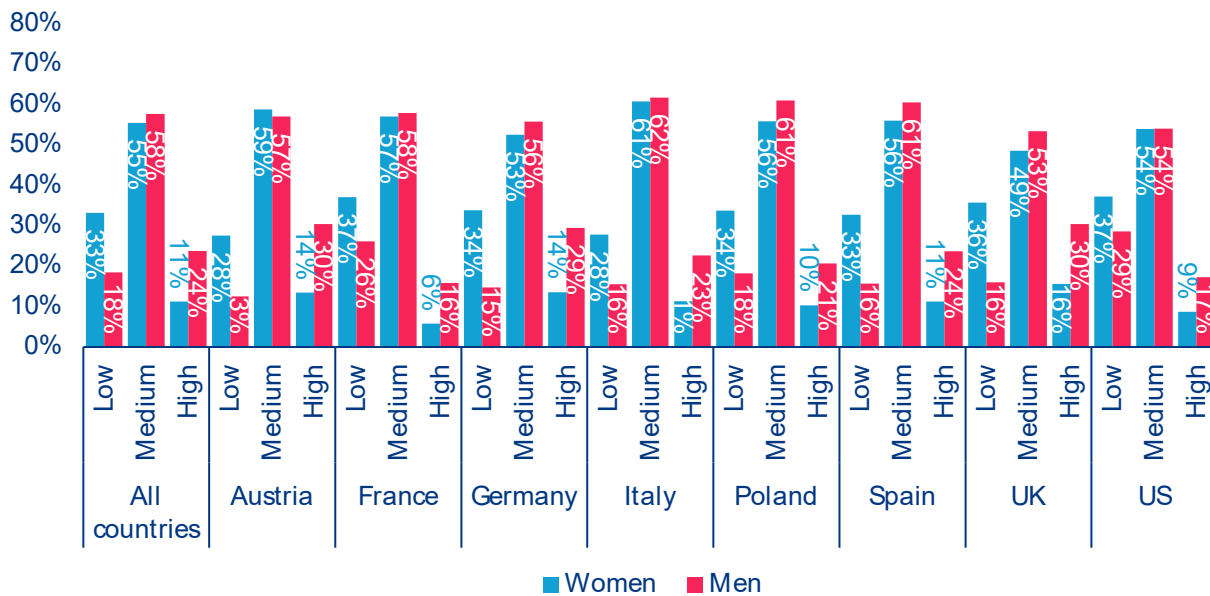


Note: Baby Boomers (birth years 1946–1964), Generation X (1965–1980), Millennials (1981–1996) and Generation Z (1997–2012).
Source: Allianz Research

⁹ Generations are defined according to the Pew Research Center: Baby Boomers (birth years 1946–1964), Generation X (1965–1980), Millennials (1981–1996) and Generation Z (1997–2012).

¹⁰ See Allianz Research (2026), "Closing the Gender Income Gap: From Paycheck to Pension," https://www.allianz.com/en/economic_research/insights/publications/specials_fmo/260305-gender-gap.html. The study estimates a lifetime income gender gap of around 16% for the 2000 birth cohort across 14 OECD countries, driven primarily by labor income, and finds that strengthening financial literacy can raise women's annual investment returns by up to 1.5pp. On pensions specifically, Eurostat (2026) reports that women aged 65+ in the EU received on average 24.5% less pension income than men in 2024.

¹¹ See: Bucher-Koenen, T., Hackethal, A., Koenen, J. and Laudenbach, C. (2025), "Gender Differences in Financial Advice," American Economic Review, 115(12), 4218–4252. Drawing on some 27,000 real-world advisory meetings at a large German bank, the authors find advisors give more self-serving advice to women, recommending expensive in-house products more often and offering fee rebates less often.

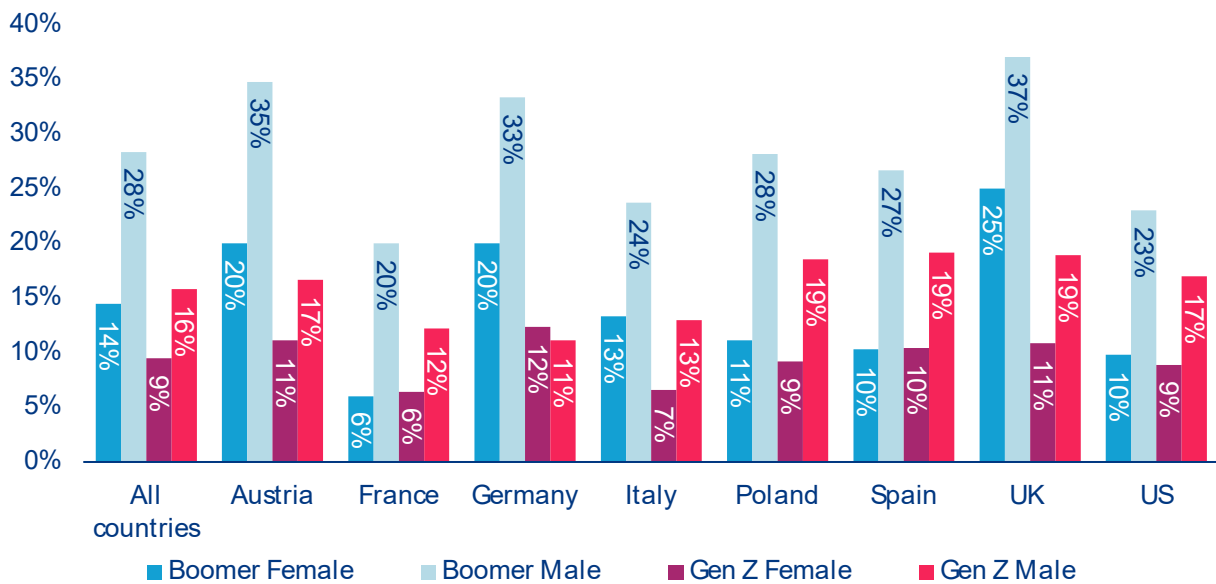
Figure 3: Gender gap in financial literacy, share of female and male respondents by country in %

Source: Allianz Research

The gender and generational divides reinforce each other. Gen Z women record the weakest results of any demographic group, with only 9.6% reaching the high-literacy band and 39.8% falling into the low band. In comparison, Baby Boomer men perform best, with 28.9% in the high-literacy band and just 13.2% in the low band (Figure 4). This pattern is particularly pronounced in Italy, where 23.7% of male Baby Boomers are highly financially literate but only 6.6% of female Gen Z – nearly a fourfold difference. The US is a notable exception, with financial literacy remaining comparatively flat across generations, echoing the earlier finding that widespread market participation does not necessarily translate into greater financial capability. Overall, the evidence points in one direction: the groups carrying the greatest responsibility for financing their own future are also the least financially prepared.

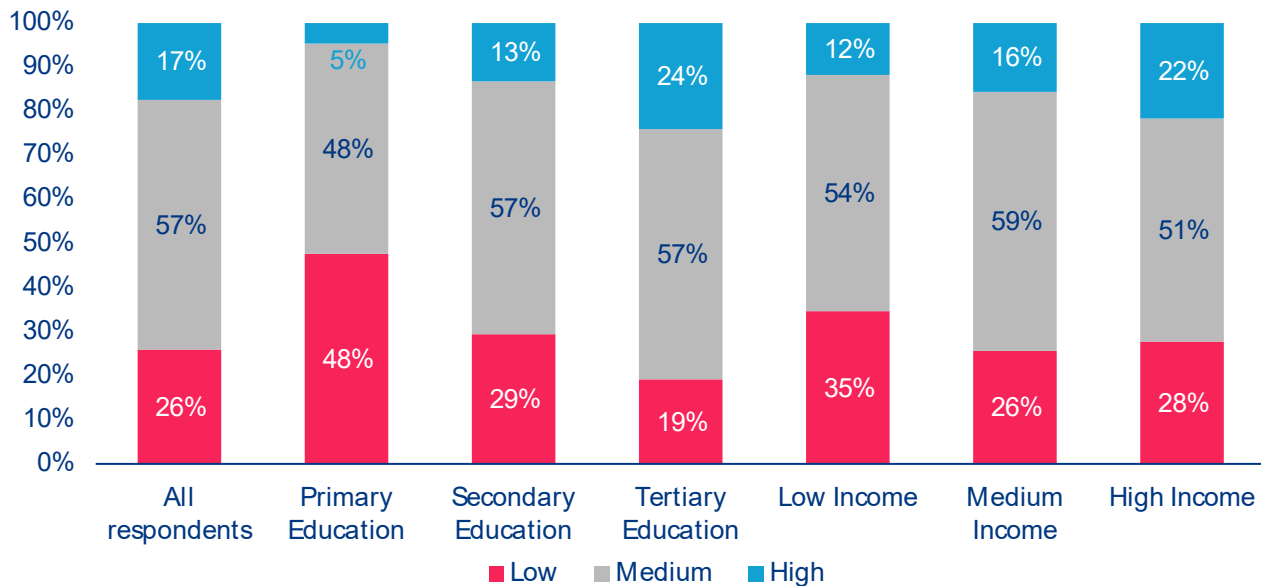
Education and income sort the remaining field, and financial literacy risks entrenching the divides they reflect. The gradients here are the steepest in the entire survey (Figure 5). Among respondents with only primary education, 47.8% fall into the low-literacy band and just 4.7% reach the high band. By contrast, 24.0% of respondents with tertiary education reach the high-literacy band – more than five times the share among the primary educated. Income shows a similar, though weaker, gradient: 21.7% of high-income respondents reach the high-literacy band, compared with 11.8% of the low-income group. The implication is that financial literacy gaps may reinforce existing economic inequalities.

Figure 4: High financial literacy generational and gender divide, share of respondents in the high literacy band by country, gender and generation in %



Source: Allianz Research

Figure 5: Financial literacy levels, share of respondents by education and income in %



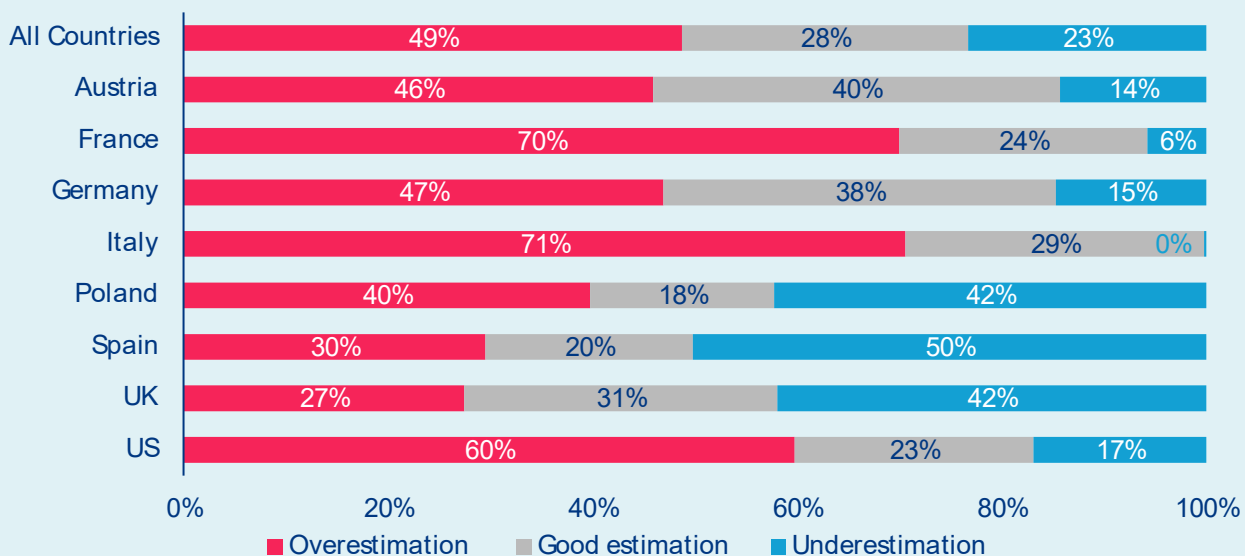
Source: Allianz Research

Box 1: Can people read the price tag?

Inflation perceptions and the cost-of-living barrier

Financial literacy is not only about understanding financial concepts but also about interpreting the economic signals that shape everyday decisions, with inflation being one of the most important. Households that overestimate inflation underestimate the real return on their savings and may conclude they have too little left to invest, reinforcing the already existing cost-of-living pressures. We therefore asked respondents to estimate the current inflation rate in their country, counting answers within one percentage point of the official rate as accurate (Figure 6). The results are sobering: only 28.0% estimated inflation correctly, while 48.7% overestimated it and 23.3% underestimated it. On average, respondents put inflation at 5.7% compared with the actual rate of 2.9%. Overestimation is particularly widespread in Italy (70.6%), France (70.0%) and the US (59.7%). Underestimation is most common in Spain (50.2%) and the UK (41.9%). Both distort financial decisions, but persistent overestimation is likely to discourage investing by leading households to underestimate the purchasing power of future investment returns.

Figure 6: Inflation perceptions, share of respondents by country underestimating, accurately estimating, and overestimating current inflation, in %

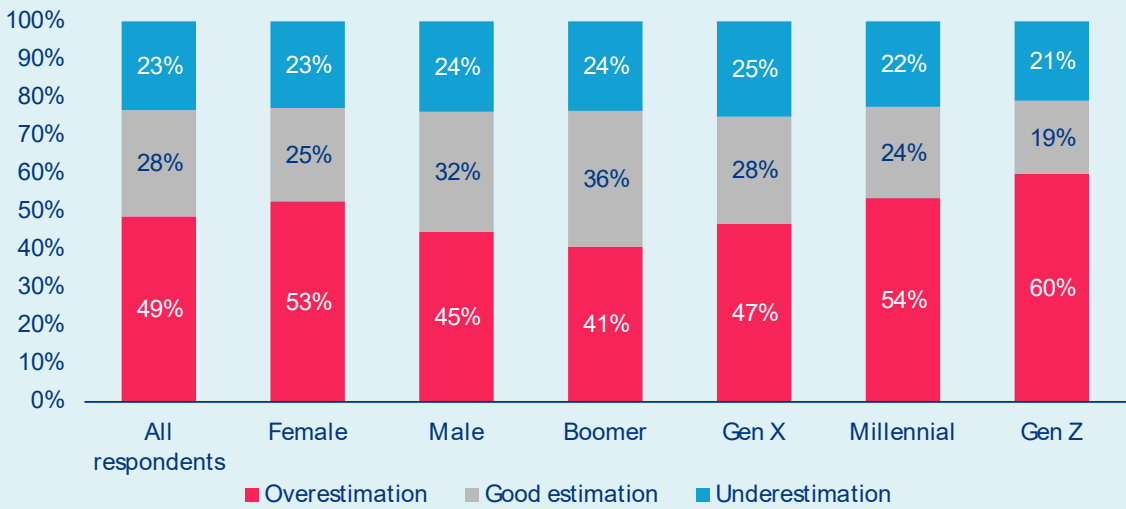


Notes: We consider estimates within a 1pp range around current inflation as a good estimate.

Sources: LSEG Datastream, Allianz Research

Inflation perceptions closely mirror the fault lines of financial literacy scoreboard. Austria (39.8% accurate) and Germany (38.4%) record the highest shares of correct estimates, while countries with weaker financial literacy also display larger misperceptions. The same pattern appears across demographic groups (Figure 7): men estimate inflation more accurately than women (31.6% vs 24.6%), and accuracy declines steadily from Baby Boomers (35.7%) to Gen Z (19.3%), who are also the most likely to overestimate inflation (59.9%). Reading the price tag, in other words, is another expression of financial literacy: the same groups that struggle most with financial concepts are also the least able to accurately assess the economic environment in which they make financial decisions.

Figure 7: Inflation perceptions, share of respondents by gender and generation underestimating, accurately estimating, and overestimating current inflation, in %



Notes: We consider estimates within a 1pp range around current inflation as a good estimate.

Sources: LSEG Datastream, Allianz Research.



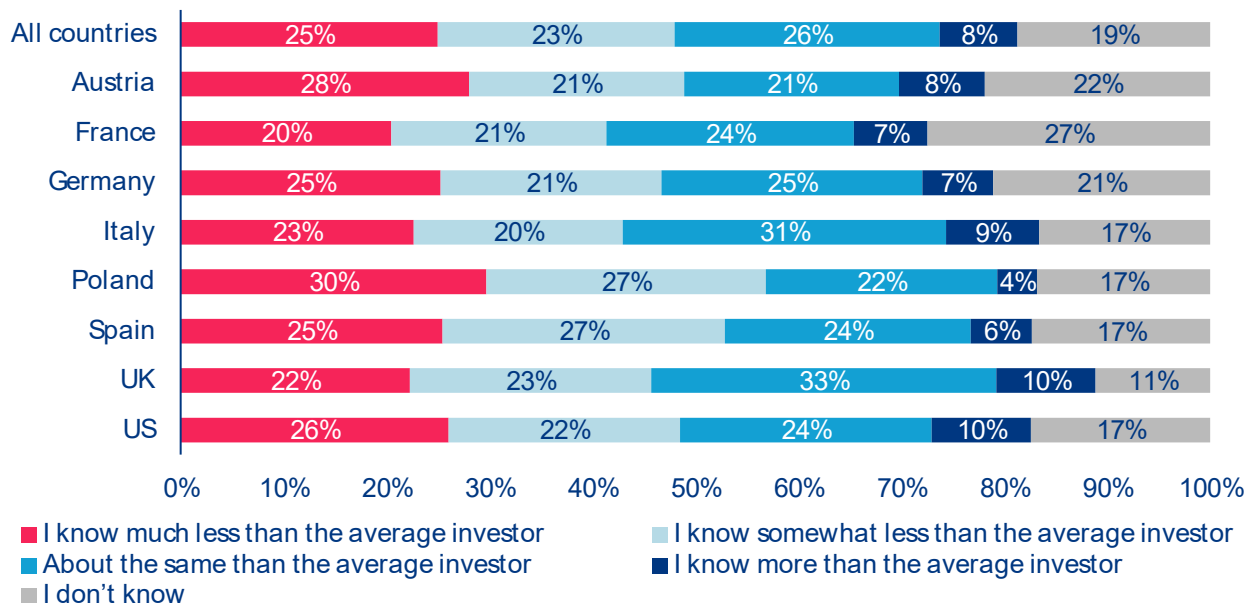
Financial confidence does not always follow financial competence

Financial knowledge is only part of the equation; people must also feel confident enough to act on it.

Asked to compare their financial knowledge with that of the average investor, respondents paint a remarkably cautious picture. Across all eight countries, one in four (25.0%) believe they know much less than the average investor, while a further 23.0% consider themselves somewhat less knowledgeable (Figure 8). Just over one-quarter (25.8%) rate themselves as about average, and only 7.6% believe they know more than the average investor. Nearly one in five respondents (18.7%) are unable to judge their own level of knowledge. Country

differences are relatively modest. Poland records the lowest level of self-confidence, with 56.9% placing themselves below the average investor and only 3.9% above, while respondents in the UK and the US express the greatest confidence, with 9.6% considering themselves more knowledgeable than average.

Figure 8: Financial confidence, share of respondents by country in %



Source: Allianz Research

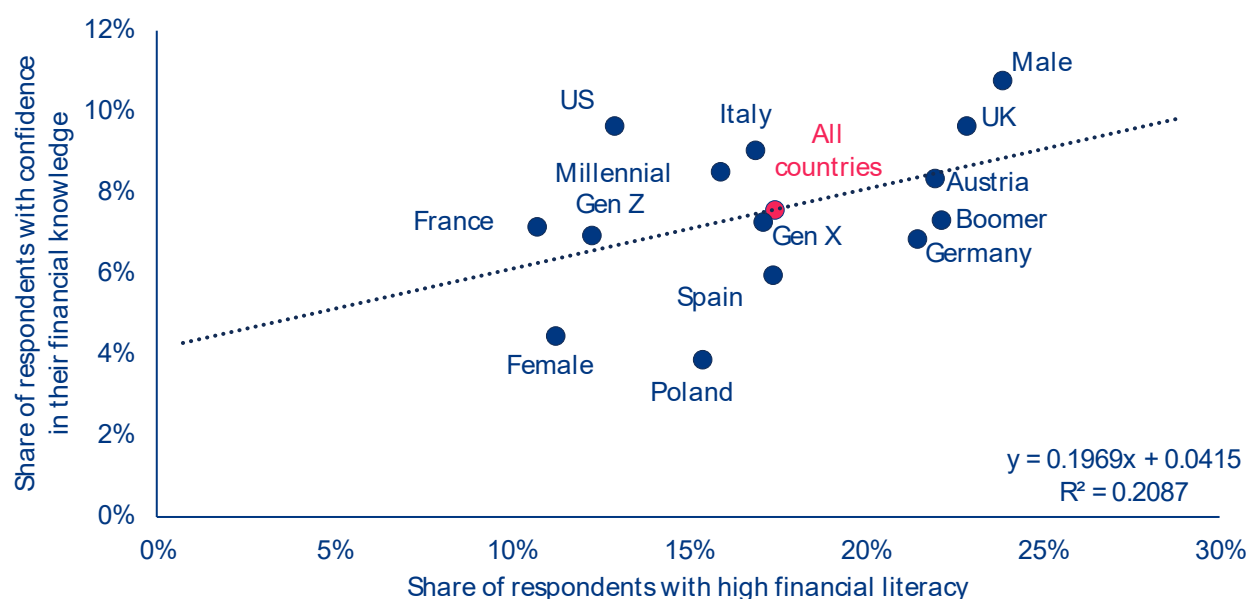
Confidence is not always an accurate reflection of competence. Across all eight countries, 17.5% of respondents reach the high-literacy band, but only 7.6% consider themselves more knowledgeable than the average investor. Conversely, almost half of respondents (48.0%) judge their financial knowledge to be below average, even though only 25.9% belong to the low-literacy band. Confidence does rise with financial literacy – the share considering themselves more knowledgeable than average increases from 4.6% in the low-literacy band to 6.0% in the medium band and 16.9% in the high band – but much more slowly than knowledge itself. The relationship also varies across countries (Figure 9). The UK and the US are the most self-assured (9.6% highly confident in each), yet while the UK combines this with one of the highest levels of financial literacy, the US records the lowest share of highly financially literate respondents (12.9%), making it the most overconfident country in the sample. Poland presents the opposite pattern: only 3.9% consider themselves above average despite a comparatively larger high-literacy group (15.4%), suggesting confidence there falls short of competence.

The gender divide is stronger than the generational divide: women’s confidence is even lower than their financial literacy. While men are overconfident relative to their knowledge – 23.9% are highly literate, but only 10.8% are highly confident – women show the opposite

and more damaging pattern. They pair an 11.3% highly literate share with just 4.5% high confidence, which are two of the widest gaps in the dataset (12.6 percentage points in literacy and 6.3 percentage points in confidence). These gaps reinforce each other and risk keeping capable women out of financial markets they could navigate perfectly well. Generations drift apart the same way: literacy climbs with age but confidence stays flat, so Baby Boomers top the literacy table (22.2%) yet claim only 7.3% high confidence, while Millennials (15.9% vs 8.5%) and Gen Z (12.3% vs 6.9%) tilt toward overconfidence. Experience, it seems, teaches humility as readily as it builds knowledge.

The mismatch between knowledge and confidence matters for financial outcomes. Overconfidence, or when confidence exceeds actual knowledge, invites costly mistakes, as people overestimate their ability to judge risk. Knowledge without confidence is just as limiting, keeping capable savers on the sidelines, postponing decisions or leaning too heavily on others. The goal of financial education should therefore be broader than raising test scores. It should help people calibrate their confidence – to trust what they know, recognize what they do not, and seek reliable advice when needed.

Figure 9: Financial literacy versus financial confidence, share of respondents by country, gender and generation in %



Source: Allianz Research



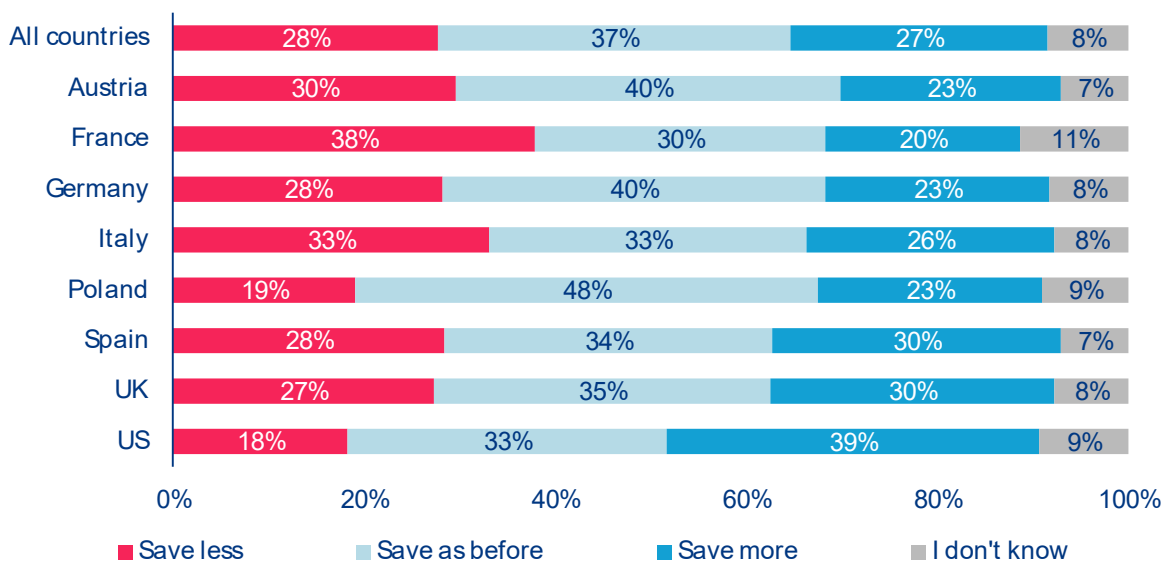
Saving intentions are stable, investment choices make the difference

Building wealth depends on two decisions: how much households save and how they invest those savings.

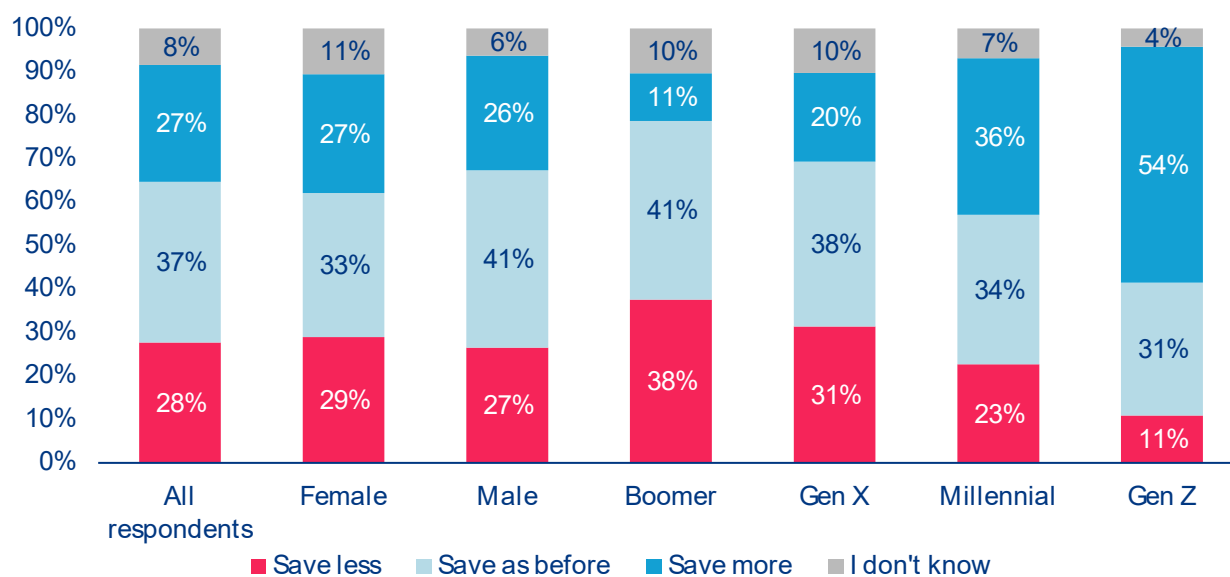
We therefore asked respondents about both their future saving intentions and the products they expect to give greater weight in their portfolios. While saving intentions also provide an indication of perceived financial constraints, they reveal remarkably little appetite for a broad shift in household saving. Across all countries, 37% expect to save as before, 27% plan to save more and 28% expect to save less, while 8% remain undecided (Figure 10). The

pattern is broadly similar across countries, with France standing out as the only country where the largest group plans to save less (38%) and the US as the only one where the largest group intends to save more (39%). These differences partly reflect national saving patterns. Household saving rates remain above their pre-pandemic levels in every country except the US, leaving American households with the most ground to recover after the post-Covid decline.

Figure 10: Will your savings behavior change?, share of respondents by country in %



Source: Allianz Research

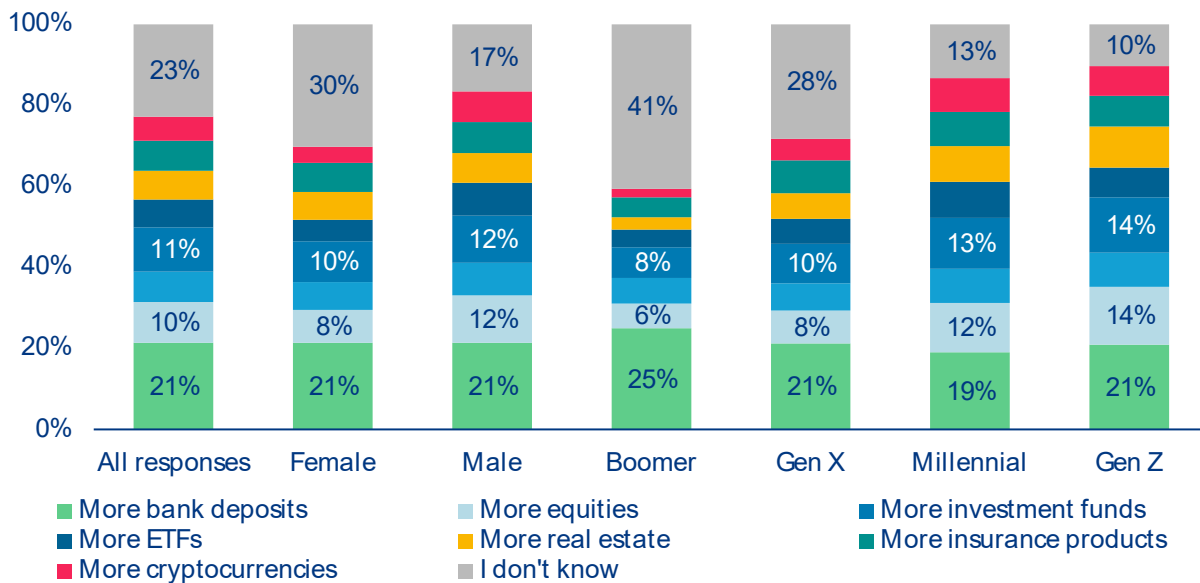
Figure 11: Will your savings behavior change?, share of respondents by country in %

Source: Allianz Research

A much clearer pattern emerges across generations (Figure 11). More than half of Gen Z respondents (53%) and over one-third of Millennials (35%) intend to save more, compared with just 19% of Gen X and 11% of Baby Boomers. Conversely, the share planning to save less rises steadily with age, from 12% among Gen Z to 38% among Boomers, closely reflecting the life-cycle hypothesis and observed saving behavior. The broader message, however, is that most households do not expect to change how much they save. If long-term wealth creation is to improve, the greater opportunity lies not in saving more, but in investing existing savings more effectively.

The survey suggests that investment choices are where the real differences begin. When asked which products they intend to give greater weight in future, men allocate 40% of their choices to capital-market instruments – equities, bonds, investment funds and ETFs – compared with 30% among women (Figure 12). Bank deposits remain the single most popular choice for both groups (21%), while real estate and insurance attract similar shares. The largest difference is uncertainty: 30% of women answer „I don't know,“ almost twice the share among men (17%). The much higher share of undecided responses echoes the confidence gap identified earlier.

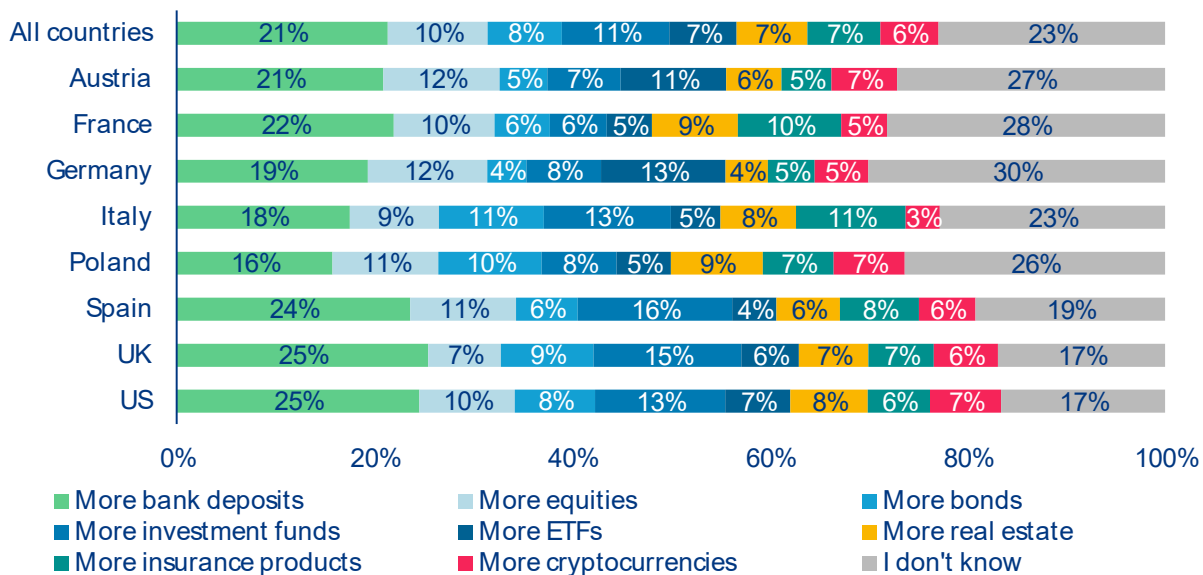
However, these results come with a caveat: They are stated intentions rather than actual portfolio allocations, but they nevertheless suggest that differences in wealth accumulation are likely to arise less from access to financial markets than from knowledge and confidence in navigating them.

Figure 12: Which savings products will you give in future more weight in your portfolio?, share of responses by gender and generation in %

Source: Allianz Research

Younger generations and male respondents have the highest appetite for capital-market investments. Those products account for 44% of the intended allocations of Gen Z respondents and 42% of Millennials, compared with 30% for Gen X and just 24% for Baby Boomers. At the same time, indecision rises sharply with age, from only 10% among Gen Z to 41% among Boomers. Even among Baby Boomers, however, securities remain the preferred choice over deposits in most countries, with the UK and France the only exceptions. Millennials also show a stronger preference for real estate, consistent with being at the life stage when home ownership is still being built. The intersection of age and gender reinforces these

differences (Figure 13). Older women are by far the most hesitant and least knowledgeable investors, with more than half answering „I don't know“ in several countries. In Poland, as many as 64% gave this answer, although younger Polish women increasingly resemble their male peers, generally preferring securities over deposits and reporting much lower levels of uncertainty. Men remain more decisive across all generations. Taken together, the findings suggest that future differences in household wealth will depend less on how much people save than on how confidently and effectively they put those savings to work.

Figure 13: Which savings products will you give in future more weight in your portfolio?, share of responses by country in %

Source: Allianz Research

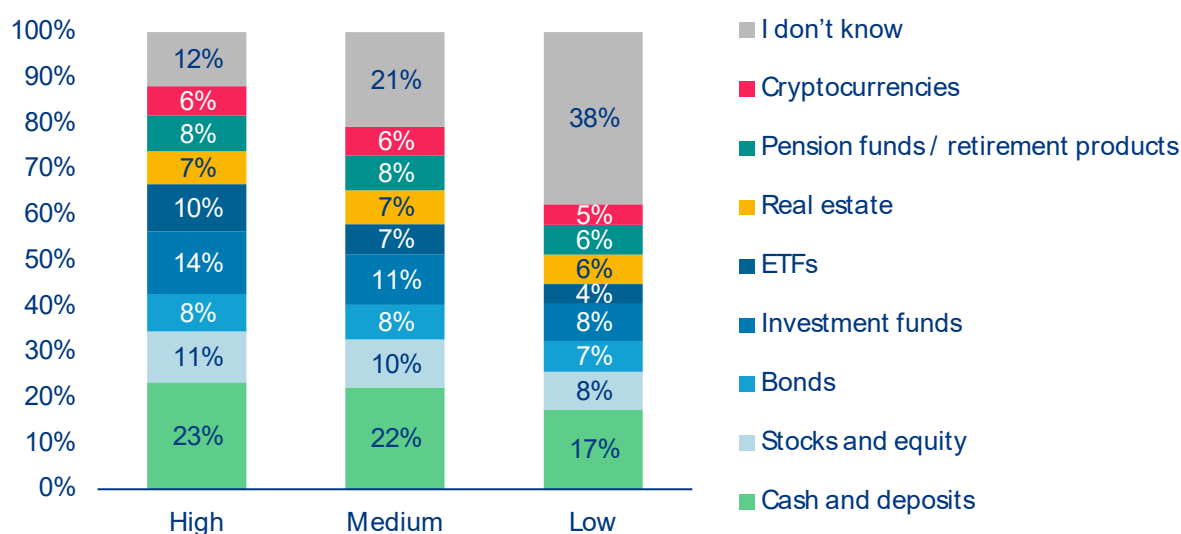


The price of not knowing is high

While many people know they should save more, a considerable share – cutting across both generational and gender lines – do not know where to put their money. When asked which savings products they plan to give more weight to in the future, and sorting the answers by financial literacy – low, medium, and high – a clear and consistent pattern emerges: financial literacy plays a decisive role in shaping where people plan to allocate their savings. While traditional products remain

broadly relevant across all groups, more sophisticated investment instruments are disproportionately favored by the financially literate (see Figure 14).

Figure 14: Which savings products will you give more weight to in your portfolio in the future?, share of responses by financial literacy level in %



Source: Allianz Research

The more you know, the more you diversify. The intention to increase exposure to equities, investment funds and particularly exchange-traded funds (ETFs) rises markedly with financial literacy. Less financially literate households show lower propensity to move into these asset classes. ETFs, long favored by cost-conscious and informed investors alike, attract nearly three times as much enthusiasm from high-literacy respondents as from low-literacy ones – reflecting not only greater familiarity with these instruments, but also a more strategic approach to long-term wealth building.

Not everything divides along literacy lines. Products such as real estate and pension or retirement-related savings display a much flatter distribution across literacy groups, driven less by financial knowledge and more by structural factors such as life-cycle considerations or institutional frameworks. Cryptocurrencies tell a similar story: the intention to increase exposure to crypto assets shows only limited variation across literacy levels, suggesting that participation in this segment reflects broader trends, accessibility or speculative motives rather than financial knowledge.

Cash and deposits tell a surprising story. Safe and liquid assets continue to feature prominently across all levels of financial literacy, suggesting that precautionary motives shape saving behavior across the board, regardless of financial sophistication. Contrary to what one might expect, it is the most financially literate respondents who express the greatest preference for cash and deposits (almost 41%), ahead of both the medium- and low-literacy groups (nearly 25% and 35%, respectively). This likely reflects not only precautionary savings but also a more deliberate portfolio management mindset – using liquid assets as a conscious strategic component rather than a default.

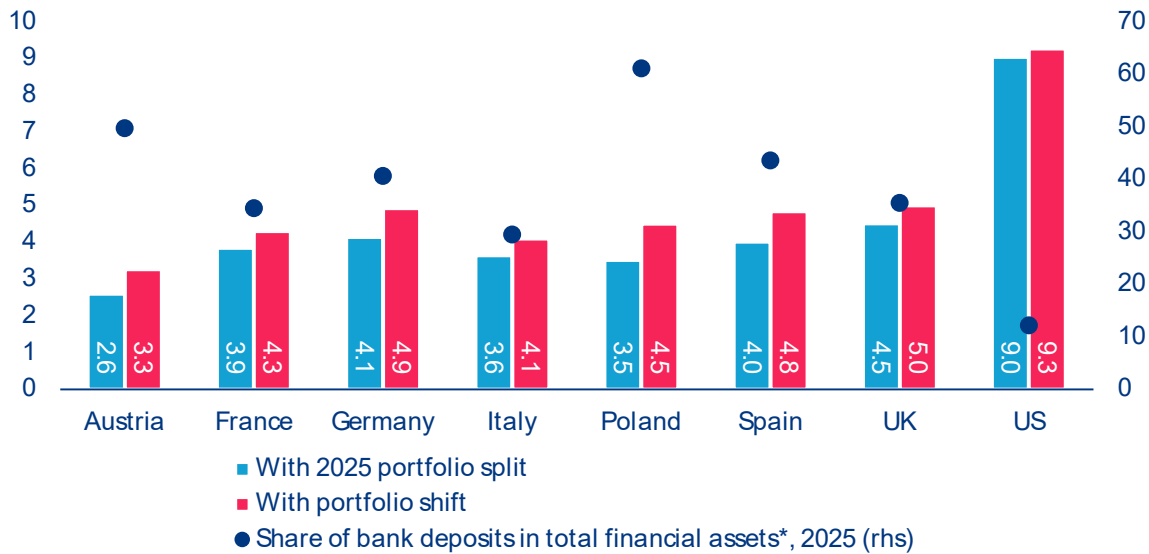
Uncertainty is the real dividing line. The starkest finding in the data is the share of respondents who answered “I don’t know.” Among low-literacy respondents, more than half (almost 53%) could not identify a savings product they would prioritize. This figure drops to roughly 33% for medium-literacy and just under 21% for high-literacy respondents. A lack of financial knowledge does not just influence what people invest in – it leaves many unable to make any informed choice at all. Rather than being put to work, savings remain parked in cash or low-interest bank deposits.

Knowing matters – but how much exactly? To put a number on it, we constructed a simple simulation based on the current average portfolio structure of households across the eight countries in our survey. If the average saver had invested their financial wealth¹² in 2005 according to their current portfolio allocation and held it for 20 years, what would their average annual return have been? We then make a key comparison: what if those same households had made one modest adjustment – redirecting half of their bank deposit holdings, in equal parts, into national bonds and equities? In every country, the answer points the same way: they would have done better. The return gap tracks closely with the share of bank deposits in total financial assets – a relationship that is as intuitive as it is consequential (see Figure 15). Poland, where deposits account for more than 60% of household financial wealth, sees the largest improvement: an actual nominal return of 3.5% per year would have risen to 4.5%. Germany and Austria, both carrying deposit shares well above 40%, would have gained 0.8 and 0.7pp respectively. The United States, where deposits represent only around 12% of financial assets and equity ownership is underpinned by the widespread use of employer-sponsored retirement plans such as the 401(k), starts from a much higher base – 9.0% per year – and has correspondingly little to gain from rebalancing. The pattern is consistent: the more a household relies on deposits, the more return it leaves behind.

Stretched over two decades, modest annual differences add up to significant wealth gains. These additional returns – and the financial surplus they generate over time – require no additional saving from current income: they arise solely from reallocating existing wealth from low- to higher-yielding assets. Countries with the highest deposit shares would see double-digit wealth gains adjusted for inflation: on average around 14% in Austria, 16% in Germany, and over 21% in Poland. In per capita terms, the nominal financial surplus, or unadjusted cash savings, ranges from an average of nearly EUR1,600 in Poland, where financial wealth is still comparatively small, to almost EUR10,300 in Austria and just under EUR17,600 in Germany (see Figure 16). Once inflation is stripped out, the range is from around EUR800 in Poland to nearly EUR6,100 in Austria and around EUR11,700 in Germany. A word about the US households: as they have far greater financial wealth than their European counterparts, even a small improvement in returns compounds to give the largest per capita wealth gain in our sample – around EUR24,800 in nominal terms and EUR15,000 in real terms.

¹² Average per capita wealth, regardless of level of financial literacy; including currency and deposits, listed and unlisted shares, investment fund shares, and insurance and pension assets.

Figure 15: More deposits, more foregone return, average annual nominal returns (2005 – 2025) and share of bank deposits (2025) by country, in %



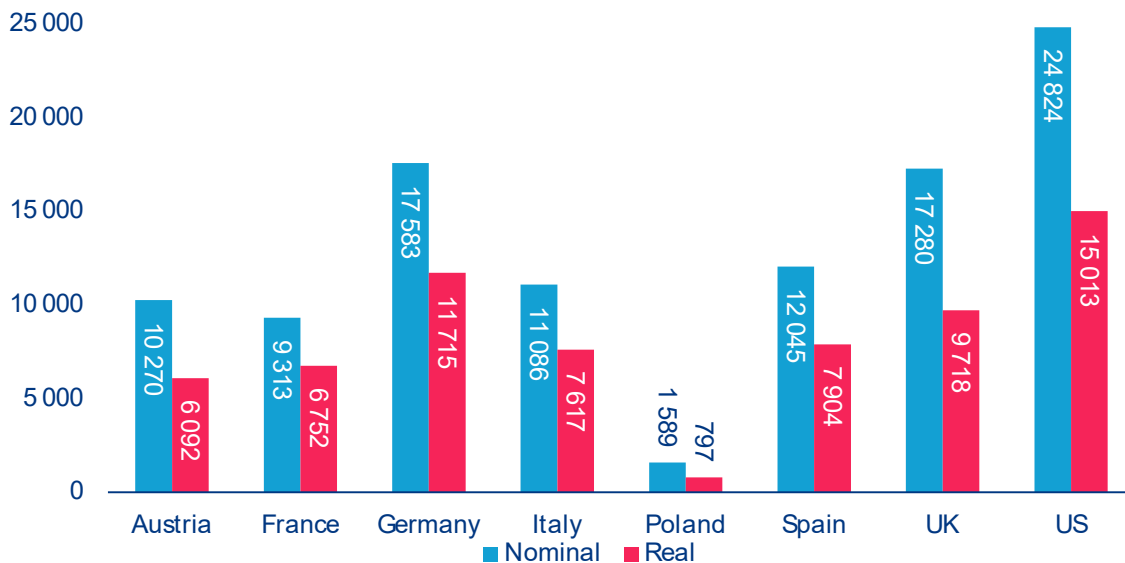
Source: Allianz Research

A single reallocation, held for 20 years, adds up.

Nominal or real, the figures point in the same direction: the cost of excess deposits is large, and it grows with every passing year. Improving financial literacy does not only shape portfolio composition, it can have a meaningful and lasting impact on long-term wealth accumulation. Yet as the high share of „I don't know“

responses reminds us, the first obstacle lies in households not knowing how to allocate their savings, leaving potential returns on the table.

Figure 16: The wealth gap, nominal and real gains from a portfolio shift, in Euro per capita



Source: Allianz Research

Box 2: From awareness to action: the case for portfolio reallocation among risk-averse households

Financial knowledge does not automatically translate into better financial decisions. Our survey shows that even households with relatively strong financial knowledge continue to hold portfolios heavily tilted towards cash. Among respondents who prioritize preserving capital and securing stable retirement income (64.9%), the dominant investment response is not greater diversification but higher cash holdings. Bank deposits are the single most frequently chosen asset, selected by 38.7% as their primary intended allocation, suggesting that many households equate lower risk with holding more cash rather than constructing a more diversified and efficient low-risk portfolio. Mapping these investment intentions onto a simplified three-asset framework implies an allocation of approximately 41% cash, 38% equities and 21% bonds, compared with an average European household allocation of 34% cash, 45% equities and 21% bonds.

The main barrier is behavioral, not informational. Almost 70% of risk-averse respondents correctly answer the survey's inflation question, the highest share of any respondent group. Yet those who understand inflation are more, not less, likely to maintain large cash balances (19.3% vs 13.1%). This suggests that the persistence of cash-heavy portfolios reflects behavioral rather than informational barriers. Our findings point to three distinct behavioral personas that arrive at similar cash-heavy portfolios through different decision-making processes, each requiring a different approach to improve financial outcomes.

Persona 1 – Liquidity seekers: Excess cash reflects precaution, not poor financial literacy. More than half of cash-heavy households (56%) expect living costs to rise (78.5%), fear their income will fail to keep pace (69.0%) and anticipate drawing on savings to finance future spending (26.3%). Their preference for cash therefore reflects precaution rather than a misunderstanding of investment principles. Their portfolio challenge is not excessive risk-taking, but distinguishing between emergency savings and long-term investments. From a portfolio perspective, moving closer to the efficient frontier would achieve either higher expected returns at the same level of portfolio volatility or lower portfolio volatility for the same expected return by replacing part of excess cash holdings with high-quality bonds.

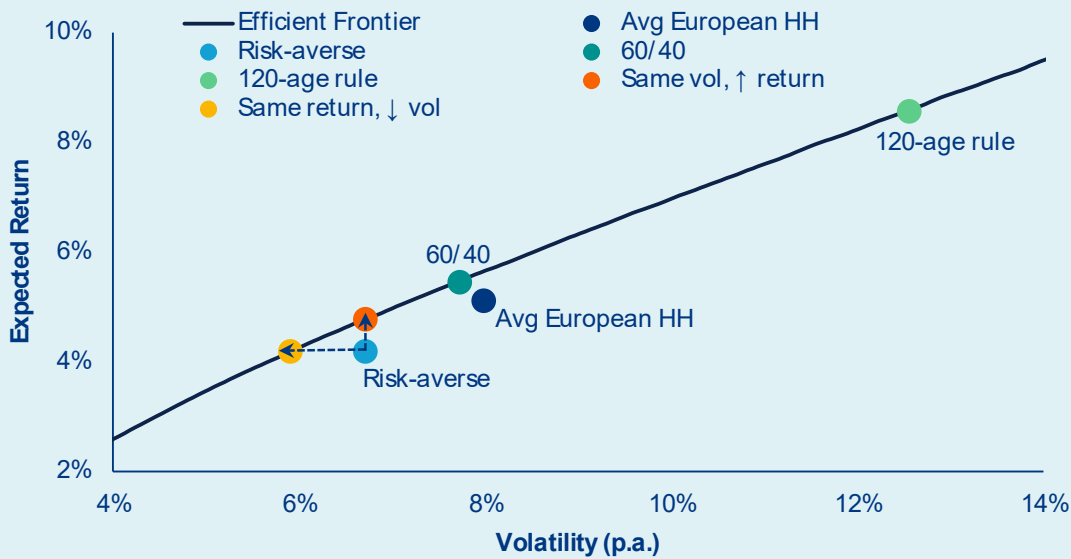
Persona 2 – Procrastinators: Knowing there is a retirement gap does not guarantee action. Nearly two-thirds (64%) expect a moderate or significant pension gap but continue to rely primarily on bank deposits. Unlike Persona 1, the issue is not immediate liquidity needs but delaying investment decisions, allowing today's portfolio to become tomorrow's default. Their portfolio challenge is not recognizing the retirement gap, but acting on it early enough to benefit from long investment horizons. Gradually reallocating long-term savings from excess cash into a more diversified mix of bonds and equities would better align their portfolios with those horizons.

Persona 3 – Inertia investors: Many households stay in cash because inertia becomes the default. Around one-fifth of cash-heavy households (19%) cannot be explained by either liquidity concerns or retirement expectations. Unlike Persona 2, they have not consciously decided to postpone investing – they simply remain with the default allocation, reflecting inertia rather than conviction. Their portfolio challenge is not selecting the optimal allocation, but moving away from an inefficient default. For many investors, a broadly diversified portfolio, such as a traditional 60/40 equity-bond allocation, would represent a more appropriate long-term starting point than maintaining excessive cash balances.

Despite different behavioral motivations, all three personas converge on the same portfolio conclusion: replacing excess cash with bonds increases long-term wealth without increasing portfolio volatility. The efficient frontier analysis quantifies the cost of excess cash holdings. Using estimated parameters from historical monthly returns, annualized at 2.5% Euro-area inflation and 100,000 simulated portfolios, both Pareto-efficient portfolios allocate zero cash, confirming that at the risk-averse group's volatility level (around 6.7% annualized), cash provides negligible volatility reduction while generating near-zero real returns¹³. Over the full period from 2002 to 2025, these more efficient portfolios compound to approximately 1.80–1.95× real wealth, compared with 1.65× for the observed risk-averse portfolio – a gain of 15–30 percentage points, driven primarily by eliminating the long-term drag from excessive cash holdings. Notably, the more efficient portfolios also hold less equity (27–33% vs 38%), illustrating that improving outcomes does not require taking more equity risk, but constructing a more efficient bond allocation (Figures 17 and 18).

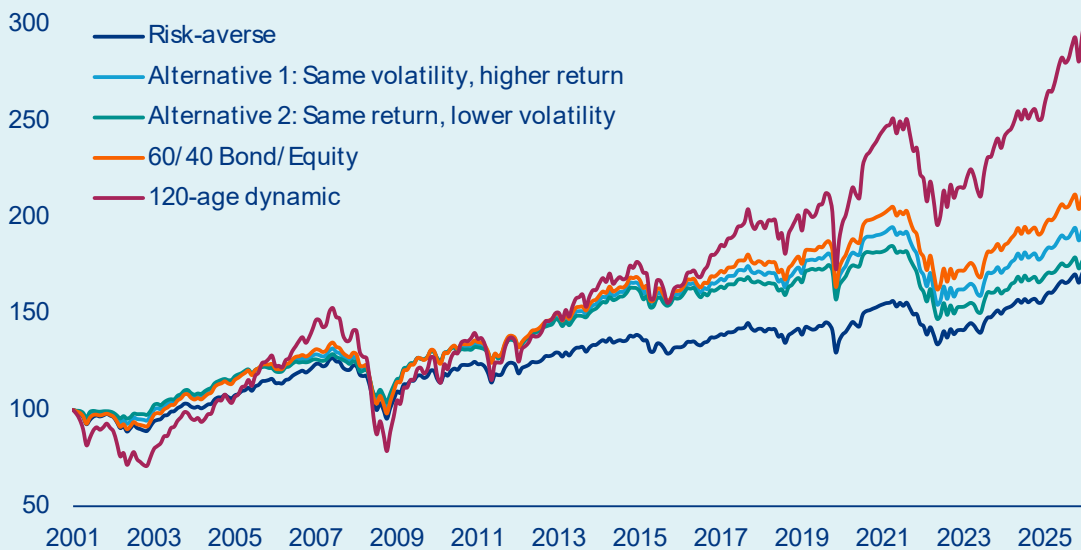
¹³ Pareto-efficient portfolios lie on the efficient frontier, meaning no other portfolio offers a higher expected return for the same level of risk or a lower level of risk for the same expected return.

Figure 17: Efficient Frontier: Improved portfolio allocation for a risk-averse household, expected return and volatility (p.a.)



Sources: LSEG Workspace, Allianz Research

Figure 18: Wealth accumulation over time: Cash-drag with significant underperformance, real return performance index (base year 2001 = 100)

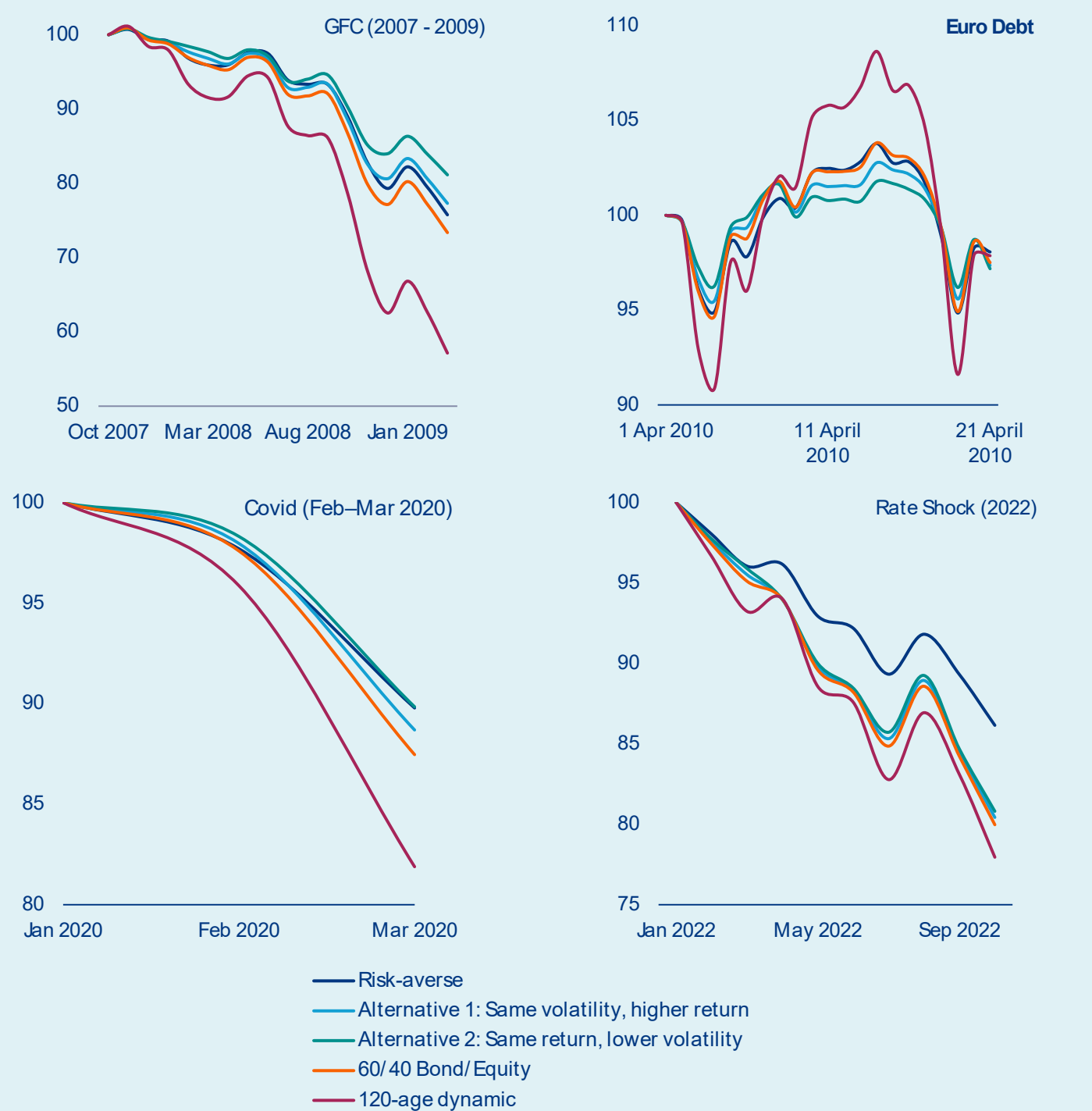


Sources: LSEG Workspace, Allianz Research

The same conclusion holds across the life cycle: reducing equity exposure with age is appropriate, but replacing it with cash rather than bonds is not. Cash hoarding rises steadily with age, increasing from 8.8% among respondents aged 18–35 to 18.1% among those aged 66 and above, with an average hoarder being 53 years old. This pattern is consistent with standard life-cycle investing principles. The widely used 120-age rule recommends reducing equity exposure over time, implying around 71% equity for the average survey respondent aged 49, falling to well below 50% for individuals in their late 60s and 70s. Historical simulations confirm that a life-cycle portfolio following this rule delivers the strongest cumulative real wealth accumulation over the full sample period, but also the largest drawdowns – including losses of around 40% during the Global Financial Crisis – making it appropriate only for younger, risk-tolerant investors. For households approaching retirement, reducing equity exposure is entirely appropriate; the mistake is replacing equities with cash, rather than high-quality bonds, which provide a more efficient balance between capital preservation and long-term wealth accumulation (Figure 19).

The portfolio solution is universal, but the behavioral barriers are not. Across all three personas, replacing excess cash with high-quality bonds improves long-term retirement outcomes. What differs is not the portfolio recommendation, but how households can be helped to implement it. Precautionary savers need reassurance that investing surplus savings does not compromise their financial buffer. Passive planners need personalized projections and trusted advice that connect today's savings choices with tomorrow's retirement income. Default investors need better investment defaults rather than another decision to make. The broader implication is that improving household wealth requires more than financial education alone. It requires decision environments that help people translate financial knowledge into action.

Figure 19: Performance during crisis: Cash-holding does not protect from losses, real return performance index (base year = 100)



Sources: LSEG Workspace, Allianz Research



From advisor to algorithm: the new financial advice ecosystem

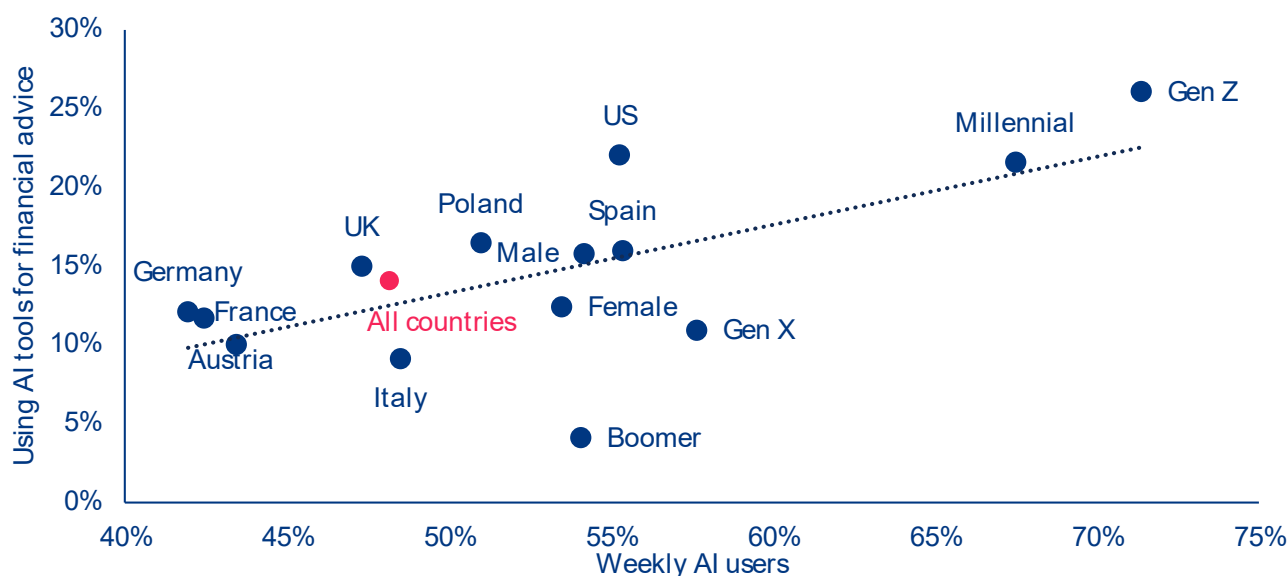
Financial advice is undergoing a quiet transformation.

A generation ago, households turned primarily to banks, professional advisors, the financial press or family when making important financial decisions. Those channels remain important today. But two new entrants have rapidly joined the ecosystem: financial influencers, who reach millions through social media, and artificial intelligence, which places an always-available assistant behind a simple prompt.

AI is already mainstream, but its use for financial

advice remains limited – for now. Only 14.1% of respondents name AI as one of their two main sources of financial guidance, despite almost half (48.1%) using AI at least weekly and more than three-quarters of Gen Z (77.1%) doing so. This suggests that most people still draw a clear line between asking AI general questions and trusting it with their money. That line is blurring fast among the young: more than one in four Gen Z

Figure 20: Weekly AI use vs. using AI for financial advice, by country, gender and generation, share of respondents in %.



Source: Allianz Research

respondents (26.1%) and more than one in five Millennials (21.6%) already count AI among their main advice sources, rising to 30.9% of young men – and past 40% in Poland and nearly 38% in the US – compared with only 22.7% of young women. Financial influencers reach almost exactly the same demographic, so that for a growing share of young households the advice ecosystem is now as much TikTok, podcasts and chatbots as a bank or a personal advisor.

The most striking finding is a mismatch between two things that ought to move together, but don't: AI users are far more confident about their financial knowledge, yet not significantly more financially literate (Figure 21). On confidence, the gap is wide and consistent. 42.5% of AI users rate their financial knowledge as average or above average, compared with 33.3% of all respondents. Only 18.8% of those who use AI for financial advice feel they know “much less” than the average investor against 25.0% of all respondents, while the share believing they know more rises from 7.6% to 11.4%. Uncertainty falls sharply too: just 7.4% of AI users answer “I don’t know,” compared with 18.7% of the full sample.

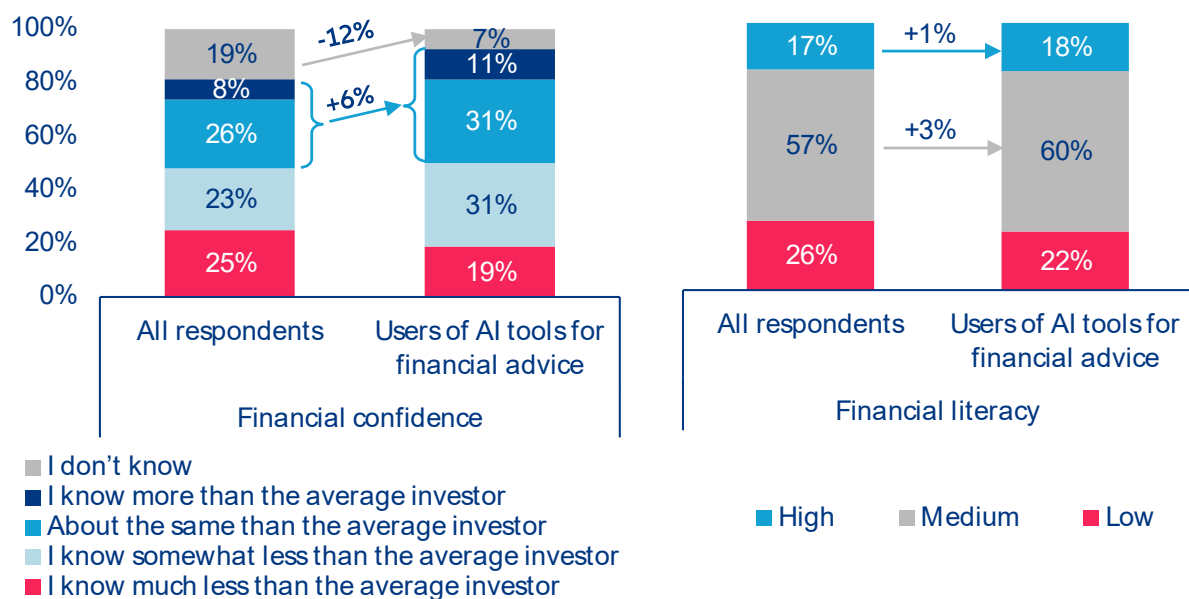
Measured literacy, however, differs little: 18.1% of AI users reach the high-literacy band, compared with 17.5% of the overall sample (Figure 21). The confidence premium therefore far exceeds the literacy premium. Whether AI genuinely deepens understanding or merely

inflates self-assurance cannot be settled from the survey, but the mismatch deserves attention: financial decisions are shaped not only by what people know, but by what they believe they know. AI can make financial information easier and cheaper to access, but it cannot substitute for the financial literacy needed to assess that information and make better financial decisions. This creates the risk of overconfidence if users become more certain without becoming better equipped to judge the quality of the advice they receive.

This does not mean AI is displacing professional advice so much as splitting along the literacy divide.

For financially literate households, AI can complement professional advice, making information more accessible and easier to process. For less literate households, however, it may instead serve as a substitute for professional advice seen as expensive, inaccessible or intimidating¹⁴. That is where the risk concentrates: the users leaning hardest on AI, trusting it most, and least able to judge its answers are disproportionately those with the least financial knowledge. The distinction will be one of the defining questions for financial inclusion over the coming decade. AI can democratize access to financial information at unprecedented scale, but access alone does not create financial capability. The task for policymakers, educators and AI providers alike is to ensure AI does not merely make advice more available, but helps people make better decisions.

Figure 21: Confidence and financial literacy, AI users vs. the overall sample, share of respondents in %.



Note: Confidence indicators move sharply in AI users' favor while literacy indicators do not – the confidence premium exceeds the literacy premium.
Source: Allianz Research.

¹⁴ Recent research on AI financial advice supports this distinction: AI-generated financial advice can move household saving, spending and investment decisions closer to standard economic benchmarks, but outcomes depend on users' financial literacy and ability to formulate effective prompts. More financially literate users received advice associated with higher simulated wealth, while less financially literate users fare worse. The authors conclude that regulators and fintech platforms might need to address disparities in AI-driven advice. See: Choukhmane, T., T. de Silva, W. Lin, and M. Akuzawa (2026): "AI Financial Advice: Supply, Demand, and Life Cycle Implications", https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6446286.

What can be done: a five-point financial literacy playbook

Our follow-up survey, three years on, delivers an uncomfortable verdict: financial literacy has improved little, while the need for it has never been greater. While the responsibility for financial security does not sit with households alone, people are being asked to take greater responsibility for their financial future, and need the tools and knowledge to make informed decisions. Financial literacy is therefore no longer simply about understanding financial concepts – it has become a prerequisite for building, protecting and growing wealth over a lifetime. Yet awareness campaigns and one-off initiatives have not moved the needle. Closing persistent literacy gaps requires a new policy playbook.

Point 1: Build financial capability throughout life.

Financial capability should be treated as a lifelong skill, embedded in financial education across school curricula while expanding opportunities for adult learning to keep up with evolving financial products, pension systems and digital technologies. Employers should reinforce these foundations through workplace financial education, while financial institutions can support learning with accessible educational content and practical decision-making tools. One example is the Allianz School for Life, a free, open-access digital learning platform designed to make financial and risk education widely accessible¹⁵. The objective is not simply higher test scores, but equipping people with the knowledge required to navigate increasingly complex financial decisions throughout their lives.

Point 2: Make future outcomes visible. Knowing financial concepts is rarely enough if people cannot see the consequences of today's choices. Tools to support decision-making should therefore become part of the financial infrastructure. Governments can accelerate the rollout of pension dashboards and personalized retirement projections. Employers can complement these with workplace pension guidance and retirement planning. Financial institutions should provide digital tools that illustrate how different saving and investment decisions affect future retirement income and long-term wealth. Making future outcomes tangible is likely to change behavior more effectively than providing additional information alone.

Point 3: Turn knowledge into action. Our findings show that the greatest obstacle is often not ignorance but inertia. Many households understand basic concepts, yet continue to hold inefficient, cash-heavy portfolios, or postpone investment decisions altogether. Policies therefore need to make good decisions easier. Employers can use automatic enrollment, regular pension reviews and financial wellbeing programs to encourage action. Financial institutions should simplify products, improve defaults, reduce unnecessary complexity and provide behavioral nudges that help households gradually move from saving, to long-term investing.

Point 4: Build confidence – and healthy skepticism

– in the AI age. Financial literacy increasingly means knowing how to evaluate advice, not simply where to find it. AI has dramatically lowered the cost of accessing financial information, but not necessarily the ability to judge its quality. Financial education therefore needs to help people calibrate their confidence, recognize when professional advice is required and understand AI's strengths and limitations. Policymakers, educators, employers, financial institutions and AI providers all have a role in promoting AI as a complement to trusted professional advice, rather than using it as a substitute.

Point 5: Reach everyone – but focus where the

gaps are largest. Financial literacy policies should be universal in reach, but targeted in delivery. The largest deficits identified in this report are concentrated among women, younger generations and adults with lower educational attainment. These groups require interventions tailored to their circumstances: early financial education before major financial decisions arise; workplace support at key life events; advice and investment tools designed to increase confidence among women; and accessible, practical learning opportunities for adults with lower educational attainment. Universal access ensures broad financial capability while targeted delivery helps close persistent inequalities.

¹⁵ See: [Allianz School For Life](#) and [Allianz MoveNow](#)

Appendix A : Survey summary statistics

The survey queried more than 8,000 respondents across eight countries: Austria, France, Germany, Italy, Poland, Spain, the United Kingdom and the United States. It was conducted as a web-based survey between 22 April and 25 May 2026. Overall responsibility lies with Allianz Research, Allianz Investment Management SE. Sample planning and fieldwork were carried out by Qualtrics.

Respondents were selected using representative quota sampling. Qualtrics was given quotas specifying how many people to survey in each country and which selection criteria to apply, with quotas set in line with official statistics for gender and age groups. A comparison with official statistics confirms that the resulting sample broadly corresponds to the population aged 18 and over in each of the eight countries.

Appendix Table 1: Sample statistics

	Total	Austria	France	Germany	Italy	Poland	Spain	UK	USA
Number of respondents	8048	1006	1006	1006	1006	1006	1006	1006	1006
Gender, in % of respondents									
Female	50.6%	50.0%	51.4%	49.9%	50.7%	50.9%	50.4%	50.9%	51.0%
Male	49.2%	50.0%	48.2%	49.9%	49.1%	49.0%	49.4%	48.9%	48.9%
Age groups, in % of respondents									
Gen Z	15.5%	15.4%	16.1%	13.6%	13.1%	15.2%	14.3%	16.9%	19.6%
Millennial	25.9%	26.8%	24.8%	25.8%	21.9%	27.0%	24.5%	28.3%	27.8%
Gen X	31.5%	34.1%	30.2%	26.2%	35.8%	33.6%	41.1%	25.0%	25.6%
Boomer	27.2%	23.7%	28.9%	34.3%	29.3%	24.3%	20.2%	29.8%	27.0%
Education, in % of respondents									
Primary (less than high school)	4.8%	2.2%	3.7%	2.6%	14.1%	4.0%	2.8%	2.0%	7.2%
Secondary (high school)	52.5%	75.2%	47.9%	69.2%	52.5%	55.2%	36.0%	43.6%	40.1%
Tertiary (university or comparable)	42.7%	22.6%	48.4%	28.2%	33.4%	40.9%	61.2%	54.4%	52.8%

Generation definition	Birth years
Gen Alpha*	2013 - 2031
Gen Z	1997 - 2012
Millennial	1981 - 1996
Gen X	1965 - 1980
Boomer	1946 - 1964
Silent**	1928 - 1945

* Not part of the survey sample.

** Not mentioned in the analysis as it was a very small portion of respondents.

Appendix B: Financial literacy questions

1. Understanding of simple interest rate. Suppose you had USD500 in a savings account and the interest rate was 5% per year. After one year, how much do you think you would have in the account if you left the money to grow?

- a. **USD 525**
- b. USD 555
- c. USD 532
- d. I do not know

2. Understanding of compound interest rate (numeracy).

Suppose you had USD1000 in a savings account and the interest rate was 2% per year, compounded annually. After 5 years, how much do you think you would have in the account if you left the money to grow?

- a. **More than USD 1100**
- b. Exactly USD 1100
- c. Less than USD 1100
- d. I do not know

3. Understanding of inflation. Imagine that the interest rate on your savings account was 5% per year and inflation was 7% per year. After 1 year, how much would you be able to buy with the money in this account?

- a. More than today
- b. Exactly the same
- c. **Less than today**
- d. I do not know

4. Understanding of risk diversification. Please tell me whether this statement is true or false. "Buying a single company's stock usually provides a safer return than a mutual fund."

- a. True
- b. **False**
- c. I do not know

5. Lotteries. Lottery A pays a prize of USD200 and the chance of winning is 5%. Lottery B pays a prize of USD90,000 and the chance of winning is 0.01 %. In either case, if one does not win, one does not get any money. Which lottery pays the higher average amount? (Please pick one option only)

- a. **Lottery A**
- b. Lottery B
- c. These two lotteries pay the same average amount
- d. I do not know

6. You can invest in two projects. Project A will either deliver a return of 10 or 6%, with either outcome equally likely. Project B will either deliver a return of 12 or 4%, with either outcome equally likely. Which of the following is true? Compared to Project B, Project A has... (Please pick one option only)

- a. Higher return and lower risk
- b. **Same average return and lower risk**
- c. Lower return and higher risk
- d. I do not know

7. You want to set aside money today for certain expenses in two years. Which of the following investment offers are suitable:

- a. A fund with an annual return of 7% in the past but also large fluctuations
- b. **A two-years fixed-term deposit with a fixed interest of 1.5% per annum**
- c. I do not know

8. How would you calculate financial net worth?

- a. **Assets minus liabilities**
- b. Liabilities minus assets
- c. Assets plus liabilities
- d. Assets divided by liabilities
- e. I don't know

9. Imagine we are throwing a loaded die (6 sides). The probability that the die shows a 6 is twice as high as the probability of each of the other numbers. On average, out of 70 throws how many times would the die show the number 6?

- a. **20 out of 70 throws**
- b. 30 out of 70 throws
- c. 35 out of 70 throws
- d. I don't know

Allianz School for Life



Allianz School For Life is a free, open-access digital learning platform designed to make financial and risk education widely accessible because better decisions start with better knowledge. The platform offers learning journeys covering budgeting, investing and risk management, tailored to different life stages – from children and teenagers to young adults – and designed to make financial education engaging, accessible and relevant to everyday life. Beyond digital learning, Allianz also supports on-the-ground delivery through local programs, partnerships and community initiatives, helping ensure that financial education reaches people wherever they are.

Check it out:



A photograph showing a variety of hands of different skin tones stacked on top of each other, resting on a rough, textured tree branch. The background is a lush green forest with sunlight filtering through the leaves. The text 'Our team' is overlaid on the image, with 'Our' in white and 'team' in yellow.

Our team

Chief Investment Officer
& Chief Economist
Allianz Investment Management SE



Ludovic Subran
ludovic.subran@allianz.com

Head of Economic Research
Allianz Trade



Ana Boata
ana.boata@allianz-trade.com

Head of Macroeconomic & Capital
Markets Research
Allianz Investment Management SE



Bjoern Griesbach
bjoern.griesbach@allianz.com

Head of Outreach
Allianz Investment Management SE



Arne Holzhausen
arne.holzhausen@allianz.com

Head of Corporate Research
Allianz Trade



Ano Kuhanathan
ano.kuhanathan@allianz-trade.com

Head of Thematic & Policy Research
Allianz Investment Management SE



Katharina Utermoehl
katharina.utermaehl@allianz.com

Macroeconomic Research



Luis Dalmau Taules
Economist for Africa & Middle East
luis.dalmau@allianz-trade.com



Maxime Darmet Cucchiarini
Senior Economist for UK, US & France
maxime.darmet@allianz-trade.com



Jasmin Gröschl
Senior Economist for Europe
jasmin.groeschl@allianz.com



Françoise Huang
Senior Economist for Asia Pacific
francoise.huang@allianz-trade.com



Maddalena Martini
Senior Economist for Southern Europe & Benelux
maddalena.martini@allianz.com

Outreach



Luca Moneta
Senior Economist for Emerging Markets
luca.moneta@allianz-trade.com



Giovanni Scarpato
Economist for Central & Eastern Europe
giovanni.scarpato@allianz.com



Heike Baehr
Content Manager
heike.baehr@allianz.com



Melissa Eddy
Content Manager
melissa.eddy@allianz.com



Maria Thomas
Content Manager and Editor
maria.thomas@allianz-trade.com

Corporate Research



Guillaume Dejean
Senior Sector Advisor
guillaume.dejean@allianz-trade.com



Maria Latorre
Sector Advisor, B2B
maria.latorre@allianz-trade.com



Maxime Lemerle
Lead Advisor, Insolvency Research
maxime.lemerle@allianz-trade.com



Pierre Lebard
Public Affairs Officer
pierre.lebard@allianz-trade.com

Thematic and Policy Research



Michaela Grimm
Senior Economist,
Demography & Social Protection
michaela.grimm@allianz.com



Patrick Hoffmann
Economist, ESG & AI
patrick.hoffmann@allianz.com



Simon Krause
Economist, ESG & Insurance
simon.krause@allianz.com



Hazem Krichene
Senior Economist, Climate
hazem.krichene@allianz.com



Kathrin Stoffel
Economist, Insurance & Wealth
kathrin.stoffel@allianz.com



Markus Zimmer
Senior Economist, ESG
markus.zimmer@allianz.com

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Director of Publications
Ludovic Subran, Chief Investment Officer & Chief Economist
Allianz Research
Phone +49 89 3800 7859

Allianz Group Economic Research
https://www.allianz.com/en/economic_research
<http://www.allianz-trade.com/economic-research>
Königinstraße 28 | 80802 Munich | Germany
allianz.research@allianz.com

 @allianz

 allianz

Allianz Trade Economic Research
<http://www.allianz-trade.com/economic-research>
1 Place des Saisons | 92048 Paris-La-Défense Cedex | France

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Allianz Research | 29 July 2026

Cracking finance: Why the quantum threat could arrive before the reward

Ludovic Subran
Chief Investment Officer and
Chief Economist
ludovic.subran@allianz.com

Olivier Salomon
Head of Quantum Hub
for Allianz SE
olivier.salomon@allianz.com

Bjoern Griesbach
Head of Macroeconomic and
Capital Market Research
bjoern.griesbach@allianz.com

Giovanni Scarpato
Economist for Central and Eastern
Europe
giovanni.scarpato@allianz.com

Jordi Basco Carrera
Head of Private Markets
Investment Strategy
jordi.basco_carrera@allianz.com

Simone Vittorio Percoco
Research Assistant
simone.percoco@allianz.com

In Summary

- **Most technologies create value before they create problems, but quantum computing may break that pattern.** Imagine two clocks ticking side by side: one measures the time until quantum computers become economically useful, the other the time until they can break the cryptography that underpins the financial system. Which clock strikes first is a genuinely open race.
- **The upside potential is real but narrowly concentrated around derivative pricing and other complex financial simulations. Beyond these areas, its impact on the financial industry is likely limited.** Quantum computing accelerates Monte Carlo methods behind financial simulations, while it shows no consistent advantage yet over improving classic methods for portfolio optimization and financial machine learning. Insurers are well positioned as simulation and optimization are at the heart of liability valuation, risk pricing, capital modelling and asset-liability management. Quantum money could use uncopyable quantum states to authenticate value and prevent counterfeiting, although commercial deployment remains remote. The rewards remain theoretical for now, threatened by error-correction overhead and a rising classical benchmark, until large machines exist. But quantum-inspired methods deliver operational value on classical hardware today, and the talent, tooling and know-how built along the way will carry over.
- **Q-day could come sooner than we think: data harvested today can be decrypted later.** The recipe for breaking today's encryption has existed since 1994 and waits only for the machine, which is also the smaller one, needing roughly 1,250–1,450 logical qubits vs around 4,700–7,500 for leading financial applications. Health records, classified government information and sensitive financial data must stay confidential for decades, but experts put the probability of Q-day, the point at which a cryptographically relevant quantum computer exists, arriving within ten years at 28–49%. Defensive standards have existed since 2024, and regulators have set dates: The US National Institute of Standards and Technology (NIST) will phase out today's algorithms after 2030, the EU requires critical financial infrastructure to migrate by the same year.
- **Digital assets make the vulnerability visible: USD400bn of Bitcoin is already at risk, and that is just the tip of the iceberg.** By spring 2026, roughly 6m bitcoins, 30% of supply, sat behind exposed public keys, and Ethereum extends the exposure to stablecoins, bridges and tokenized assets. Spillover to mainstream finance would be systemic: The BIS puts losses above 1% of GDP over 15–20 years, while a severe Hudson Institute scenario reaches up to USD3.3trn in GDP losses (3% of global GDP) over 6 quarters.
- **Both private and public sectors are visibly chasing the reward, while concerns about the threat play out mostly behind closed doors.** Investors bet mostly on the hardware: about 97% of each dollar funds the machine and under 1% the applications, on an estimated USD12.6bn raised by quantum startups in 2025. Public efforts reveal a geographic imbalance: The US prepares for reward and threat via parallel executive presidential orders, while Europe, which draws just 5% of private quantum investment versus over half for the US, pursues its own response through fragmented policy instruments, risking a repeat of its AI experience as adopter and regulator of a technology it does not own. Preparation for the threat is purely an internal IT budgeting decision, competing resources with AI, and only about 35% of large organizations completed a cryptographic inventory. Institutions need crypto-agile architectures, prioritized inventories, migration roadmaps and tested fallbacks before the clock reaches zero: preparing for only one clock is the wrong strategy, regardless of which strikes first.

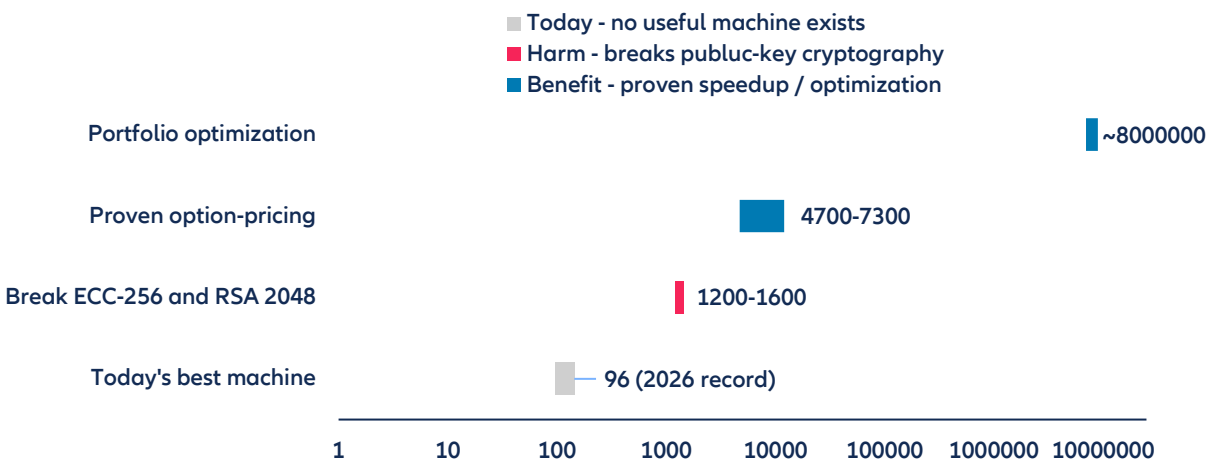
Two clocks ticking at different speeds

Most technologies create value before they create problems. The automobile delivered mobility before congestion became a policy concern; the internet created a digital economy before it created cybercrime. Even when the benefits arrived slowly – electrification took decades to lift factory productivity, and computers were once, as Robert Solow famously observed, visible "everywhere but in the productivity statistics" – the direction was the same: the rewards came first, and the risks followed. Quantum computing may break that pattern. Imagine two clocks ticking side by side. One measures the time until quantum computers become economically useful; the other, the time until they can break the cryptography that underpins the financial system. There is a credible case that the threat clock is running at least as fast as the reward clock, and that institutions, drawn to the visible upside, are quietly preparing for only one of the two.

To see why the two clocks can run at different speeds, start with the machine itself. A quantum computer is not simply a faster version of a conventional computer, nor does it "try every answer at once". It is a different class of machine that can solve a narrow set of mathematical problems in ways that remain intractable for conventional systems. Its basic unit is the qubit, the quantum counterpart of the classical bit. Whereas a conventional bit must be either 0 or 1, a qubit can occupy a quantum state that allows information to be processed differently. The difficulty is that qubits are fragile. The physical qubits built in today's laboratories are sufficiently error-prone that long computations quickly dissolve into noise. Useful large-scale quantum computing therefore depends on error correction: bundling large numbers of physical qubits into a much smaller number of stable logical qubits. Logical qubits, rather than raw physical-qubit counts, are the more meaningful measure of what a machine can actually compute. Even then, quantum advantage is highly specific. Most operations performed by banks, insurers and asset managers would see little benefit from quantum hardware. Yet one of the problems for which quantum computers are exceptionally well suited is factoring and discrete logarithms, the mathematical foundations of much of today's public-key cryptography.

On paper, and only against the most space minimized estimates, the code-breaking machine can be the smaller one. Attacking the elliptic-curve cryptography behind a Bitcoin key has been estimated to require roughly 1250–1,450 logical qubits, depending on circuit design and the trade-off between qubit count and runtime. The lowest estimates for breaking RSA-2048, a distinct problem from the elliptic-curve case, have fallen to a similar order of magnitude as architectures and algorithms have improved, though the standard single-shot figure stays several times higher. By comparison, a leading fault-tolerant proposal for pricing and risk using amplitude estimation requires roughly 4,700–7,500 logical qubits, while a general and economically relevant portfolio-optimization advantage may require far more resources and has not yet been demonstrated (Figure 1).

Figure 1: Estimated number of logical qubits required for different tasks



Sources: Babbush et al. (2026), Gidney (2025), Gong et al. (2026), Dalzell et al. (2023), Allianz Research. Note: The estimation of the number of logical qubits for portfolio optimization comes from a study using a method called the quantum interior point method (QIPM). Under this method, the number of qubits scales as $800n^2$, where n is the number of assets in the portfolio.

A simple count of logical qubits, however, does not settle the matter. Gate counts, fidelities and circuit depth matter as much as qubits and the deeper asymmetry is competitive rather than technical. For breaking cryptography, a quantum computer is essentially the only tool that can do the job; there is no classical shortcut, so the code-breaking machine only must exist. For finance, quantum must compete: classical hardware and algorithms already perform these calculations and keep getting better, so even a machine that is smaller on paper must clear a far higher bar to be worthwhile. Cryptographic attacks thus appear to be among the smallest clearly specified fault-tolerant applications, while the best-developed financial use cases demand larger and faster machines and must still beat a moving classical benchmark that the attacker never faces. The same asymmetry shapes the economics of building the machines: money spent chasing a financial edge competes against cheaper classical improvements, whereas money spent on defense buys protection against something only quantum can do.

The decisive asymmetry, though, is about recipes rather than machines. Shor's algorithm, the quantum procedure published by Peter Shor in 1994 that efficiently factors large numbers and solves the related discrete-logarithm problem on which public-key cryptography rests, has existed already for three decades. It is proven, published and has a clearly defined target: given sufficiently capable fault-tolerant hardware, it can recover the private keys behind the public-key cryptography, which authenticates and encrypts nearly every digital exchange, whether a bank transfer, a private message or a social network post. The remaining uncertainty is about the time when a machine with scale, speed and error correction sufficient to run the algorithm will exist.

Financial quantum algorithms are not absent, but their route to economic value is far less settled. Quantum amplitude estimation offers a firm mathematical basis for accelerating Monte Carlo calculations, while other approaches target optimization, machine learning and linear algebra. Yet for amplitude estimation the potential gain is quadratic rather than exponential, and for optimization and machine learning many claimed advantages remain heuristic or highly problem specific. The destructive use case already has its recipe and waits mainly for the machine. Productive uses need two things at once: the very hardware the adversary is also waiting for, and financial algorithms that still have to be found, implemented and shown to outperform strong classical methods. However, the two clocks compete on different dimensions, both worth exploring, and it is hard to say which clock will strike first.

The reward clock

The strongest case for quantum computing in finance starts from a simple observation: the industry already devotes enormous computing capacity to simulating uncertainty. Derivative pricing, market-risk measurement, economic-capital calculations and stress testing rely heavily on Monte Carlo methods – repeated simulations of thousands or millions of possible future outcomes. Quantum amplitude estimation can, in principle, achieve a given level of statistical accuracy with fewer evaluations than classical Monte Carlo. The potential result is not a wholly new calculation but the ability to perform existing calculations faster, more frequently or at greater resolution: compressing overnight risk runs, refreshing capital positions more often and evaluating a broader range of scenarios. Because Monte Carlo methods are used for a variety of applications, like pricing complex derivatives, calculating Value-at-Risk and expected shortfall, estimating counterparty exposure or measuring economic capital, the industry is unusually well aligned with one of the few quantum applications supported by a firm mathematical speedup.

The gain should nevertheless not be overstated. A quadratic acceleration can be valuable, especially for deeply nested simulations, but the moving classical benchmark takes concrete form here: specialized chips, GPUs, distributed cloud infrastructure, variance reduction and continually improving classical algorithms. A commercially useful quantum system must outperform them after incorporating error correction, data loading, state preparation and the cost of the full workflow. This is the single most important test any financial quantum application has to pass, and it is why even a genuine speedup on paper may not translate into a genuine advantage in practice.

Insurers sit at a competitive intersection: the one quantum advantage with the strongest mathematical foundation maps directly onto both their heaviest regulatory calculations and parts of their investment process. On the underwriting and capital side, valuing life, health and long-tail property-and-casualty liabilities can require simulations extending decades into the future. Those valuations feed reserving, stress testing, economic-capital calculations and Solvency II coverage ratios, often through layers of nested simulations that become

computationally expensive. Amplitude estimation targets exactly this kind of workload: best-estimate liability valuation, solvency-capital and Value-at-Risk computation and the economic-scenario generation that feeds them. Faster simulation could improve efficiency, but more importantly it could change the cadence of risk management, moving capital and solvency monitoring from a periodic exercise toward more frequent updates and more dynamic steering and hedging. The same techniques carry over to the asset and asset-liability management side. Evaluating a proposed allocation across thousands of economic scenarios, generating those scenarios, and measuring tail and counterparty risk all belong to the same simulation-intensive family that amplitude estimation accelerates.

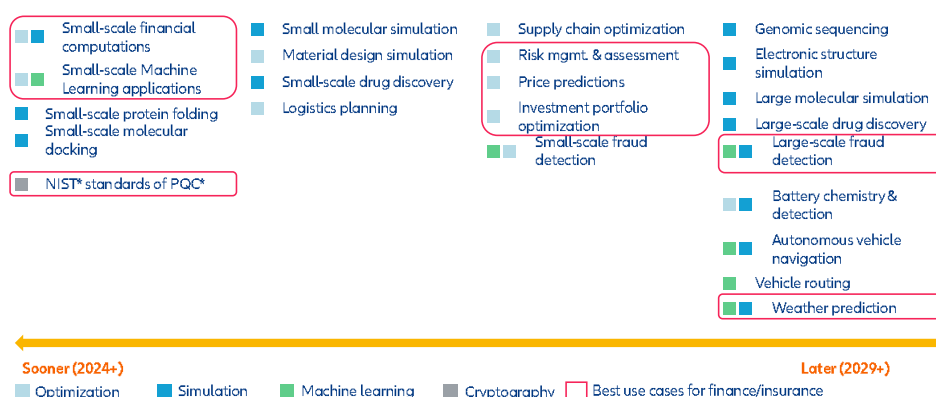
This is also where the wider industry is concentrating its experiments. Goldman Sachs has explored quantum Monte Carlo for derivative pricing; JPMorgan has tested optimization routines for portfolio and collateral problems and Crédit Agricole has piloted hybrid credit-risk models. The honest split is that evaluating an allocation is the part with a proven speedup, whereas choosing the optimal allocation is a combinatorial-optimization problem for which quantum methods have yet to show a robust advantage. The credible opportunity therefore lies first in evaluating allocations faster and with richer scenarios, rather than in expecting a quantum machine to discover the optimal portfolio automatically.

Beyond simulation, the list of proposed applications is long, but the evidence base less so. Corporate and investment banks are testing collateral allocation, balance-sheet optimization and faster pricing systems. Asset managers are exploring portfolio construction, asset modeling and scenario analysis. Payments businesses see potential applications in fraud detection, compliance and, further out, quantum-secure settlement. Retail banks may eventually use quantum or quantum-inspired methods for credit assessment and large-scale decision problems (Figure 2).

At the most experimental end lies quantum money. First proposed by Stephen Wiesner and later developed by Charles Bennett and Gilles Brassard, the concept does not describe a new currency comparable with cryptocurrencies or central-bank digital money, but rather payment tokens encoded in quantum states that cannot be perfectly copied without disturbing them. In principle, quantum mechanics could therefore provide a new way of authenticating value and preventing counterfeiting. Literal quantum banknotes remain technologically remote, however, as they would require fragile quantum states to be stored, transferred and verified reliably, while secure public verification remains an open research problem. More recent experiments have shifted toward quantum payment tokens and one-time cryptograms, using quantum communication to make payment authorization harder to forge without requiring users to preserve a quantum state indefinitely. These approaches remain far from commercial deployment, but they illustrate a broader potential reward.

Most of these opportunities are less certain than the Monte Carlo case. Optimization and machine-learning applications face intense competition from classical solvers that have improved rapidly over the past decade, and a result demonstrated on a small or synthetic problem is not necessarily relevant to an institution operating with noisy data, latency constraints, regulation and legacy infrastructure. The likely economic impact is therefore concentrated in a narrower set of applications than the long lists of potential use cases suggest.

Figure 2: Quantum use cases by expected time to maturity



Sources: Allianz SE Cybersecurity & NextGen IT Think Tank, Arthur D. Little, Allianz Research

Even within that narrower set, value is available today with no quantum hardware advantage. The most dependable gain comes from reformulating the model itself. When you take an actuarial, Asset-Liability-Management or capital model and rewrite it as a QUBO, Ising or tensor form¹, you force an explicit restatement of variables and constraints, surfacing stale constraints and dead risk factors whose corrections feed back into classical production models. Named quantum-inspired solvers are already in commercial use: Multiverse Computing and Crédit Agricole CIB report tensor-network pricing that matches accuracy with fewer parameters and faster training, and Toshiba's Simulated Bifurcation Machine has run low-latency pairs-trading at the Tokyo Stock Exchange. Independent benchmarks find plain simulated annealing comparable in some regimes, and the edge of these solvers comes from specialized hardware rather than a quantum effect. None of this proves a speedup against the moving classical benchmark; the value is real but operational.

In fact, the reward clock has two horizons. A nearer horizon already exists in hybrid quantum-classical workflows and "quantum-inspired" methods that borrow mathematical techniques from quantum computing while running on conventional hardware; platforms such as NVIDIA CUDA-Q couple GPUs with quantum processors and speed up circuit design, simulation and hybrid execution. These gains are operational rather than proof of a quantum speedup, but they lower experimentation costs and help institutions build talent, tooling and credible benchmarks. The far horizon is fault-tolerant quantum advantage: a machine that performs a complete financial task more efficiently than the best classical alternative. Putting a credible number on that long-run reward is not yet possible. The upside is that the reward can therefore arrive the way most technology rewards do: gradually, as capability, tooling and know-how spill over from experimentation into production, with modest gains already available today and a larger advantage possible later. The threat keeps no such schedule. A cryptographic break would not diffuse gradually through the economy, but rather arrive all at once, and its countdown has already begun. It is to that clock that we now turn.

The threat clock

What Shor's algorithm can break extends far beyond any single market. RSA and elliptic-curve schemes secure key exchange, digital signatures, authentication and ownership across the financial system and, in fact, across the wider digital economy that finance depends on. Symmetric cryptography such as AES is also weakened by quantum search but can generally be protected by increasing key lengths; public-key infrastructure requires a more fundamental migration.

The machine does not need to exist for the exposure to exist: adversaries can harvest now and decrypt later.

An adversary can collect encrypted information today and store it for future decryption once suitable hardware arrives, a practice known as "harvest now, decrypt later". For data that must remain confidential for a decade or more, like merger discussions, customer records, trading strategies and health information, the clock has effectively already started. Quantum threat is unusual because part of the future exposure sits in the past: information transmitted years before a cryptographically relevant machine exists may later become readable.

Is Q-Day coming? Expectations of when that machine arrives have moved materially. No general-purpose machine has yet run Shor's algorithm at a cryptographically meaningful scale, but recent studies have sharply reduced estimates of the physical and logical resources required to attack commonly used schemes, and expert opinion has shifted accordingly: The Global Risk Institute's 2025 *Quantum Threat Timeline Report*, answered by 26 experts, assigns a 28–49% probability to a cryptographically relevant quantum computer within ten years, depending on the definition used. The range is wide and should be treated as expert judgment rather than an objective probability, but it is sufficient to make early migration to quantum-safe infrastructure economically rational. Cryptographers formalize the logic as Mosca's inequality: whenever the time needed to migrate plus the required secrecy lifetime of the data exceeds the time until a capable machine arrives, the rational moment to start is already now. The cost of starting too early is a phased technology program; the cost of starting too late could be the loss of trust in critical infrastructure.

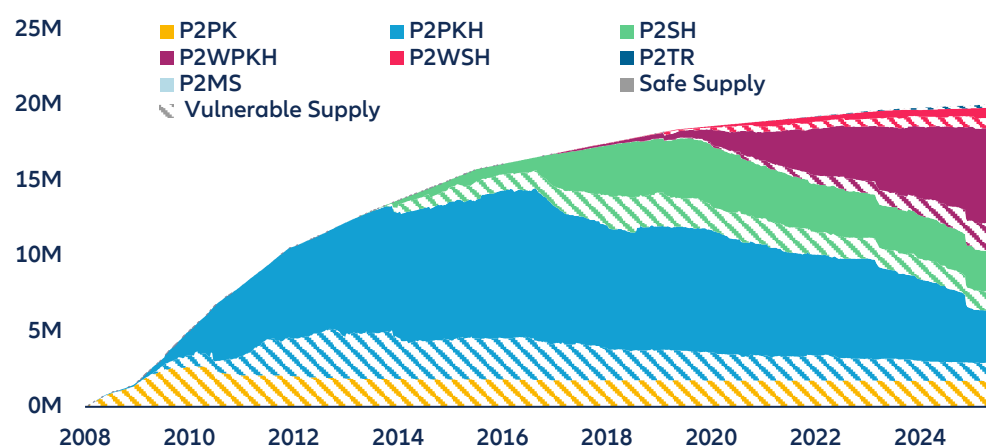
¹ Mathematical forms originally designed for quantum or quantum-inspired solvers, where QUBO stands for Quadratic Unconstrained Binary Optimization, essentially a way of expressing a problem as binary yes/no choices with trade-offs between them.

The defensive mathematics is already settled: In August 2024, the US National Institute of Standards and Technology (NIST) finalized the first post-quantum cryptographic standards, including ML-KEM for key establishment and ML-DSA for digital signatures. Regulators are attaching dates to the transition: NIST plans to deprecate today's vulnerable public-key algorithms after 2030 and disallow them after 2035, turning the migration deadline into a compliance fact rather than a physics forecast. The key uncertainty is therefore no longer whether replacement algorithms will exist, but whether institutions can identify, test and replace their cryptographic dependencies before the threat becomes operational. The new algorithms carry real engineering costs. Post-quantum public keys, ciphertexts and signatures are often larger than their current counterparts, creating challenges for bandwidth, storage, processing and constrained devices. When the Eurosystem and the BIS tested post-quantum cryptography in Project Leap, message sizes expanded and verification times increased. The increase was manageable in many applications, but material in high-volume or latency-sensitive infrastructure. Industry migration has begun in parallel: JPMorgan has operated a production quantum-secured, crypto-agile network since 2024.

The lesson is organizational rather than scientific. Institutions must inventory every system that relies on cryptography, identify data requiring long-term confidentiality, rank systems by risk and migration difficulty, test vendor readiness and build crypto-agility so that algorithms can be replaced without redesigning the entire application. This resembles a multi-year rebuild of trust infrastructure, not a conventional software patch.

Public blockchains could show what a cryptographic migration looks like when no one is in charge of the migration. Bitcoin provides the clearest illustration. By spring 2026, roughly 6mn Bitcoin, or around 30% of total supply, worth roughly USD400bn at current prices, sat behind public keys already visible on-chain. Some of that exposure is operational and can be mitigated by moving funds to fresh addresses. Some is structural. Early pay-to-public-key outputs and a large share of Satoshi-era coins reveal public keys by design, while the corresponding private keys are widely believed to be lost. The chart below (Figure 3) illustrates how that vulnerable supply has evolved across script types over time. The precise number is less important than the principle: a meaningful share of the world's largest digital asset already sits behind cryptography that a future quantum computer could challenge.

Figure 3: Bitcoin supply by script type and vulnerability



Sources: Google Quantum AI, Allianz Research. Note: Labels refer to Bitcoin address types, from the oldest (P2PK, 2009–11) to the newest (P2TR "Taproot", since 2021). What matters is the split: safe supply (solid areas) are coins whose cryptographic key is still hidden and thus cannot be attacked; vulnerable supply (striped areas) are coins whose key is already visible on the blockchain and could in principle be stolen by a powerful future quantum computer. Glossary of the legend: P2PK = Pay-to-Public-Key (key always visible); P2PKH = Pay-to-Public-Key-Hash; P2SH = Pay-to-Script-Hash; P2WPKH / P2WSH = SegWit formats; P2TR = Pay-to-Taproot (key visible by design); P2MS = bare multisig (keys visible)

Ethereum broadens the issue from digital assets to digital infrastructure. The exposure extends beyond user balances to administrative keys controlling stablecoins, tokenized assets, bridges and layer-two networks. As tokenization expands, the same cryptography that secures blockchains will increasingly secure representations of real world financial assets. What begins as a cryptocurrency issue therefore becomes a custody, settlement and investor protection issue for mainstream finance. Industry projections envision tokenized real world assets reaching trillions of dollars over the coming decade; those assets would inherit the same cryptographic vulnerabilities as the networks on which they reside.

Systemic implications

A quantum-enabled cryptographic systemic break would differ from most cyber events for which financial institutions currently plan. Most attacks affect a particular firm, service or network and can eventually be contained and restored. A break in a widely used cryptographic method could spread much further because banks, payment systems and market infrastructures rely on the same small set of standards.

The risk could extend well above the disclosure of confidential information. If public keys and digital signatures could no longer be trusted, an institution might be unable to establish whether a certificate is genuine, whether a message came from the stated counterparty or whether a payment instruction was authorized. An attacker could potentially impersonate a customer or institution, alter instructions or distribute malicious software under an apparently valid signature. The problem would then move quickly from data security to the integrity of payments, settlement and ownership records.

The systemic channel is confidence. Central banks and financial institutions are connected through common payment, messaging and settlement infrastructures, all of which depend on reliable authentication. A problem at one institution would not automatically bring down the system, but uncertainty over whether messages and signatures remained genuine could cause counterparties to delay transactions or suspend activity while the breach was investigated. The stability of the financial system is fundamentally dependent on cryptographic security. Maintaining trust and confidence is therefore central to the response, not merely recovering the data or systems initially affected.

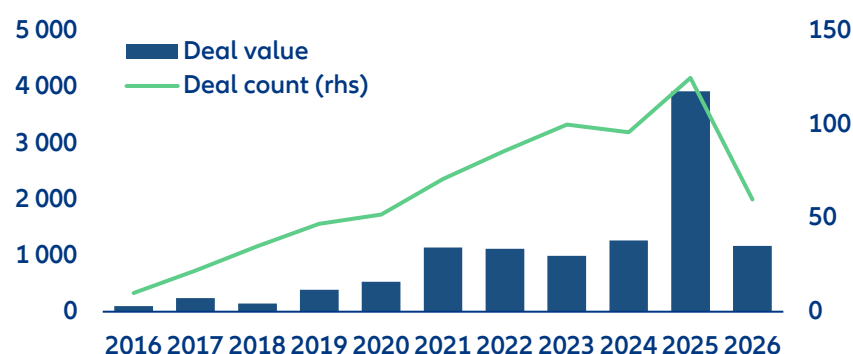
The available loss estimates, though not comparable with each other, point in the same direction. The BIS puts a restrained probability-weighted figure on the danger: financial-system losses equivalent to roughly 0.1% of GDP within five years of a quantum-enabled attack, rising above 1% over a 15- to 20-year horizon, with potentially larger effects in emerging markets. The Hudson Institute examines a more severe tail scenario, modelling a quantum-enabled cyberattack on the Fedwire interbank payment system using a two-stage approach: first calibrating the payment-network contagion effects of an attack on a systemically important bank (drawing on Federal Reserve researchers' Fedwire impairment simulations), then feeding that shock into Oxford Economics' Global Economic Model to trace the macroeconomic path. Their central estimate is a 10–17% decline in US real GDP, equivalent to USD 2.0–3.3trn in GDP at risk (3% of World GDP). Hudson separately offers a cruder, earlier estimate — USD 726bn–1.95trn — for a single-day disruption to one of the five largest US banks' Fedwire access, extrapolated from prior work on quantum threats to the power grid and cryptocurrencies rather than from the full macro model; Hudson itself flags this figure as preliminary and imprecise. The assumptions and methodologies differ substantially across and even within these studies, so the numbers should not be read as forecasts or combined into a single estimate. Their common message is that a failure of financial cryptography would be a macro event with sizeable spillover in the economy, not an IT incident.

Follow the money: capability is accelerating faster than readiness

If the systemic stakes are this high, the natural question is whether investment is flowing toward the machines, the defenses or both. The two are not equally visible: capability spending shows up cleanly in venture and public market data, while defending from the threat is happening behind closed doors. It concerns financial institutions' internal IT budgets, cybersecurity vendors and infrastructure-replacement programs, rather than in flows classified as quantum investment. So, the answer the data can give is necessarily partial and biased. Capital is flowing rapidly into quantum capability: Quantum-technology start-ups raised an estimated USD12.6bn in 2025, with more than

90% directed toward computing rather than sensing or communications. Within that total, venture investors deployed roughly USD3.9bn across around 125 rounds, more than triple the previous year (Figure 4).

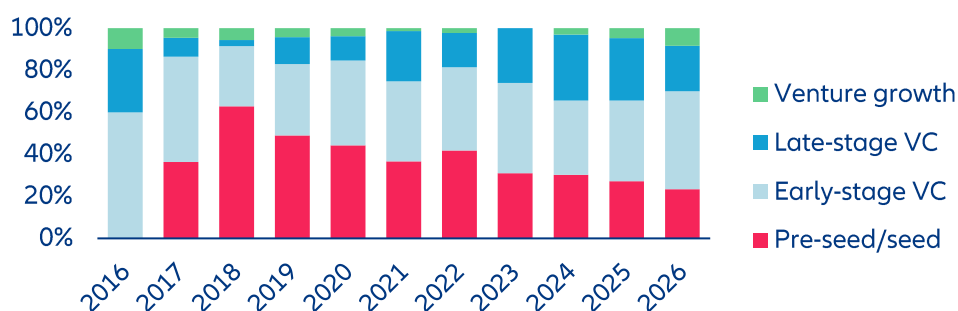
Figure 4: Quantum-computing VC deal activity by quarter, deal value (USD mn) and deal count



Sources: PitchBook, Allianz Research. Note: 2026 is YTD values.

The composition of that capital has changed even faster than its size. As recently as 2024, late-stage and venture-growth rounds were a rounding error; in 2025, venture growth alone jumped from around 1% to more than 30% of deal value, the largest single-year stage shift in the dataset, while mega-rounds of USD25mn and above quadrupled to roughly USD3.5bn. The investors writing those checks are no longer specialist quantum funds taking science risk but crossover institutions, strategic corporate balance sheets and sovereign wealth, some of the same names that anchor late-stage AI and semiconductor rounds. In capital markets terms, quantum has graduated from venture curiosity to institutional allocation. (Figure 5).

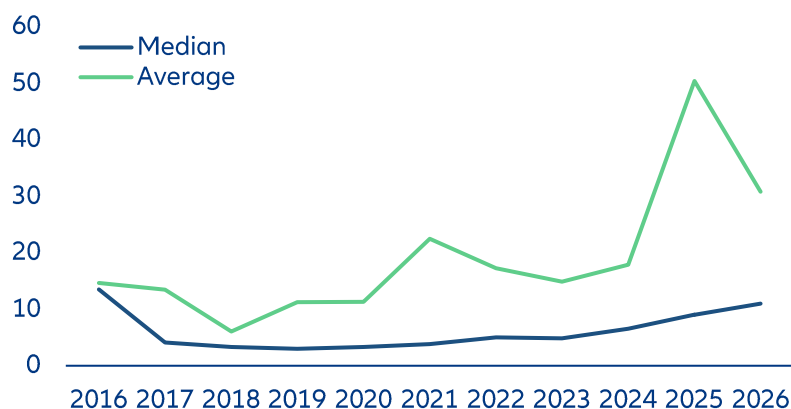
Figure 5 Share of quantum-computing VC deal value by stage



Sources: PitchBook, Allianz Research

The surge in capital has not made the market any broader. The category is governed by an unusually steep power law: the five largest deals account for roughly 27% of all venture value, and the five largest investors for more than 70%. The two largest rounds of the year were struck at pre-money valuations of roughly USD10bn and USD6bn, against a 2025 median pre-money valuation of under USD35mn. That gulf between the average and the median is the signature of a winner-take-most market, and it carries a direct portfolio consequence: the exposure is far easier to obtain than to underwrite. The listed proxies make the same point: pure-play quantum equities trade with betas of 2.5–3x the broad market, and the leading thematic ETF saw assets swell from around USD1bn to nearly USD6bn in 18 months. This is a beta bet on a narrative, not yet a cash-flow story (Figure 6).

Figure 6: Median and average quantum computing VC deal value, USD mn

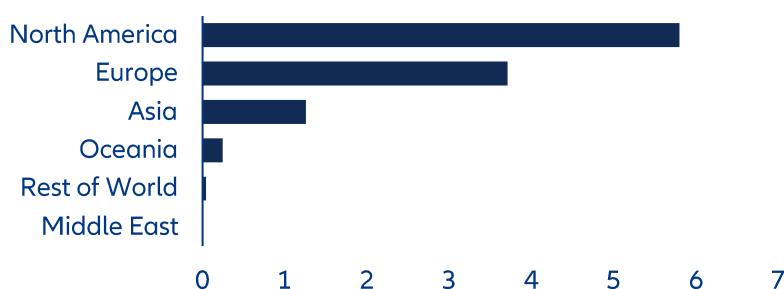


Sources: PitchBook, Allianz Research

The harder truth sits on the other side of the holding period. Quantum is deep-technology investing in its purest form, because it takes a decade or more from laboratory to liquidity and that horizon has so far left little room to exit: public-market debuts have generally underwhelmed, the underlying businesses are too young and too dependent on talent and intellectual property to suit a leveraged buyout and realized proceeds were close to nil as recently as 2024. The partial reopening in early 2026, a single quarter that produced more exit value than the previous three combined, reads more like a release of pent-up pressure than the arrival of a functioning exit market. The real conclusion is about duration rather than returns: the reward clock asks investors for a ten- to fifteen-year lockup, while the threat clock may strike inside a decade.

The narrowness is geographic as well as financial. Since the category took shape, roughly five of every six venture dollars have been committed in North America and Western Europe, while much of Asia's effort runs through state programs that never surface in venture statistics. Even within the West the split is lopsided: Europe attracts only around 5% of global private quantum investment, against more than half for the US, which is positioned to convert its advantages in capital markets, semiconductors and cloud infrastructure into technological leadership. The companies raising the money are also older than they used to be, with the median age at financing nearly doubling over the past decade from around two to four years. The picture is of a market that is deep but not wide: a concentrated pool of large, late and geographically clustered bets whose returns will be decided by a handful of outcomes rather than by the average (Figure 7).

Figure 7: Quantum-computing VC deal value by region, 2016-2026 (USD Bn)



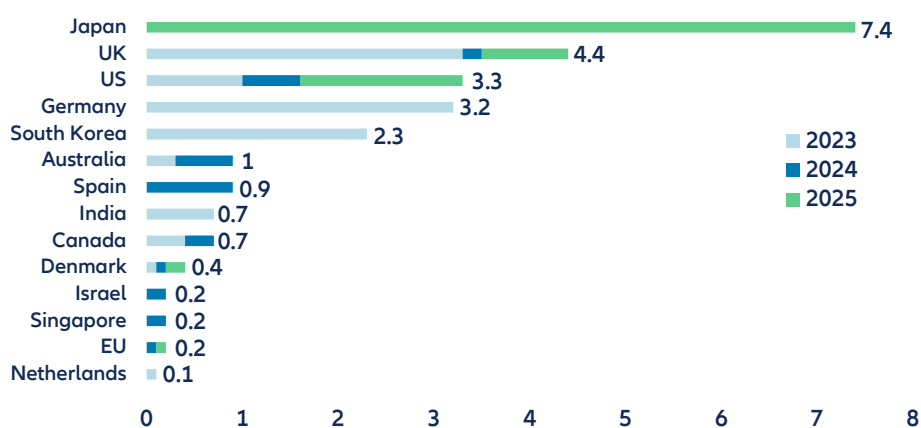
Sources: PitchBook, Allianz Research

Most revealing of all is where the money goes. Of every venture dollar committed to quantum, around 97% has gone to the technology stack – the qubits, the control systems, the platform layer – and less than 1% to financial-services applications. Private capital, in other words, is funding the machine's capability, not the applications that would benefit finance directly. The investor base financing that capability is, to a large extent, the same base that ultimately relies on the cryptographic foundations of the financial system, so the industry has a clear and shared

interest in seeing resilience advance in step with capability. For now, investment in building the technology is running well ahead of investment in preparing for its consequences.

Governments remain important sponsors, and in Washington, policy now addresses both clocks at once. Around USD11.4bn of new public quantum-funding commitments were announced in 2025, led by Japan, the US, the UK and Spain, while earlier multi-year programs in Germany, South Korea and China remain substantial (Figure 8). The US executive order *Ushering in the Next Frontier of Quantum Innovation* seeks to accelerate computing, sensing and networking; strengthen supply chains and workforce development; expand public-private partnerships and deploy a scientific quantum computer at a Department of Energy facility. A parallel order, *Securing the Nation Against Advanced Cryptographic Attacks*, directs agencies to appoint migration leads, inventory cryptographic assets, pilot post-quantum systems and move high-impact federal systems toward post-quantum key establishment and signatures around the turn of the decade. The US is thus addressing capability and defense through a single, coordinated federal instrument.

Figure 8: Government announced commitments on quantum by year, USDbn



Sources: McKinsey, Allianz Research. Note: Figures reflect press-search coverage of announced government commitments only, Jan 2023– Dec 2025; France's national quantum strategy (around €1.8B, Plan Quantique) falls outside this window as it was announced in 2021, and additional country-level commitments may be missing where they run through EU-level programs (e.g., Quantum Flagship) or lacked sufficient press coverage/breakdown to meet inclusion criteria. Figures should be read as directional signals of announcement activity, not a comprehensive accounting of national quantum investment.

Europe is not standing still, but its response is structurally more diffuse. The European Commission's Quantum Europe Strategy, published in July 2025, targets European leadership by 2030, and a proposed Quantum Act, expected in 2026, would try to turn that ambition into industrial policy and coordinated funding. On the defensive clock, at least, Europe has set dates: The EU's coordinated post-quantum roadmap calls on member states to adopt national transition strategies by end-2026 and for critical infrastructure, including finance, to complete migration by 2030. On the capability side, however, the difficulty is fragmentation, which the strategy itself identifies as the central obstacle: European support is spread across Horizon Europe, EuroHPC, the Chips Act, national strategies and a dozen other instruments, without a comparably deep single market for scale-up capital. Where Washington can act through parallel executive orders, Brussels must first coordinate member states and programs. After artificial intelligence, quantum technologies risk becoming another arena in which the US converts a unified market and deeper capital into a durable lead, while Europe legislates and regulates a technology it does not yet own.

The larger gap, however, is not between regions but inside institutions. Only around 35% of large organizations have completed a comprehensive inventory of cryptographic assets, and a smaller share combines executive governance, budgets and a credible migration roadmap. Post-quantum migration is not principally a quantum-research project. It is a large IT and engineering program involving inventory, procurement, architecture, testing, certificate management, hardware replacement and coordination with vendors and counterparties. And the gap runs on both sides of the ledger: just as few institutions have a credible migration roadmap, few have moved beyond

scattered pilots to the systematic talent, benchmarks and integration capacity needed to capture the reward when it arrives. Prepared institutions are the exception on either clock.

Its binding constraints are therefore budget, project capacity and executive attention, the same resources being absorbed by artificial intelligence. The competition is concrete and largely fought within the same IT budget line: the engineers, cloud spend and board time that could fund a cryptographic-migration program are the ones being directed toward AI, which offers more visible and immediate returns. Post-quantum defense risks losing that internal contest not because the threat is unknown, but because its payoff is an avoided loss rather than new revenue, and its deadline remains uncertain.

None of this weakens the opportunity. The investment data show genuine conviction, growing institutional participation and the early development of an industrial ecosystem, and the Monte Carlo case gives insurers in particular a legitimate reason to experiment on both sides of the balance sheet. But the discipline applied to capturing the upside should be matched by equal discipline on timing and resilience, because the two clocks reward different behavior. Capturing the reward is a wager on when quantum beats a moving classical benchmark, and it can afford to wait for evidence. Managing the threat cannot: Harvest-now-decrypt-later means today's traffic is already exposed to tomorrow's machine, and migration itself takes years. That is why quantum preparedness is best understood as an IT-resourcing decision rather than a bet on an arrival date: Crypto-agile architectures, prioritized inventories, migration roadmaps and tested fallbacks pay off across almost every timeline, while waiting pays off only in the one where the machine never comes. What to watch from here: a credible demonstration of factoring at cryptographic scale, logical-qubit counts scaling past error-correction thresholds, the first mandated live post-quantum payment rail, credible business advantages to the financial sector, enforcement of the 2030 depreciation deadlines and the annual shift in the expert threat timeline. Institutions do not need to know which clock will strike first to know that preparing for only one is the wrong strategy.

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Allianz Research | 29 July 2026

Diversification: Why bonds no longer hedge equities

Ludovic Subran
Chief Investment Officer and
Chief Economist
ludovic.subran@allianz.com

Nils Bradtke
Senior Investment Strategist
nils.bradtke@allianz.com

Jordi Basco Carrera
Head of Private Markets Investment
Strategy
jordi.basco_carrera@allianz.com

In Summary

- **The 60/40 portfolio has quietly stopped working.** The risk-adjusted return investors could reasonably expect from the standard balanced portfolio has fallen to roughly a third of its long-run level, from a Sharpe ratio of 0.47 to just 0.14, as bonds and equities have stopped hedging each other for the first time in a generation. Since mid-2022, the two have moved together rather than apart, with the equity-bond correlation flipping to a positive +0.77, and no comparable episode in the past quarter-century has behaved this way for anywhere near this long.
- **The culprit is policy uncertainty, not only inflation.** The reassuring story that 2022 was a one-off inflation shock and that everything returns to normal once prices settle does not survive contact with the data. Inflation still matters, but it is no longer the master variable. Tariff policy, geopolitical fragmentation and the slow erosion of central bank credibility are now doing the heavy lifting.
- **Gold, private credit and infrastructure are the diversifiers doing the work now.** Where government bonds have failed to catch equity drawdowns since 2022, three asset classes have quietly stepped into their place. Gold has delivered positive risk-adjusted returns in nearly every regime tested and is the closest thing to an all-weather hedge. Private credit is posting its strongest showing on record. Infrastructure has held up across both this cycle and calmer ones before it, a rare cross-regime performer.
- **A regime-robust 'No-Regret' portfolio, built to work across cycles rather than optimized for the current one, roughly doubles the risk-adjusted return of a standard global benchmark.** It delivers a Sharpe ratio of 1.33 against the benchmark's 0.59, compounds an initial dollar to roughly 9x its starting value over two decades against just 5x for the benchmark and cuts the deepest drawdown in the sample from -24.6% to -15.3%. The composition marks a substantial departure from tradition: government bonds are cut from roughly 20% of the benchmark to under 4% and public equities from 43% to 13%, funding materially larger allocations to gold (16%), private credit (29%) and private equity (16%), with a smaller residual position in infrastructure.
- **The historical hedge will return only once policy uncertainty resolves and inflation is fully re-anchored, and rate cuts alone will not do it.** Only two of the five preconditions for a durable return to negative equity-bond correlation are true binding constraints, and neither is something a central bank can deliver on its own. Genuine resolution likely requires a durable Iran settlement and a more stable US trade regime, and both look like multiyear propositions rather than near term fixes. The remaining conditions, on monetary and liquidity settings, equity valuations and the yield curve, follow mechanically once the first two are in place.

How 2022 snapped the 60/40 strategy

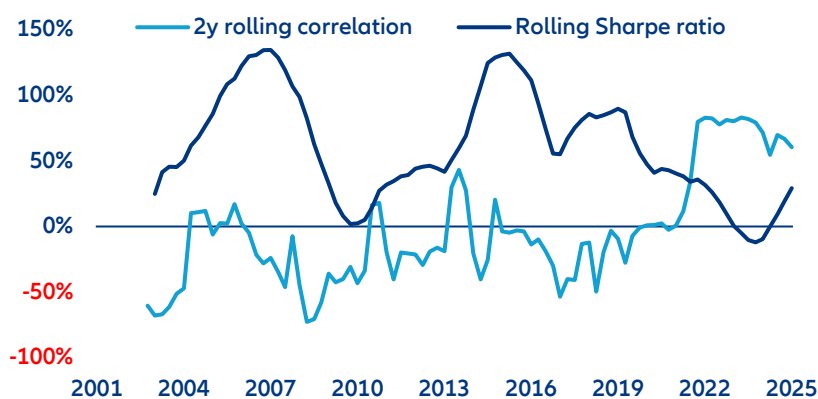
For two decades, the 60/40 portfolio did something rather remarkable: it made the difficult look dull. The formula was straightforward. Sixty cents of every dollar bought equities, which supplied the long-run growth. The other 40 went into government bonds, which stood by, patiently, waiting to catch the fall. When equities dropped, as they periodically did, bonds duly rose. Growth scares brought rate cuts, rate cuts pushed bond prices up and the 40% sleeve absorbed a decent share of the equity pain. Investors could speak, without embarrassment, of a portfolio that returned around 7% a year with the volatility of a mid-cap stock and the drama of an accountancy conference.

But it was a convention, not a law of nature. Yet it held often enough, and for long enough, that the entire architecture of modern retirement finance came to be built on top of it. Liability-matched pension books assumed it. Target-date funds mechanized it. An industry of insurance companies, endowments and family offices arranged themselves around the same quiet promise, that bonds would zig when equities zagged.

Then, in 2022, the convention snapped and the wound is still open. Between the early 2000s and 2021, a 60/40 book delivered a Sharpe ratio of roughly 0.47, a measure of how much excess return a portfolio earns per unit of risk taken (above 0.5 is respectable, above 1.0 enviable). It was unglamorous, but the sort of number pension trustees could take to their boards without flinching. In 2022, the hedge failed at the moment it was most needed. Equities and bonds sold off in unison, and the balanced portfolio delivered its deepest drawdown in a generation. A manager reviewing the year's returns could be forgiven for a moment of vertigo: the diversification premium, dependable for so long that it had ceased to feel like a premium at all, had simply vanished.

Four years on, markets have not forgotten. The Sharpe ratio implied by today's pricing on a 60/40 book sits closer to 0.14, less than a third of its long-run average. Portfolios that had been sold, quite reasonably, as boring have been forced to justify themselves anew (Figure 1).

Figure 1: When bonds stopped catching the fall (%)



Sources: LSEG Datastream, Allianz Research

At this point the industry has, predictably, split into two camps. The optimists take a relatively benign view. 2022 was an inflation shock and inflation shocks have always played havoc with the correlation between bonds and equities. Once inflation settles, and provided the world manages to avoid a second, 1970s-style episode, the old negative correlation should quietly reassert itself, as ordinary growth shocks reclaim the driver's seat. On this reading, patience is the only strategy required. The pessimists point to something rather harder to unwind. Fiscal and monetary policy have become, if not fused, then uncomfortably entangled. Geopolitical fragmentation is running at levels not seen since the end of the cold war. And central bank credibility, that intangible asset which underwrote so much of the previous cycle, has been slowly, publicly, corroded. On this view, the positive correlation

can persist long after inflation itself has been tamed, because what has changed is not the price level but the framework around it.

In this paper we apply both theories to post-2022 data and find that the pessimists have the better of it, though with an important wrinkle. The dominant force behind today's positive correlation is not the monthly inflation print, awaited each month with such nervous attention on Wall Street, but policy uncertainty itself. Inflation still matters, of course. It shows up in the regression, alongside tighter liquidity, a flatter yield curve and stretched equity valuations, as a genuine driver of correlation. It is simply not the master variable. The implication is uncomfortable, particularly for those who have spent the past two years waiting for a single, clean catalyst. A first rate cut, however loudly cheered, will not on its own restore the hedge. Policy uncertainty is the binding constraint, and the bar for its removal is a good deal higher than markets are currently pricing. Investors counting on a return to the old regime may find themselves counting for some time.

Where are we in the cycle?

Five regimes, not one endless cycle. The temptation, when confronted with the current mess, is to reach for the nearest historical analogue and declare it a rerun of the 1970s. This is almost always a mistake. Instead of assuming the past 25 years amount to a single continuous cycle, we sifted the macro data into five distinct regimes, each with its own equity-bond correlation and, if one is honest, its own personality. The first, which we call liquidity and rates normalization, ran from 2001 to 2008 and produced the workhorse negative correlation of -0.38, not spectacular, but exactly the diversification investors had learned to expect. (Correlation runs from minus one, a perfect mirror image, through zero, no relationship at all, to plus one, perfect lockstep; the more negative the number, the more reliably bonds cushion equity losses.) Crisis and deleveraging, the label for the acute shocks of 2008-09 and 2020, was brief but violent, and pushed the correlation to -0.69, the strongest hedge in the entire sample; when the world briefly ended, in other words, bonds worked harder than ever. Post-crisis reflation, the recovery years of 2009 to 2012, offered a more modest -0.19, with wide credit spreads and a steep yield curve. Then came the long risk on expansion of 2012 to 2022, a decade in which valuations rose, volatility slept and correlations grew lazy, flattening the hedge to -0.12.

And then, in May 2022, the arrangement changed. Today's regime is the odd one out. Elevated rates, stubborn inflation, high policy uncertainty and a compressed yield curve have combined to push the equity-bond correlation to a +0.77, a figure without precedent anywhere in the past 25 years.

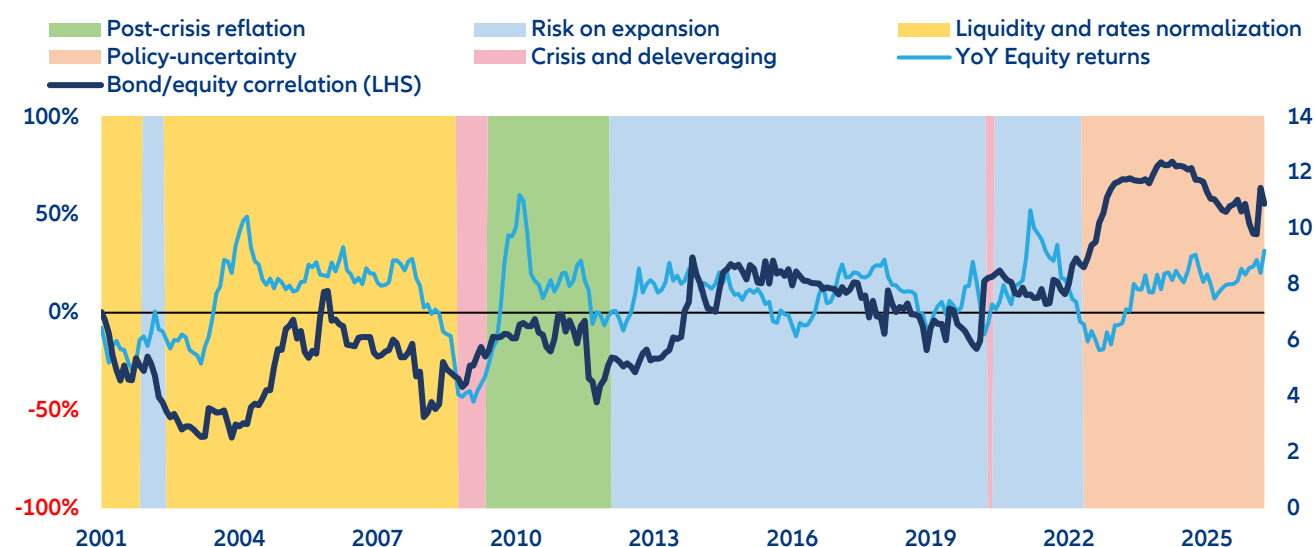
Overlay the five regimes onto realized equity returns and the rolling bond-equity correlation and a familiar picture emerges: The hedge holds through every "normal" regime and breaks only when inflation becomes the dominant shock. Through the dot-com bust and the subsequent recovery of 2003 to 2007, the correlation stays firmly negative, touching -60% even as equity returns swing from -40% to +40%. Rates and the shape of the curve did most of the work. The Fed entered 2001 with the funds rate still at 6.5%, cut aggressively through the dot-com bust to 1.75% by year-end and, because short rates fell faster than long ones, watched the curve steepen sharply even as growth expectations collapsed. That combination, elevated starting rates giving the Fed room to cut and a yield curve that could steepen on cue, is precisely the mechanism that lets bonds rally when equities sell off. It is, in other words, a textbook illustration of why bonds do their job as shock absorber at exactly the moment equities need them to.

Then came 2008. The global financial crisis was, above all, a credit and volatility episode. Those two forces dominated all other drivers, sentiment and growth expectations collapsed to sample lows and the recovery of 2009 and 2010 was a sharp, V-shaped affair driven by quantitative easing, which briefly, and unusually, pushed both equities and correlation higher together. What followed was the decade-long stabilization regime of 2011 to 2020, in which correlation settled into a shallow, range-bound negative-to-neutral band and equity returns unfolded in the calmer, low-rate manner that came to define the era. The Covid-19 shock of 2020 was narrow and sharp, marked by the highest readings in the sample for monetary and liquidity conditions as central banks intervened at unprecedented scale. It, too, resolved quickly.

The regime that breaks this pattern is defined by policy, not growth, and that is precisely why we do not expect it to resolve on its own. Since 2022, the correlation between bonds and equities has decoupled from its historical negative range and climbed into the 60% to 80% band, even as equity returns themselves have been comparatively muted. Unlike 2001, the Fed entered this shock with rates near zero, and there was, therefore, no room to cut when inflation and growth diverged. Instead, consumer price inflation and commodity inflation reached their highest levels in the sample, the dollar strengthened, equity valuations sold off, and hiking rather than cutting pushed bonds and equities in the same direction.

By 2024, the character of the shock had rotated. Inflation itself cooled, but policy uncertainty and interest rate levels took over as the elevated forces, while monetary and liquidity conditions sank to sample lows. Tariff policy and geopolitical shocks now sustain the same positive-correlation dynamic that inflation started. This, more than anything, is why the current regime should be treated as structurally different, not merely a noisier version of the stabilization years that preceded it (Figure 2).

Figure 2: A quarter-century in five acts (%)



Sources: LSEG Datastream, Allianz Research; Note: 2y correlation

The current regime is structurally different, not simply unusually long. Three features set it apart from every prior episode. The first is duration and consistency, every single quarter since May 2022 has fallen into this regime, and the average correlation has stayed stubbornly above +0.40 throughout. There is no precedent in the sample for that kind of persistence. The second is the identity of the dominant signal. It is not merely inflation, as many assumed at the outset, but policy uncertainty specifically. Our Economic Policy Uncertainty index, expressed as a z-score (that is, in standard deviations away from its long-run average, so that +1 means one standard deviation above trend and anything above +2 is unusually high), sits at +1.7 as of May 2026. It has been elevated since 2024, and briefly touched a remarkable +5.4 on “Liberation Day”. The third, most awkward of the three is that this regime has never ended. Every quarter since May 2022 has remained inside it, with no observed exit anywhere in the historical record.

There are two plausible exits from here, and one rather less likely one. A transition into the risk on expansion regime, in which policy uncertainty resolves and liquidity conditions ease, is one credible route; a transition into the liquidity-normalization regime, in which rates settle into a predictable path and broad liquidity returns to the system, is the other. Both are structurally consistent with the way prior regimes have unwound. A direct jump into an acute crisis regime is also possible, though far less probable, and historically brief. Those episodes have, on average, lasted only around five months before resolving into something calmer.

What is driving the change in correlation? An anatomy of a broken hedge

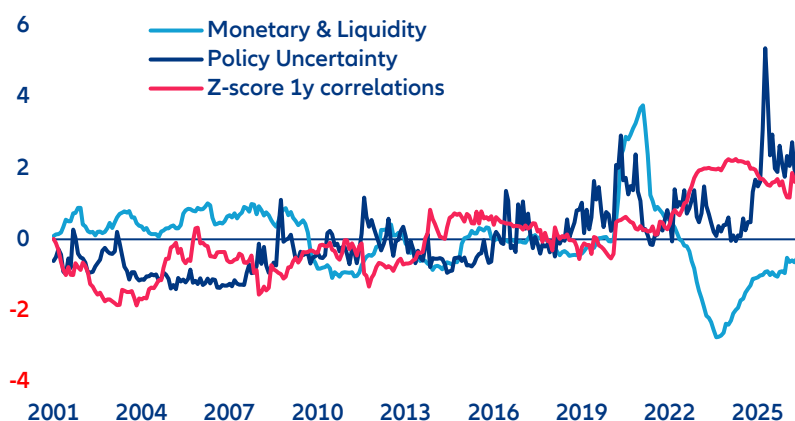
Policy uncertainty is the biggest driver, but inflation matters too. To work out which macro forces have actually been moving the equity-bond correlation, we regressed 25 years of rolling correlation data against five candidate factors. The results are unambiguous. Roughly two-thirds of the variation in correlation can be traced to those five drivers, and all five clear the usual statistical thresholds comfortably. Policy uncertainty emerges as the dominant force, pushing correlation firmly into positive territory whenever it rises. Monetary and liquidity conditions carry an almost identical, but opposite, effect, looser conditions restore the traditional hedge, tighter ones destroy it. Equity valuations turn out to matter more than most would guess. A stretched market pushes correlation up, a calmer one pulls it back down. A steeper yield curve, the mark of a "normal" rate environment, works quietly on the negative side. And inflation, contrary to those hoping the story is a pure policy-uncertainty morality play, also pushes correlation up in its own right, an effect that has been especially pronounced since 2022.

That inflation still matters is a corrective to the pure policy uncertainty story. The two forces have moved together, often, over the past two decades. In 2008 and 2020, the monetary and liquidity indicator surged to nearly 3.8 standard deviations above its long-run average, and correlation briefly flipped positive alongside extreme easing. In 2021, correlation turned positive ahead of the inflation shock itself. The two then rose together as consumer prices climbed, with CPI reaching 3.9 standard deviations above trend in 2022 while correlation's z-score ran up to a peak near 2. The post-2021 regime, in other words, was inflation-driven before it became uncertainty-driven.

The character of the shock rotated after that. In 2022, monetary and liquidity conditions collapsed to almost 2.8 standard deviations below trend as the Fed hiked, yet correlation stayed elevated: growth was no longer the master shock, inflation had taken over. Inflation then receded faster than correlation, which anchored near 1.5 to 2 through 2024 and 2025 as policy uncertainty stepped in as the marginal driver, consistent with the tariff-driven uncertainty of the second Trump administration. The sharpest reading in the sample, policy uncertainty touching 5.4 in early 2025, captures the "Liberation Day" shock. Uncertainty then remained stubbornly high through the escalation to "major combat operations" in the February 2026 Iran conflict, a longer and more consequential episode than the brief Israel-Iran strikes of June 2025 (which had only pushed Brent crude from around USD65 to the low USD80s before subsiding). All of it feeds the same inflation-and-uncertainty channel that has kept bonds and equities correlated near their sample highs.

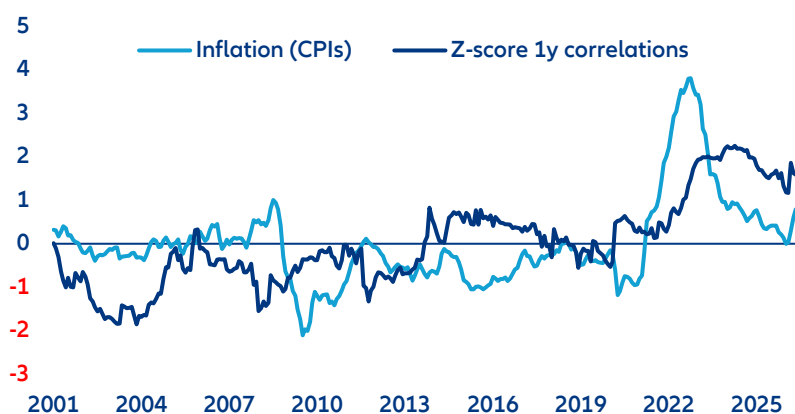
The model, reassuringly, still works. Its predictions match reality most closely when one or two forces dominate the regime, and most poorly when short, sharp shocks temporarily overwhelm the underlying macro drivers. The fit is at its best during the 2001 to 2008 and 2022 to 2026 windows, precisely when a small number of factors have been doing most of the driving, rates and the yield curve in the early 2000s, and policy uncertainty and monetary policy together from 2022 onwards. It tracks worst around the 2009 to 2010 rebound and the 2020 Covid-19 dip, when actual correlation overshot the model in both directions faster than macro variables could move. Those, unsurprisingly, are precisely the moments when positioning, liquidity dislocations and forced deleveraging drive realized co-movement independent of the fundamentals the model measures. The gap widens for a while, then both series converge again. Feeding May 2026 inputs into the model produces a predicted correlation reading of 1.05 standard deviations above the long-run average, against an actual reading of 1.59. Policy uncertainty is the largest contributor at +0.62, with equity valuations, monetary and liquidity conditions and inflation each contributing between +0.11 and +0.14, and a re-steepening yield curve chipping in a marginal +0.04. The gap of +0.54 between prediction and reality is larger than in prior months, consistent with a degree of regime persistence the underlying factors alone do not capture. The correlation has become, in effect, self-reinforcing. Portfolio managers have already cut their duration exposure in response to it, and their reduced participation keeps realized co-movement elevated even as the underlying drivers ease a fraction (Figure 3 and 4).

Figure 3: Where the correlation comes from (Z-scores)



Sources: LSEG Datastream, Allianz Research

Figure 4: The model, put to the test (Z-scores)



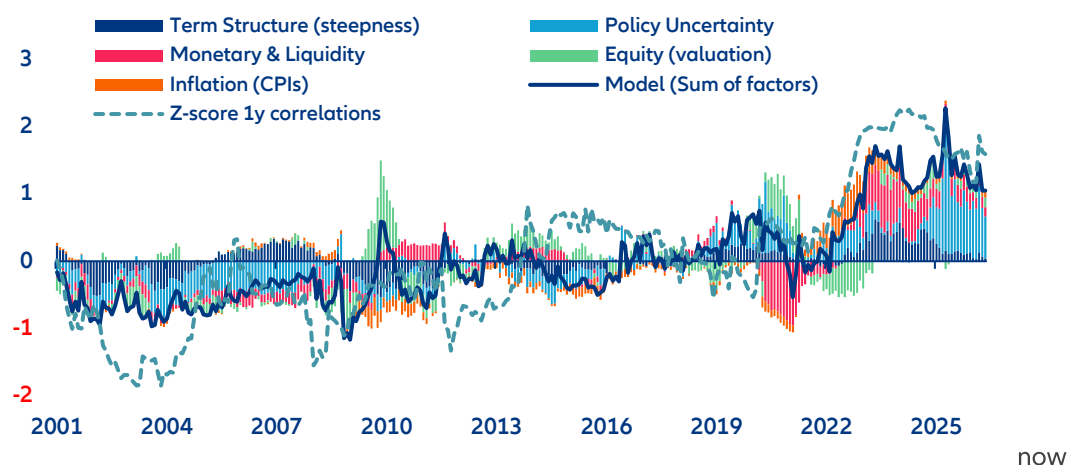
Sources: LSEG Datastream, Allianz Research

What has to happen before the hedge normalizes? A durable return to negative bond-equity correlation requires five conditions to line up. The good news is that only two of them are genuine constraints. The other three are largely endogenous to a normal easing cycle and should follow, mechanically, once it is underway.

The hard ones first. Policy uncertainty must recede, and inflation must remain on a credible disinflation path. Neither is something a central bank can deliver on its own, which is why we do not expect the hedge to return simply because the Fed or the ECB begin cutting. Policy uncertainty today is anchored by an unresolved geopolitical and trade backdrop, with the Iran conflict's ceasefire collapsing again in July 2026 and tariff policy still being litigated through the courts and alternative executive authorities. Genuine resolution likely requires a durable Iran settlement and a more stable US trade regime, and both look like multi-year propositions rather than near-term fixes. Inflation, for its part, has retraced most of its 2022 surge, but tariff pass-through and Gulf-driven energy volatility keep the disinflation path only partially secured.

The remaining three conditions should follow of their own accord. Monetary and liquidity conditions will loosen as the new Fed leadership under Kevin Warsh, who was expected at his May 2026 installation to lean toward easier policy but has so far held a more neutral stance, continues to cut rates. Equity valuations, a channel of similar size to policy uncertainty and liquidity in the updated regression, are likely to moderate as the AI capital-spending cycle cools or broadens into green-transition and infrastructure investment rather than through an outright shock. And the yield curve should steepen as a natural consequence of rates normalizing once the first four conditions are in train. Investors waiting on the first cut, in short, are waiting on the wrong thing (Figure 5).

Figure 5: The road back to a working hedge (Z-scores)



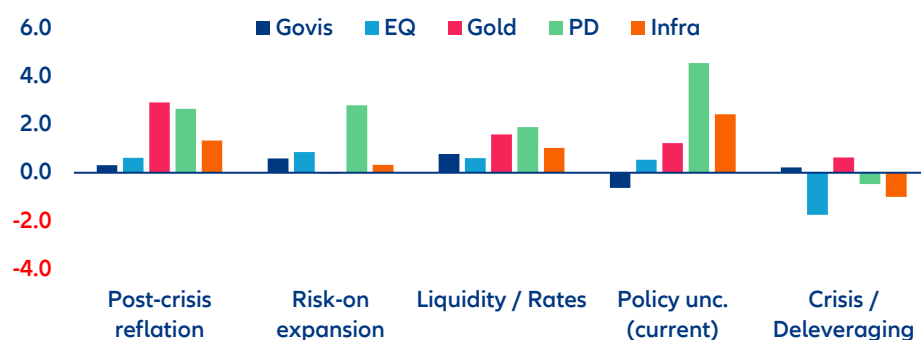
Sources: LSEG Datastream, Allianz Research

What are the implications for portfolio construction?

Some asset classes are earning their keep. In the current regime, the 15 asset classes we track are being sorted rather ruthlessly. Government bonds, agencies, investment-grade credit and inflation-linked bonds have all posted negative risk-adjusted returns in Regime 4 (the current policy-uncertainty and inflation regime, running since May 2022), the direct consequence of losing the very negative correlation that once made them portfolio anchors. Government bonds alone have swung from a positive Sharpe ratio in every other regime (between +0.22 and +0.78) to -0.64 in the current one. Public equities have held up, more or less, but without the tailwind of a hedge behind them: a modest +0.54, well short of their strongest showing during the risk-on expansion of the 2010s (+0.86) but nowhere near the collapse they suffered in the crisis and deleveraging regime (-1.75).

Three asset classes stand out as the genuine diversifiers of this cycle. Gold has delivered a positive Sharpe ratio in nearly every regime we tested, including the current one at +1.22, and its finest hour came in the post-crisis reflation of 2009 to 2012, at +2.92. It is the closest thing to an all-weather hedge in the dataset. Private credit has produced its best risk-adjusted performance of any regime in this one, at +4.55, quietly stepping in as a source of income where traditional fixed income has been failing. Infrastructure, meanwhile, has been the standout real asset, reaching +2.43 in the current regime while also holding up in calmer, more traditional regimes such as liquidity and rates normalization (+1.03) and post-crisis reflation (+1.34). It is one of the very few exposures with strong performance across both the current regime and the more typical ones that came before: a genuinely cross-regime diversifier rather than a regime-specific bet.

Figure 6: Where risk pays and where it doesn't (Sharpe Ratio (rf=2%, annualized))



Sources: LSEG Datastream, MSCI, Allianz Research

In the face of unexpected regime shifts, it makes sense to plan for the transition, not just the present. Regime 4 has a perfect track record, in our sample, of persisting rather than resolving. The more useful planning question, then, is not whether it will eventually end but what should change once it does. The two structurally plausible exits both point in the same direction. As policy uncertainty fades and liquidity normalizes, government bonds and investment-grade credit should regain a meaningful share of their traditional hedging role, while the current outperformers – private credit, infrastructure and gold – would see their relative advantage compress back toward more typical historical levels. That argues for holding today's tilts with conviction, but not permanently.

The No-Regret portfolio we define is built to perform across regimes, not just this one. A brief word on how we got there. The optimization blends three ingredients in a way that deliberately avoids over-relying on any single one. It starts from a market-capitalization-style prior allocation, which anchors the portfolio to the actual composition of investable capital in the world. It then nudges those weights using the return signal implied by market pricing itself, a standard technique known as Black-Litterman, which avoids the notorious instability of feeding raw historical averages straight into an optimizer. And it adds a modest tilt from a machine-learning model trained on each asset's own momentum and drawdown history. Diversification, turnover and tail-risk penalties are built into the objective directly and the final result is stress-tested by resampling across thousands of simulated histories rather than optimized against a single historical path. The final weights, in short, are not simply the one combination that happened to work best in hindsight.

The portfolio calculations are a mathematical exercise rather than a bespoke recommendation for any individual investor. As discussed, the optimizer does embed a series of modeling choices we set directly rather than resulting from historical return data alone. However, it does not account for market depth, transaction costs, asset manager fees, taxes or an investor's specific liquidity needs or mandate constraints. We assume a risk-tolerant, unbiased and unrestricted investor purely relying on mathematical concepts, which is rarely the case in practice.

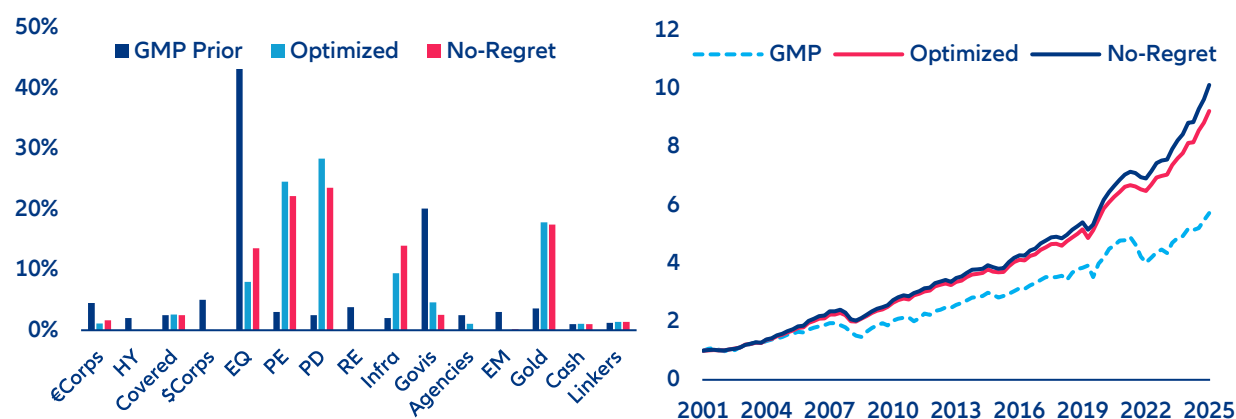
As such, the output should be read as an analytical starting point rather than an end point: a way to stress-test conventional allocations and surface where the mathematics disagrees with convention, with the actual translation into an investable portfolio left to an investment professional who can weigh it against real-world constraints.

On a full-sample, risk-adjusted basis, the No-Regret portfolio outperforms both a traditional benchmark and a portfolio optimized purely for the current regime. Optimizing a portfolio for the current regime works beautifully if the regime persists indefinitely, and rather less well the moment it does not. Instead, we constructed a blended No-Regret portfolio, using an equal-weighted average of all five regime correlations, with the current regime given double weight to reflect its higher-for-longer persistence. The blended equity-bond correlation embedded in the construction comes to +0.037, essentially neutral, against -0.032 using the full-sample quarterly correlation and +0.77 using the current regime alone. The result is a portfolio anchored around private markets and gold, with a reduced equity allocation and a small residual government bond position kept specifically as insurance for the eventual regime transition.

The shift away from the Global Market Portfolio (GMP) prior, which is roughly the market-capitalization-weighted mix of all investable assets globally and serves as the optimizer's neutral starting point, is large, consistent across methodologies and has paid off for two decades, not just recently. Private debt rises from a 2.5% prior weight to 28.5% in the No-Regret allocation, private equity from 3.0% to 15.9% and gold from 3.6% to 16.0%, while government bonds fall from 20.1% to just 3.6% and equities are cut from 43.2% to 12.6%. Reassuringly, an alternative construction built directly on the unconditional full-sample correlations rather than the regime blend lands in a broadly similar place on most positions. The tilt toward private markets and gold, in other words, does not appear to be an artefact of regime-weighting choices, it holds up even without leaning on the current regime at all.

The payoff has been persistent, not a recent flourish. Both the optimized and No-Regret portfolios compound from 1.0 to roughly 9x over the sample, clearly separating from the GMP benchmark's roughly 5x outcome by the mid-2010s and holding that gap through both the 2020 and 2022 drawdowns, consistent with Sharpe ratios of 1.35 and 1.33 against the benchmark's 0.59.

Figure 7: Two decades of compounding (allocation weights – left , cumulative NAV – right)



Sources: LSEG Datastream, MSCI, Allianz Research

On that basis, the No-Regret portfolio posts the second-highest Sharpe ratio of any portfolio we tested (1.33), behind only the full-sample optimized portfolio (1.35), which has the significant benefit of hindsight, and comfortably ahead of the Global Market Portfolio benchmark (0.59). Its maximum drawdown, at minus 15.3%, is meaningfully shallower than the benchmark's minus 24.6%, and notably better than a portfolio built purely around current-regime assumptions (minus 18.9%). That last portfolio, despite looking attractive on paper today, offers the weakest downside protection of the group once tested across the full cycle.

Table 1: Risk-adjusted performance across portfolio constructions

Portfolio	Sharpe ratio	Annualized return	Max drawdown
Global Market Portfolio (GMP)	0.59	7.1%	-24.6%
Optimized for current regime (R4)	1.07	10.0%	-18.9%
Full-sample optimized	1.35	9.61%	-12.9%
No-Regret portfolio	1.33	9.9%	-15.3%

Sharpe ratio uses $r_f = 2\%$. Figures based on quarterly data, 2001–2025 (97 observations).

Relative to a typical diversified benchmark, the resulting No-Regret mix makes three deliberate trade-offs: less public equity, materially less government bond duration and a substantially larger allocation to gold and private markets, particularly private credit and infrastructure. The optimizer switches public for private counterparts: Private equity offers higher returns with lower volatility than public equity; private debt carries slightly more volatility than government bonds but far higher returns. None of these tilts is extreme or all-in. They represent a rebalancing of emphasis toward the exposures that have actually protected portfolios through this cycle, without abandoning the assets that will matter again once the regime eventually turns.

The No-Regret allocation looks private. The optimizer overweight private markets and gold for a straightforward reason. These exposures combine higher risk-adjusted returns with genuinely low correlation to a standard 60/40 book, which is precisely the combination a mean-variance optimizer is built to chase. The diversification case is visible in the data. US private equity, private debt, and infrastructure move through distinctly different cycles rather than in lockstep: private equity's mid-cycle strength around 2016 to 2019 arrives as infrastructure is flat or negative, infrastructure's sharp 2016 and 2022 spikes come while private debt is subdued and the four series only really converge during the 2008 to 2009 and 2020 shocks, when almost everything sells off together.

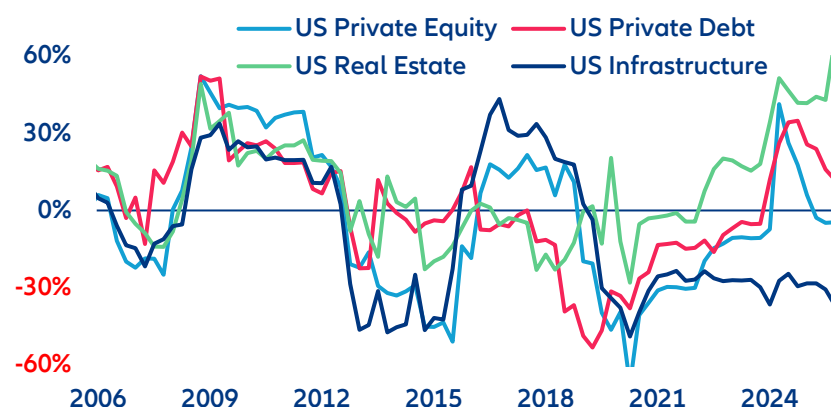
Table 2: Recommended allocation versus a typical diversified benchmark

Asset class	GMP (typical mix)	No-Regret (recommended)	Difference
Public equities	43.2%	12.6%	−30.6pp
Private equity	3.0%	15.9%	+12.9pp
Private credit	2.5%	28.5%	+26.0pp
Infrastructure	2.0%	13.5%	+11.5pp
Government bonds	20.1%	3.6%	−16.5pp
Gold	3.6%	16%	+12.4pp

GMP reflects a market-capitalization-weighted global benchmark spanning public and private asset classes. Figures are illustrative allocations, not personalized investment advice.

That pattern of usually uncorrelated, occasionally-jointly-stressed returns is precisely what shows up in the regime-level Sharpe ratios. Private debt posts its best risk-adjusted performance of any regime in the current one (+4.55). Infrastructure holds up across the current and calmer regimes alike (+2.43 today, and between +1.03 and +1.34 in earlier non-crisis eras). Gold, meanwhile, is positive in nearly every regime we tested. Higher Sharpe plus low correlation to the traditional stock-bond mix is, in a word, what the optimizer is rewarding when it reallocates away from equities and government bonds and into these exposures.

Figure 8: Private assets, dancing to their own beat (three-year rolling correlation of private assets with 60/40 portfolio)



Sources: MSCI, Allianz Research

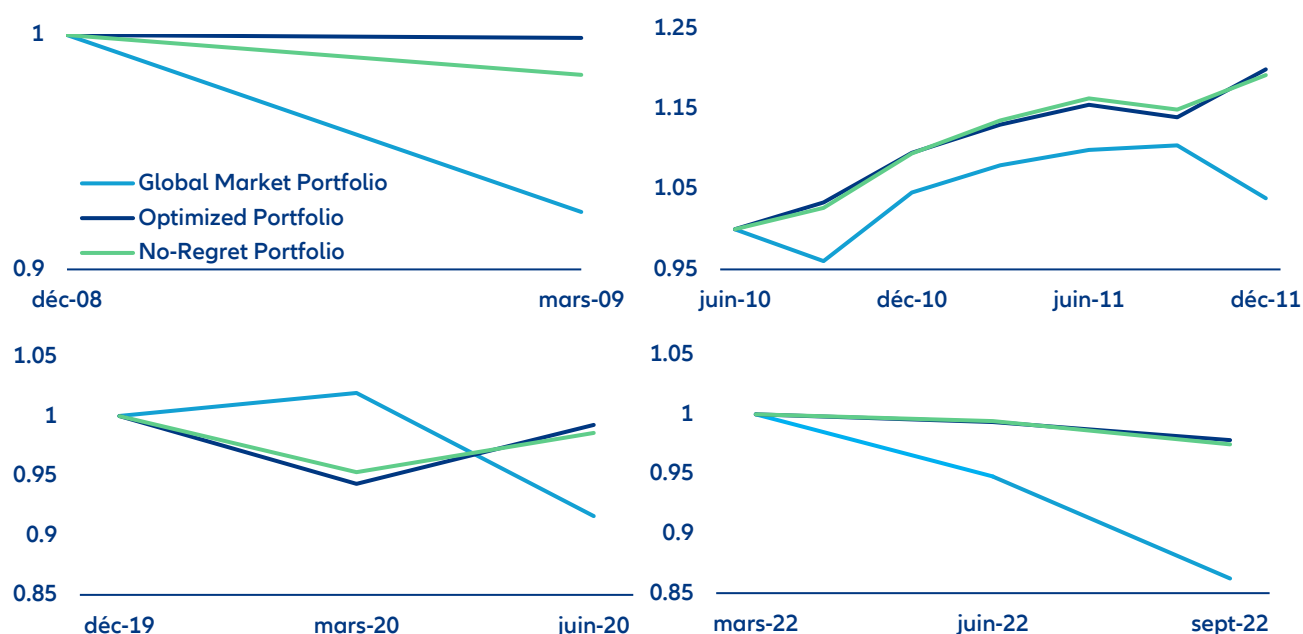
One important caveat. The shift into private assets is only appropriate for investors with genuine capacity to bear illiquidity risk. Private markets cannot be rebalanced or exited on short notice the way public equities and government bonds can. The allocation therefore assumes a long investment horizon, limited near-term liquidity needs and a governance structure able to tolerate multi-year lock-ups and unpredictable capital calls. Where that capacity is absent, the same diversification logic points to a smaller private-markets tilt, funded from public equity rather than from the liquid fixed income that may still be needed for near-term cash flow.

The cost of choosing robustness over point-optimality is small and well worth paying. The fully-optimized, full-sample portfolio edges out the No-Regret construction on both Sharpe ratio (1.35 versus 1.33) and drawdown (minus 12.9% versus minus 15.3%). But it does so only by implicitly assuming that the benign full-sample

correlation structure persists, which is precisely the assumption this paper's diagnosis calls into question. Giving up 0.02 of Sharpe and roughly two additional points of drawdown, in exchange for a portfolio that is not quietly betting on the current regime never ending, is an easy trade.

The crisis-period behavior makes the case for that insurance plain. In the global financial crisis, the No-Regret portfolio held its NAV within two points of the fully-optimized one, while the Global Market Portfolio benchmark lost roughly 8%. In the 2022 rate shock, the one crisis actually built on the current regime itself, both No-Regret and the optimised construction held up far better than the benchmark, even as the portfolio built purely for today's regime actually gained ground. That last outcome is a useful reminder: the regime-specific optimum wins only if the regime in question is the one that actually materializes.

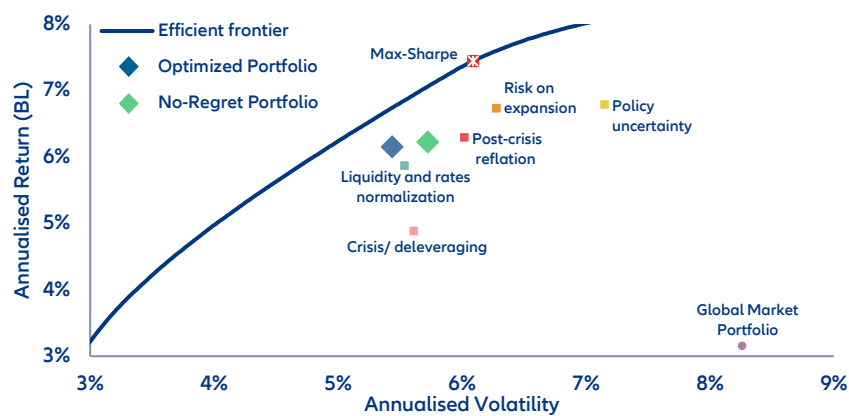
Figure 9: How each portfolio held up when it mattered (return rebased to 1)



Sources: LSEG Datastream, MSCI, Allianz Research

The Monte Carlo frontier confirms this is not a free lunch dressed up as one. No-Regret sits close to the efficient frontier with a Sharpe of 0.74, just behind the fully-optimized point (0.76) and ahead of every single-regime-optimal portfolio except Regime 2's risk-on construction (0.75), including the Regime 4-specific optimum (0.67 Sharpe, though R4 does carry the highest raw return among the regime portfolios, reflecting a bet on today's regime persisting). It sits well clear of the Global Market Portfolio benchmark (0.14). The frontier's true mathematical ceiling sits higher still, at a maximum Sharpe of 0.89, but only by concentrating nearly 70% of the portfolio in private debt alone, with gold, private equity and infrastructure making up almost all of the remainder and ten of the fifteen asset classes receiving no allocation at all. That portfolio is a mathematical artefact, not something any real mandate would hold. Taken together, the evidence says the No-Regret portfolio gives up only a modest amount of Sharpe (both against the fully-optimized point and against the true theoretical ceiling) in exchange for a portfolio that is actually diversified, and for materially better protection against downside risk and the one scenario this whole paper is built around: the current regime eventually ending.

Figure 10: Robustness, for a sliver of Sharpe



Sharpe ratios on this chart use Black-Litterman-blended (BL) returns, a shrinkage-adjusted version of historical averages. This is why the same portfolios (e.g. GMP, No-Regret) show different Sharpe ratios here than in Table 1.

Sources: LSEG Datastream, MSCI, Allianz Research

These assessments are, as always, subject to the disclaimer provided below.

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The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed or implied in such forward-looking statements.

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Allianz Research | 23 July 2026

The price of political fragility in Europe

Ludovic Subran
Chief Investment Officer & Chief
Economist
ludovic.subran@allianz.com

Patrick Krizan
Senior Investment Strategist
patrick.krizan@allianz.com

Pierpaolo Fiore
Research Assistant
pierpaolo.fiore@allianz.com

In Summary

- **With elections ahead in France, Spain, Greece, and Italy in 2027, political risk will be at the center of European government bond markets. Our high-frequency Political Fragility Index (PFI) shows that Europe is at a peak level of political fragility.** Our PFI, is built on polling data across four drivers — fragmentation, disaffection, polarization and governability. The Netherlands and Belgium sit high on fragmentation (votes splintered across a rising number of smaller parties), while France tops the list on polarization (rising voting share for far-left, far-right, Eurosceptic and populist parties). Germany and the UK show the sharpest increases in fragility since 2020 both suffering from weakening governability.
- **Sovereign risk premia (defined as asset swap spreads or ASW) have never been more sensitive to political fragility, and institutions matter more than the level of fragility.** The effect of political fragility on risk premia is a one-way ratchet: once triggered, it stays in the equation even after the trigger fades. This link, latent under QE, is now structural, with sensitivity at a record high (+1 std. dev. in the PFI adds +53bps to Italy's ASW). The risk premium is purely fiscal. Note that despite rising Eurosceptic votes, markets have priced out redenomination or euro breakup risk. Institutionally, majoritarian systems (UK and France) carry more political tail risk than their fundamentals imply for a given rating. This argues for a wider hedge around their electoral and budget calendar, while risk premia from consensus system issuers (e.g. Netherlands) tend to stay anchored through political noise.
- **To date, and since the end of QE in 2022, political fragility has resulted in an additional interest expense of around EUR 98bn (partly paid and partly still to be paid), when we sum up Italy, France, Spain, Belgium and the UK. On average, this represents an extra 3% of their annual debt servicing costs past and future.** This burden is concentrated in the UK (EUR 41bn which represents roughly 2.8% of additional debt service costs per annum over the maturity of the debt) and Italy (EUR 41bn; 4.9%), followed by Spain (EUR 14bn; 4.5%) and France (EUR 11bn; 2.3%). In France's case, most of this additional cost stems from the dissolution of parliament in 2024. Safe haven countries like Germany, the Netherlands and Austria show no structural fragility premium yet. Their shift from a negative to a positive ASW was mainly driven by a repricing of their fiscal fundamentals. However, given their underlying fragility this could change quickly, a structural fragility premium could add 1-2% to annual interest expenditure. Portugal and Greece were excluded given their bailout-distorted financing conditions.
- **France's presidential elections (April 18 and May 2, 2027), followed by Italy's general election due by 2027, are the next big rendezvous between political fragility and the bond market.** Our PFI shows France's sovereign premia are structurally sensitive to political fragility (+1 std. dev. adds +38bps to the ASW). This channel may keep pushing the OAT-Bund spread toward 90bps by year-end, with normalization contingent on a parliamentary majority in June 2027. In Italy, fragility is the single most sensitive channel in our sample (see supra) but it is currently dormant. Our baseline holds the BTP-Bund spread near 80bps, though a Eurosceptic turn on either flank, amplified by the proposed majoritarian electoral reform, is a fatter tail that could reawaken the redenomination premium France's spread no longer carries.

Political fragility is a major yet hidden factor for pricing European government bonds

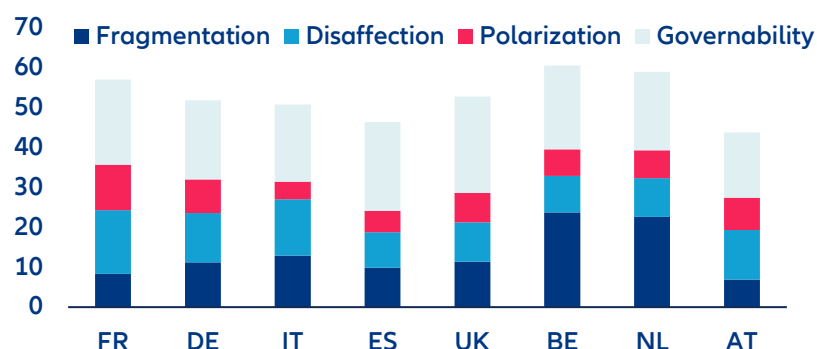
With upcoming elections in France (spring 2027) and Italy (latest in December 2027), political risk will be at the center of European government bond markets. But it is notoriously difficult to measure.

Political risk has played an important role in European government bond prices since the euro crisis of 2012, influencing them in two ways: Political risk can amplify or dampen the risk premia implied by economic and fiscal fundamentals depending on whether the political environment is perceived as fragile or stable, and it can also reflect the probability of an intentional Euro exit. This so-called “redenomination risk” is closely related to the fragmentation of Europe’s political systems under the ascent of populist, euro-sceptic movements. But political risk is notoriously difficult to measure, especially on a market-relevant frequency, and often remains under qualitative assessment prone to personal biases. To bring this hidden factor into the light of objective measurement, we develop a high-frequency index to quantify the impact of political fragility on European government bond prices, and ultimately set a fiscal price tag on political fragility in Europe.

Our Political Fragility Index (PFI) turns a country's politics into a single weekly number. Our base assumption is that we can observe the risk inherent in political systems by observing the dynamics within the party system. This is a crystallization of the underlying social cleavages and can therefore serve as a proxy for the political system as a whole, using weekly aggregations of election polls, which are available for almost all EU countries plus the UK. In this analysis, we will focus on eight countries: France, Germany, Italy, Spain, the UK, Belgium, the Netherlands and Austria. We leave out Portugal and Greece, whose bailout and debt-restructuring programs distorted their financing dynamic and, with them, our results. Based on the polls data, we can further separate four dimensions of political fragility:

- 1) *Fragmentation (30%)*: arises when votes splinter across a rising number of smaller parties (rescaled Laakso effective number of parties).¹
- 2) *Disaffection (20%)*: occurs when voters abandon the traditional governing parties, which usually indicates a weakening of traditional institutions (100 – traditional-party share).
- 3) *Polarization (20%)*: as the result of a rising voting share for far-left, far-right, Eurosceptic and populist parties (the average of far-left-plus-far-right, Eurosceptic and populist vote shares).²
- 4) *Governability (30%)*: is affected when the government itself leans on political forces on the extreme right and left of the political spectrum (100 – the share of not extreme governing parties).

Figure 1: The Political Fragility Index (July 2026)



Notes: The Political Fragility Index is a weekly, poll-based index on a 0–100 scale (higher = more fragile), built from smoothed (Kalman filtered) poll-of-polls vote shares. Sources: Wikipedia, Allianz Research

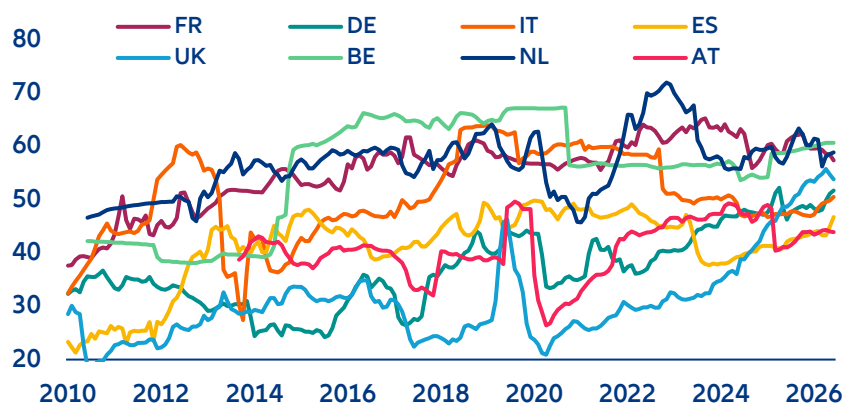
¹ Laakso, M. & Taagepera, R. (1979)

² Party-family classifications (populist, far-left, far-right, Eurosceptic) follow The PopList (EiQCC Database) and, like government composition, are time-aware. Source: Rooduijn M. et al. (2024)

This separation determines the dominant sources of the political fragility in each country. The latest data shows that all countries exhibit an elevated score in terms of governability. Belgium and the Netherlands structurally score higher in fragmentation while polarization is stronger in France, Germany and Austria where populist vote concentrates on one or two parties. France, Germany and Austria along with Italy also score high when it comes to the decline of traditional parties measured by disaffection (Figure 1).

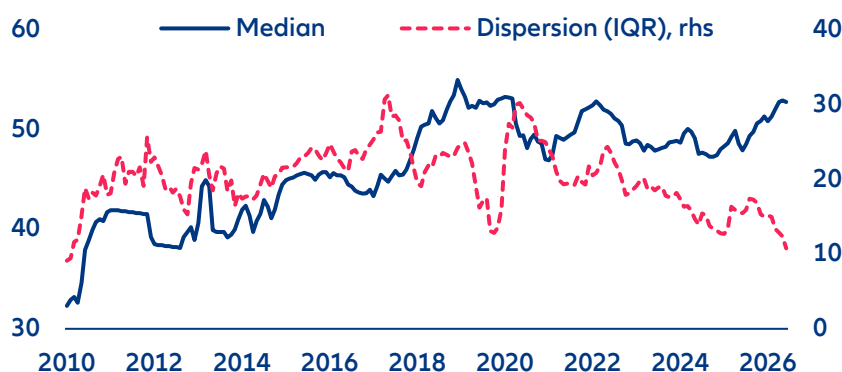
Political fragility has ratcheted higher in Europe since 2010, levels still differ from country but they are converging as safe haven erode. We can distinguish three phases in the evolution of political fragility in Europe since 2010. The post-GFC hangover (2010-2012) drove an economic and fiscal fragmentation that opened a gap between a politically fragilizing periphery (Italy, Spain) and a politically stable core (Germany). That built into the 2012 euro sovereign crisis where fiscal austerity and reform fatigue seeded a durable rise of Eurosceptic and populist parties that has since raised fragmentation and polarization. The third phase opens in 2020 with the Covid-19 shock. After a first compression fragility rebounds to fresh peaks due to eroded institutional trust and economic uncertainty, with the noticeable difference that the periphery-core spread is narrowing as fragility migrates into the core. Indeed, Germany and the UK show the strongest increases in fragility since the Covid-19 pandemic (Figure 2). The combination of rising median fragility and decreasing dispersion indicates that safe haven are eroding as Europe become equally fragile (Figure 3).

Figure 2: The Political Fragility Index, 2010–2026



Notes: The Political Fragility Index is a weekly, poll-based index on a 0–100 scale (higher = more fragile), built from smoothed (Kalman filtered) poll-of-polls vote shares. Sources: Wikipedia, Allianz Research

Figure 3: The Political Fragility Index, 2010–2026

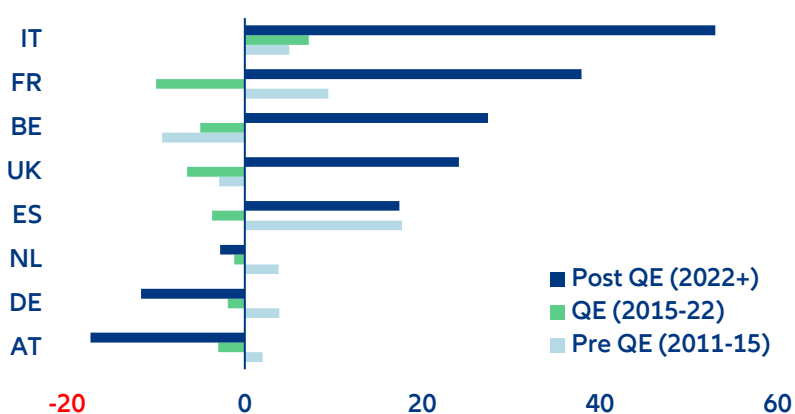


Notes: The Political Fragility Index is a weekly, poll-based index on a 0–100 scale (higher = more fragile), built from smoothed (Kalman filtered) poll-of-polls vote shares. Sources: Wikipedia, Allianz Research

Sovereign risk premia have never been more sensitive to political fragility

The translation of this political fragility to European Government Bond risk premia is anything but **mechanical**. Fragility can remain latent for years without leaving a clear imprint on sovereign debt prices. On the other hand, risk premia can reprice strongly on news flow even though the underlying political fragility remained unchanged. To measure political fragility as marginal driver of sovereign risk premia, we determine the part of each 10y asset-swap spread (ASW) that is not explained by country fundamentals (growth, fiscal balance, debt stock-flow, current account and competitiveness) and see how these residuals react to the PFI Index in different monetary regimes (Pre-Quantitative Easing (QE), QE and Post-QE). Before QE (2011–2015) changes in political fragility is not a driver for sovereign risk premia as the euro crisis effect prevails. Under QE (2015–2022), when central banks became the marginal buyer of government bonds, the sensitivity of sovereign risk premia to political fragility was suppressed and mostly negative (only Italy as exception as ASW surged under Salvini government despite QE) (Figure 4).

Figure 4: Sensitivity to political fragility (10y ASW, bps per 1sd of PFI Index)



Notes: The pricing analysis uses the surprise in fragility: the PFI's deviation from its own two-year moving average, lagged one week. Sources: Wikipedia, LSEG Datastream, Allianz Research

When QE purchases stopped in 2022 after the Covid-19 pandemic, Europe found itself in a weak fiscal position and was also politically strongly fragilized. The sensitivity of sovereign risk premia to political fragility has increased materially. A one-standard-deviation increase in PFI Index now adds 53bps to the Italian and 17bps to the Spanish ASW. The effect also extends to the UK (+24bps), Belgium (+27bps) and France (+38bps). Germany, the Netherlands and Austria continue to show no positive response (Table 1).

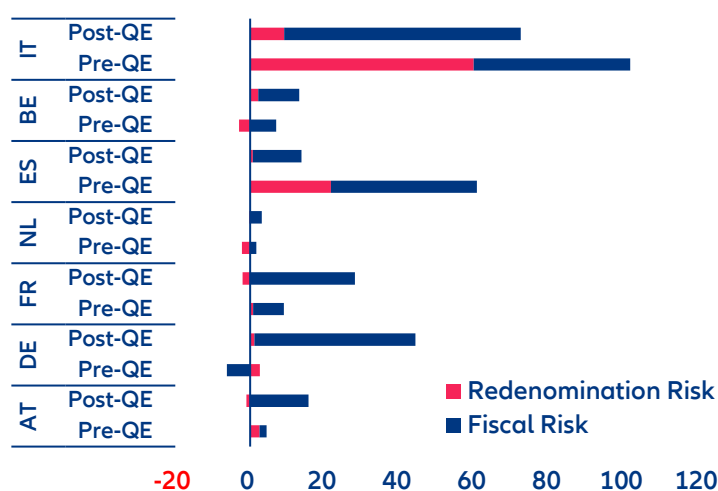
Table 1: PFI levels, range, volatility and sensitivity to spreads

Country	Current PFI	Max PFI	Min PFI	1 std	Sensitivity to ASW per 1 std (Post-QE)
Belgium	55	67.2	38.0	9.8	+27bps
Netherlands	57	72.0	43.9	5.7	-3bps
France	54	65.7	37.3	7.2	+38bps
United Kingdom	31	56.3	18.5	7.8	+24bps
Germany	36	52.3	24.0	7.4	-12bps
Italy	49	63.9	26.7	9.5	+53bps
Spain	39	50.2	18.6	9.1	+17bps
Austria	41	49.8	26.2	5.3	-17bps

Sources: Wikipedia, LSEG Datastream, Allianz Research

European government bonds are pricing out euro break-up risk almost entirely, despite a rise in Eurosceptic votes. The risk premium on Eurozone government bonds embeds two distinct risks: a fiscal risk tied to economic fundamentals and a "redenomination risk" reflecting the probability of a country exiting the euro, whether by political intent or market accident. By decomposing the risk premium into these two components and examining to which extent they respond to changes in the PFI Index, we can establish whether markets price political fragility primarily as a fiscal risk or as a euro break-up risk. The result is clear. Before the ECB launched QE in 2015, redenomination risk was present and accounted for almost half of the spread sensitivity for Italy and Spain. Today, sensitivity is driven almost entirely by fiscal risk. Redenomination risk has totally disappeared since QE, suggesting the emergency mechanisms put in place by the ECB and the EU are viewed as credible; an exit by market accident is no longer seen as plausible. An intentional exit, whether through a Eurosceptic populist government or a referendum, remains a tail risk. But it is one the market no longer appears to price. The Eurosceptic risk seems to have shifted entirely from a mix of fiscal and implied currency risk to a risk of populist fiscal profligacy (non respect of domestic and European budget rules) (Figure 5).

Figure 5: Sensitivity to political fragility (10y ASW, bps per 1sd of PFI Index)

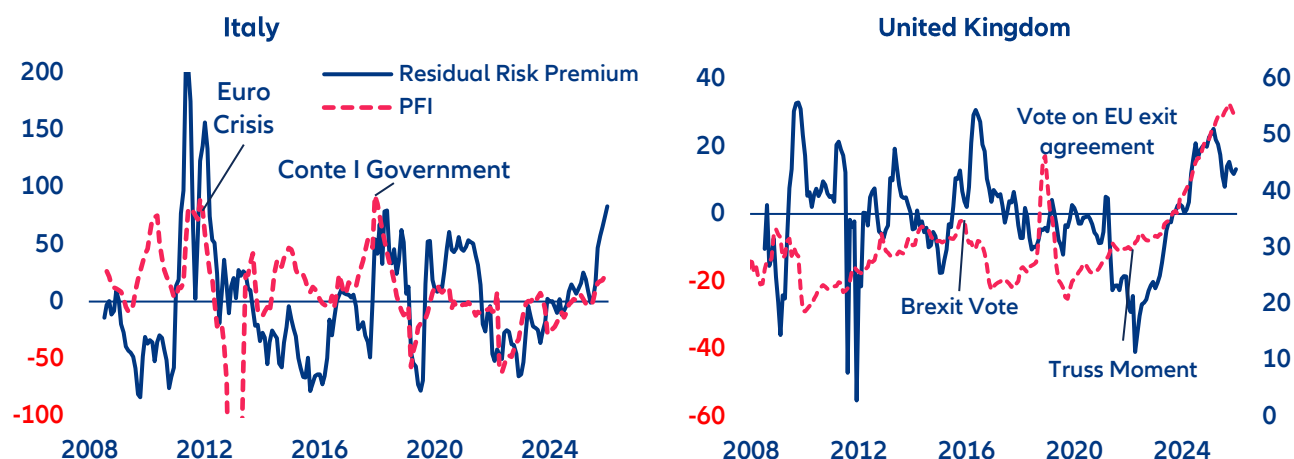


Notes: We use a variance decomposition model to separate the risk premia, methodology can be found [here](#).

Sources: Wikipedia, LSEG Datastream, Allianz Research

The relation between political fragility and sovereign risk premia has become structural. Two conclusions follow. First, political fragility is not priced into sovereign premia automatically. A threshold effect appears to be at work. But once priced, the effect persists. Second, current sensitivity levels are the highest observed to date. Italy and the UK offer instructive case studies. The Italian risk premium has long shown elevated sensitivity to political fragility. Historically, however, this relationship was episodic, driven by news flow and distinctive political events. Therefore, we observe sensitivity peaks during the 2012 euro crisis and again with the formation of the Salvini-led populist government in mid-2018 (notably, despite ongoing QE). Since 2022, the relationship has become structural, marked by a sustained elevated correlation between risk premium and political fragility. A similar structural change is observable in the UK. The 2022 Truss moment established a durable link between political fragility and UK sovereign risk premium, marking the onset of a synchronous rise. Prior to this, UK risk premium were largely insensitive to political fragility, even the Brexit referendum and the protracted EU exit negotiations left little discernible mark (Figure 6).

Figure 6: Sensitivity to political fragility (10y ASW, bps per 1sd of PFI Index)



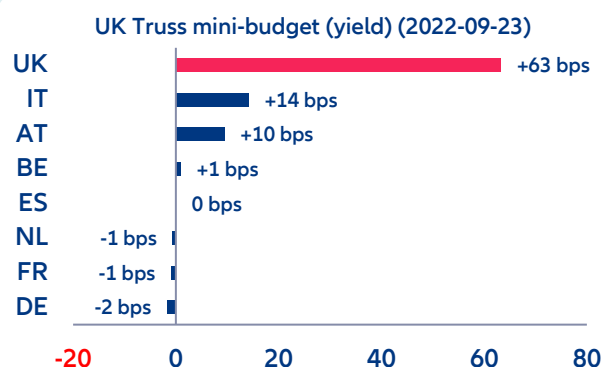
Sources: Wikipedia, LSEG Datastream, Allianz Research

Four case studies – How Political Fragility reacts to distinctive events

The UK is the cleanest switch-on in the sample.

Fragility went unpriced for a decade, then repriced violently at the September 2022 mini-budget. Around GBP45bn of unfunded tax cuts met a fragile mandate, the 30y Gilt yield jumped roughly 120bps in three days and a forced-selling loop in liability-driven pension funds left some “hours from collapse” until the BoE stepped in with emergency long-gilt purchases. The gilt premium has tracked political stress ever since. The moment fragility started to matter is dated almost to the day.

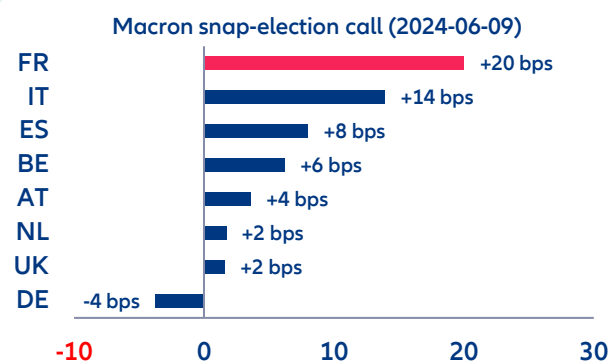
10y ASW change, post 4w – pre 8w mean, bps



France has crossed from core to periphery, and the market charges it for polarization.

The premium opened at the June 2024 dissolution: When Macron called a snap election, the 10-year OAT-Bund spread widened from about 45bps to as much as 80bps, its widest since the 2011–2012 crisis. France's headline fragility actually fell as the far right consolidated, so the composite score understates the risk while polarization is what the market prices. The correlations confirm the shift: Before the dissolution the French spread tracked Germany and the Netherlands (+0.61) more closely than Italy and Spain (+0.51); since the snap election

10y ASW change, post 4w – pre 8w mean, bps

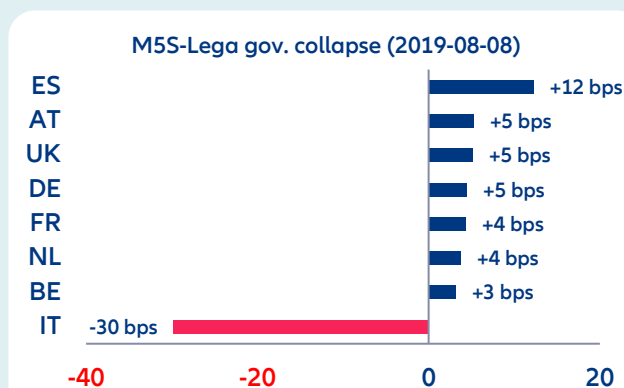
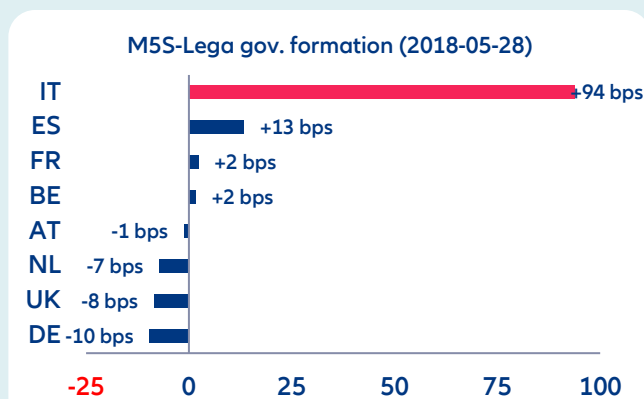


that ranking flipped, with France co-moving with the periphery (+0.69) than the core (+0.57).

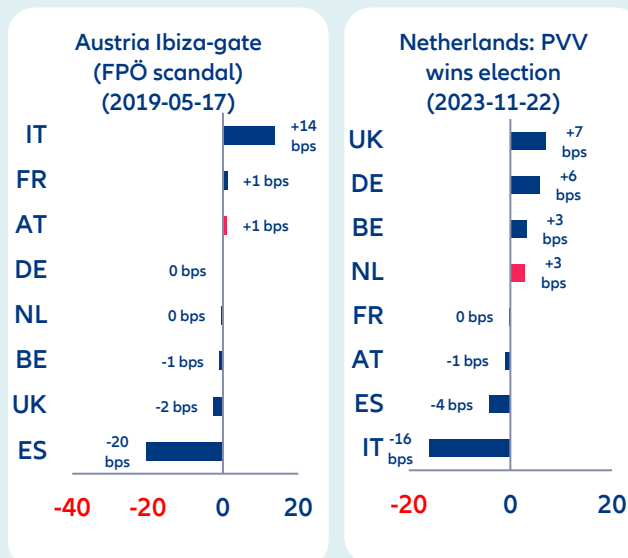
Italy is the mirror image of France, and the clearest proof that the price is state-dependent. Italy is Europe's archetypal political-risk pricer: The 2018 League-Five Star coalition added some 60bps of redenomination premium, and the 2019 collapse handed it back. Yet under Meloni the BTP-Bund spread (Italian over German government bonds) has fallen from about 250bps in late 2022 to a 15-year low below 90bps in 2025, alongside fiscal consolidation (the deficit cut from 4.3% towards 2.8%) and a string of rating upgrades. Two caveats keep us honest: Part of the narrowing reflects higher Bund yields (Germany's debt and defense package) and French instability rather than Italian strength alone, and our index shows the political channel is dormant, not gone. When fragility recedes, so does its price.

Germany, the Netherlands and Austria are the negative controls. None show a robust positive response to fragility once QE ends. Germany needs a caveat: Its recent spread regime is partly supply-driven, owing to the fiscal package and heavier issuance, so the political reading there is an upper bound, not a signal.

10y ASW change, post 4w – pre 8w mean, bps



10y ASW change, post 4w – pre 8w mean, bps



Note: UK shown on yield because in the 2022 Truss moment the SONIA swap dislocated more than the gilt, compressing the gilt-swap ASW. Sources: Wikipedia, LSEG Datastream, Allianz Research

Institutional design matters: Why the same fragility costs more in some countries

The country the market punishes most for fragility is not the most fragile one. The Netherlands looks more fragile than the UK, yet the market charges it almost no premium while the UK pays political premium. We see one explanation: Investors take into account the stability and reliability of the country's institutions. Political science separates two virtues: decisiveness (the power to change policy) and resoluteness (the power to commit and not reverse it), which trade off against a system's number of veto players.³ A centralized majoritarian system like the UK or France concentrates power and is decisive but not resolute, so one election can especially rewrite the budget. A federal consensus system like the Netherlands or Germany spreads power, so no single result moves fiscal policy far and its commitments are more credible. Investors may take into account that a shock change in Dutch fiscal policy is likely to dissipate in coalition and corporatist bargaining whereas a British one will hit immediately.

In majoritarian systems, investors hedge around the electoral calendar as they carry more political tail risk. Within our political fragility framework we score our eight countries according to the democracy type, the active cleavage behind its political party system and the type of welfare state acting as a cushion against political fragility (Table 2).

Table 2: Institutions, not the level of fragility, decide the bill

Country	Democracy type ⁴	Active cleavage ⁵	Welfare regime ⁶	Political fragility	Cost when fragility rises	Profile
Netherlands	Consensus	Functional	Corporatist / social-dem.	57	-3bps	High volatility
Belgium	Consensus (federal)	Territorial	Conservative-corporatist	55	+27bps	High active fragility
France	Majoritarian	Functional	Conservative / social-dem.	54	+38bps	Acute crisis risks, recoverable
Italy	Consensus → majoritarian	Territorial + functional	Mediterranean	49	+53bps	Low now, high latent
Austria	Consensus (corporatist)	Functional	Conservative-corporatist	41	-17bps	Rising corporatist erosion
Spain	Consensus-hybrid	Territorial + functional	Mediterranean	39	+17bps	Territorial-cleavage strain
Germany	Consensus (federal)	Functional + territorial	Conservative	36	-12bps	Resilient but fragmenting
United Kingdom	Majoritarian	Functional	Liberal	31	+24bps	Majoritarian with high latent risk

Notes: Democracy type: majoritarian = power concentrated, one bloc can govern alone; consensus = power shared across coalitions, chambers and courts. Active cleavage: the social divide driving politics, territorial (region, language, identity) or functional (class, economy). Welfare regime: liberal = thin and market-based; conservative = contributory and status-preserving; Mediterranean = family-based and dualistic, generous to insiders but thin for outsiders. Sources: Wikipedia, LSEG Datastream, Allianz Research

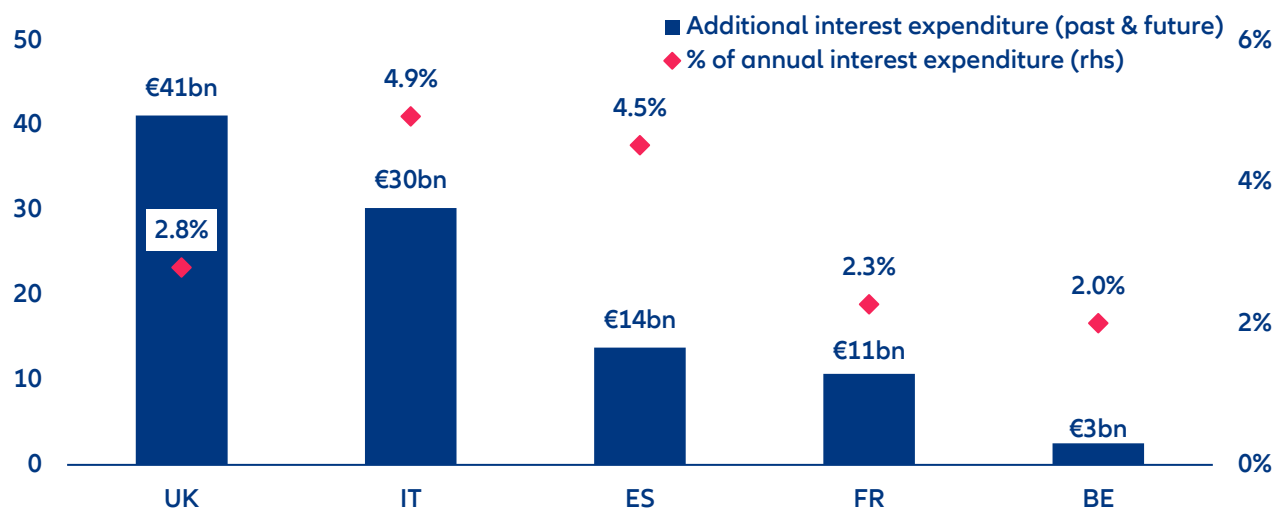
³ Cox and McCubbins(2001); Tsebelis (2002); ⁴ Lijphart, A. (2012); ⁵ Rokkan, S. (1970); ⁶ Esping-Andersen, G. (1990) and Ferrera, M. (1996)

For a given rating, majoritarian issuers carry more political tail risk, while consensus-system spreads tend to stay anchored through political noise. In our estimates, a rise in fragility adds around +31bps to the premium in the majoritarian systems, UK and France. Among the consensus democracies, the safe havens (the Netherlands, Germany and Austria) barely move. The bond market seems not to charge for how fragile the political system is – as measured by the changes in electoral polls via the PFI, but rather how likely this fragility will translate to financial turmoil and/or affect fiscal sustainability. The threshold for majoritarian systems seems to be lower especially when associated with deep active cleavages. The welfare state does not have a clear cushioning effect.

Political fragility accounts for 2% to 5% of annual debt service, but only for some countries

Political fragility can increase annual debt service by 3.3% on average, but the effect remains still unevenly distributed among European countries. We show that increases in political fragility can increase sovereign risk premia (asset-swap spread or ASW) which translates into higher government funding costs. Since the end of central bank QE in 2022, we estimate that political fragility has added a premium on newly issued European sovereign debt, raising past and future interest expenditure for Italy, France, Spain, Belgium, and the UK by a combined EUR 98bn. In relative terms, this premium represents on average about 3.3% of these countries' annual debt service, ranging from 2% to 5% (Figure 7). The burden is concentrated in the UK and Italy, with a cumulated effect of EUR 41bn and EUR 30bn (2.8% and 4.9% of annual debt service, respectively). Spain and France follow, with a cumulated effect of EUR 14bn and EUR 11bn. For France, the increase in debt service can be almost entirely attributed to the June 2024 parliamentary dissolution. For safe haven countries like Germany, the Netherlands or Austria political fragility has not yet become a structural driver of their sovereign risk premia. Their shift from a negative to a positive ASW was driven mainly by a changed perception of fiscal supply, such as the German Sondervermögen, not by shifting perceptions of political stability. However, given their underlying elevated fragility, it cannot be ruled out that this changes ahead of uncertain elections or during a government crisis. Should the premium become structurally priced in, it could quickly amount to 1-2% of annual interest expenditure.

Figure 7: The cost of political fragility since 2022 (EUR bn, % interest expenditure)



Notes: We estimate, for each country and monetary regime, the sensitivity of the political premium to fragility (γ , bps per one-standard-deviation move in the excess Political Fragility Index). The committed cost is based on the post-QE regime (from 2022), when fragility became a priced risk factor: the attributable premium is taken as $\max(\gamma \times z, 0)$, retained only where γ is positive and capped at the total non-fundamental (ASW) premium. Because most sovereign debt is fixed-rate, this premium is locked into coupons at issuance and paid until maturity. The committed cost then applies this index-attributable premium to the debt as it rolls over across its maturity structure (average residual maturity of 7 to 14 years): following standard debt-sustainability practice, the premium feeds into the interest bill gradually as maturing debt is refinanced, rather than repricing the entire stock at once. UK converted to EUR. Sources: Wikipedia, LSEG Datastream, Allianz Research.

The 2027 elections: upside risk to political fragility in France and Italy

We see a risk that the OAT-Bund spread widens further into the French presidential election (April 28 and May 2, 2027), particularly if polling continues to confirm the strength of the far-left and far-right blocs. The market is already positioning for this: The 10y OAT-Bund spread has trended clearly wider over the past two months and we expect it to reach 90bps by year-end (from 80bps today). Whether this widening proves structural or reverts quickly to fair value (that we see around 60bps), as has typically been the case in prior episodes, will depend not only on whether a moderate candidate wins the presidency, but critically on whether that candidate can also secure a parliamentary majority (elections in June 2027). Without a majority, France's majoritarian system heavily passes political fragility through to the sovereign risk premium. This makes this French election differs from prior ones. In the past, market risk repricing around French elections was effectively resolved after the two rounds of the presidential vote. This time, full resolution may not come until after four rounds, following the legislative elections as well. However, the nature of the French sovereign risk premium remains entirely related to fiscal sustainability, the probability of a "Frexit" is not priced - at least not for now.

We see upside risk to the Italian spread around the next general election, due by 2027, even though our baseline keeps the 10y BTP-Bund spread broadly stable at around 80bps over the next two years, on the back of the fiscal credibility the current government has built. The risk sits on both flanks: on the left, a win for the opposition coalition would return to the government the historically Eurosceptic, higher-spending Five Star Movement; on the right, the rise of Futuro Nazionale, a far-right party founded by the MEP and former army general Roberto Vannacci, could pull the governing coalition into chasing him on Eurosceptic ground. What makes this more than the usual noise is the institutional backdrop. The new

"Stabilicum" electoral law hands a large majority bonus (58% of the seats) to any coalition that clears 42% of the vote; and because the President of the Republic is elected by parliament, that same reinforced majority would come close to choosing the next president in 2029, dissolving the counterweight the president has exercised before, as in 2018, when he vetoed a Eurosceptic economy minister in the first Conte government. A more Eurosceptic politics on either flank, transmitted through a majoritarian system stripped of its circuit-breaker, is what could reawaken the redenomination component that France's premium lacks and that Italy prices most acutely at the short end. It is not our base case, but it is a fatter tail than today's tight spread implies.

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed

or implied in such forward-looking statements.

Such deviations may arise due to, without limitation, (i) changes of the general economic conditions and competitive situation, particularly in the Allianz Group's core business and core markets, (ii) performance of financial markets (particularly market volatility, liquidity and credit events), (iii) frequency and severity of insured loss events, including from natural catastrophes, and the development of loss expenses, (iv) mortality and morbidity levels and trends,

(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

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Cash at risk: Inventories are driving a new cycle of working capital strain

The structural rise in inventory financing needs could build up credit pressure



16 July 2026

Allianz Research

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Executive Summary



Ana Boata
Head of Economic Research
ana.boata@allianz-trade.com



Maxime Lemerle
Lead Advisor, Insolvency Research
maxime.lemerle@allianz-trade.com



Leo Michel
Public Affairs Intern
leo.michel@allianz-trade.com

The global cash conversion cycle (CCC) rose again in 2025 — and it is stuck at a high plateau. The global CCC – how long it takes for a unit of cash spent on operations to be converted into cash collected from sales – lengthened by a moderate half a day to above 67 days of turnover, some +3 days above its 10-year average and close to its 2023 high of 68. The past four years now run about +4 days longer than before 2020 (63): not a spike, but a structurally higher plateau that shows no sign of easing.

Asia stands out with the longest cash conversion cycle. The cash conversion cycle lengthened in Western Europe (+1.8 day), the Middle East & Africa (+1.6 days), Asia (+1.2 days) and South America (+1 days), but shortened in North America (-2.2 days) and Eastern Europe (-1.7 days). Asia carries the longest cycle at 70 days, driven by a persistent commercial gap: long customer terms (59 days of Days Sales Outstanding) that a low Days Payable Outstanding (44 days) cannot fully offset. It is followed by a tightly-clustered Western Europe, North America and South America (all 63 days), while Eastern Europe and the Middle East & Africa run the leanest.

The real driver is inventories in a world shaped by geoeconomics. Days Inventory Outstanding (DIO) now explains almost 80% of the CCC level and more than 80% of its change over 2014-2025 – up from 68% of the variation before 2021 to 90% since. This reflects a fundamental change in corporate behavior: Companies are moving away from “just-in-time” efficiency toward “just-in-case” resilience, with larger inventories now a strategic hedge against geopolitical uncertainty, supply-chain disruptions and trade fragmentation. In other words, supply chains are no longer optimized only for cost; they are increasingly designed for security, resilience and optionality.

Sector global dispersion is extreme: 25% of firms sit below 43 days and 25% above 107, suggesting an emerging bifurcation between strategic sectors and the rest of the economy. Twelve of 20 sectors extended their cash conversion cycle – automotive suppliers (+4 days) and paper, metals and textiles (+3 days each) leading – while eight compressed, notably transport equipment (-6 days), computers & telecom (-4 days) and energy (-3 days). The sectoral divergence in working-capital dynamics reflects the uneven impact of securonomics. Industries most exposed to supply-chain fragmentation – particularly upstream manufacturing and industrial inputs – have increased inventory buffers, resulting in structurally higher working-capital requirements. By contrast, several strategically important sectors, including energy, transport equipment and digital infrastructure, have shortened their cash conversion cycles despite heightened geopolitical uncertainty. This likely reflects a combination of stronger pricing power, robust cash generation, supportive industrial policies and sustained structural demand. Strategic relevance, therefore, does not necessarily translate into higher inventory intensity; it increasingly provides the financial flexibility and operational leverage needed to strengthen resilience while preserving working-capital efficiency.

For 2026 we expect a contained rise in CCC. First, the US-Iran conflict is expected to follow a “lighter” version of the 2022 supply-chain shock template: faint in listed firms’ financials in H1 2026, as flows take time to normalize, and more tangible in H2 as the disruption feeds through supply chains with a lag. As a result, we do not expect a full replication of the 2022 surge (+5 days of CCC, of which 81% came from inventories). The shock will not land evenly across sectors. Electronics, pharmaceuticals, textiles, automotive suppliers, metals and paper face the most direct pressure: already inventory-heavy and running elevated cycles, they have the least room to absorb a further DIO rise without tipping their financing needs into distress territory. Construction and machinery & equipment carry the largest absolute cycles (~103 days) and are unlikely to escape a broad inventory rebuild. Second, the shock should be partly offset by continued private-sector spending on AI infrastructure and data centers, which supports computers & telecoms and software & IT, keeping a meaningful share of the economy on a compressing or at worst flat trajectory. On these assumptions, measures to reassess energy security, strategic inventories and supply-chain resilience, together with the direct fallout of the conflict, should push global DIO up by around +2 days. Given our estimated elasticity, each additional day of DIO translates into +1.16 days of CCC globally.



Definitions & Methodology

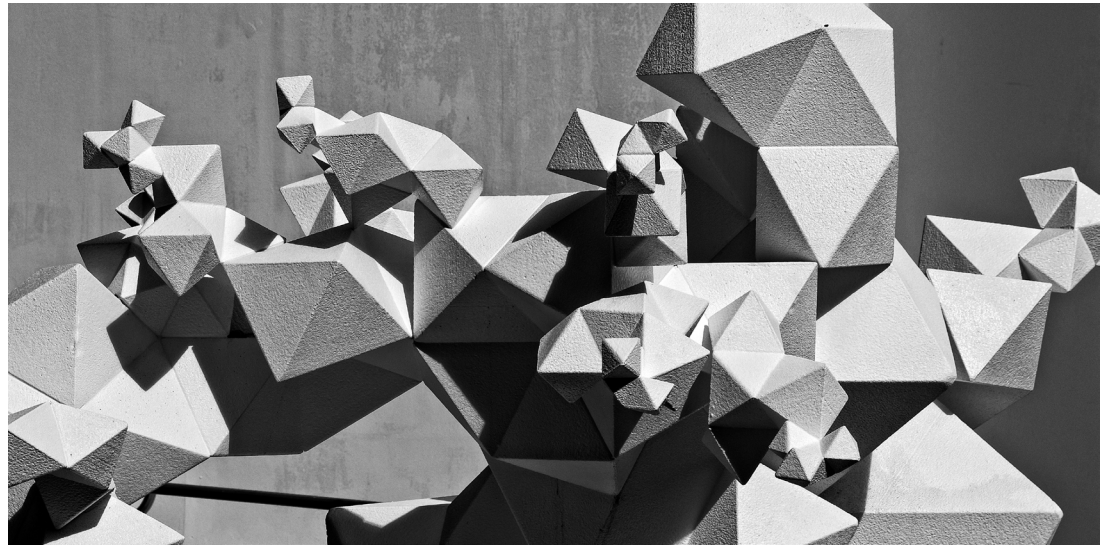
Definitions.

DSO (Days Sales Outstanding) measures the time taken to collect cash from customers; DPO (Days Payable Outstanding), how long the company takes to pay suppliers; DIO (Days Inventory Outstanding), how long cash is locked in stock. CCC (Cash Conversion Cycle) is a time metric expressed in days – how long it takes for a unit of cash spent on operations to be converted into cash collected from sales; it is the implied result of the computation $DSO - DPO + DIO$. WCR (Working Capital Requirement) is its monetary counterpart, expressed in currency: how much money is frozen in the operating cycle at a given point in time.

Methodology.

Our computation is based on the financials of listed firms available on LSEG for 55 countries, i.e. around 25,000 firms, focusing on companies that release extended financials (income statement plus balance sheet). Annual levels and changes are computed from the succession of quarterly figures, using a four-quarter rolling average rather than year-over-year to smooth accounting distortions such as window-dressing and end-of-period bulk supply arrivals. Compared with past editions we have switched from a mean to a median approach to limit the impact of outliers (lowering most figures) and aligned the breakdown with our internal 20-sector nomenclature (from 22). Global and regional figures aggregate all firms across all countries, rather than averaging regional or country means, to better reflect business and competitive reality. For 2025 the sample splits 53% Asia, 19% North America, 15% Western Europe, 7% Middle East & Africa, 4% South America and 2% Eastern Europe. All figures are as of 1 June 2026.



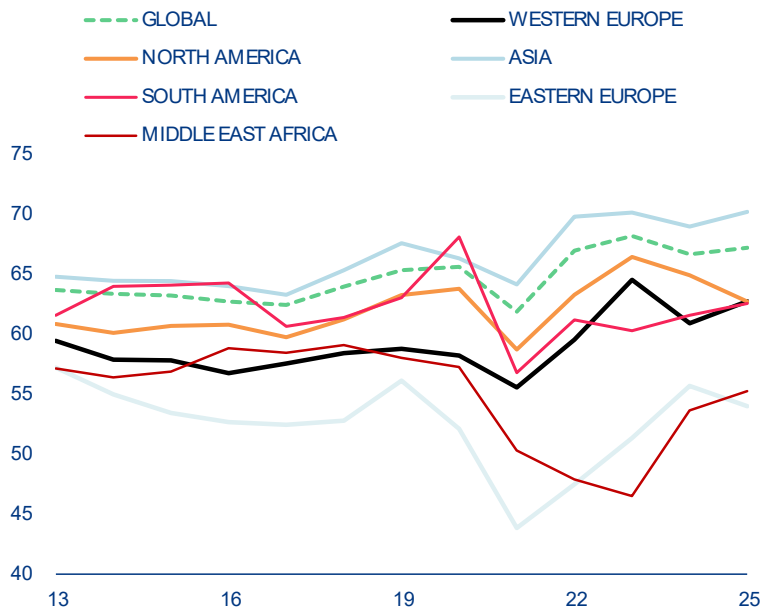


A fragmented picture

The global cash conversion cycle rose by +0.5 in 2025 to above 67 days of turnover (annual average) – about +3 days above its 10-year average and close to the 2023 high of 68. The past four years now average some +4 days more than before 2020 (63). Yet the headline

figure is not representative of the underlying diversity of situations: the 2025 landscape hides asymmetric views, with noticeable regional differences layered over country- and sector-level fragmentation, and uneven drivers beneath.

Figure 1: Global and regional CCC, annual levels, in number of days of turnover

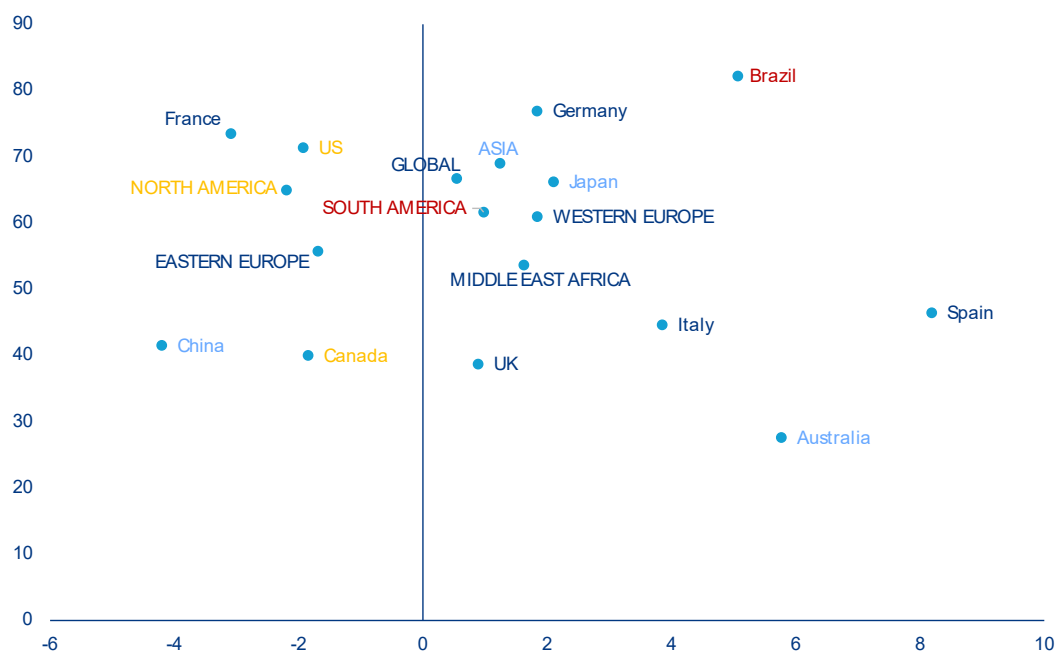


Sources: LSEG, Allianz Research

Diverging trends, uneven levels. The cash conversion cycle lengthened across most regions – Western Europe (+1.8d), Middle East & Africa (+1.6d), Asia (+1.2d) and South America (+1d) – but shortened in North America (-2.2d) and, to a lesser extent, Eastern Europe (-1.7d). Levels are just as uneven: Asia runs the longest cycle (70 days), ahead of Western Europe, North America and

South America (all ~63d), while Eastern Europe and Middle East & Africa sit at the lean end. The shift is structural, too: The last four years (2022-2025) stand clearly above the 2013-2019 average in Asia (+5d), Western Europe (+3d) and North America (+3d).

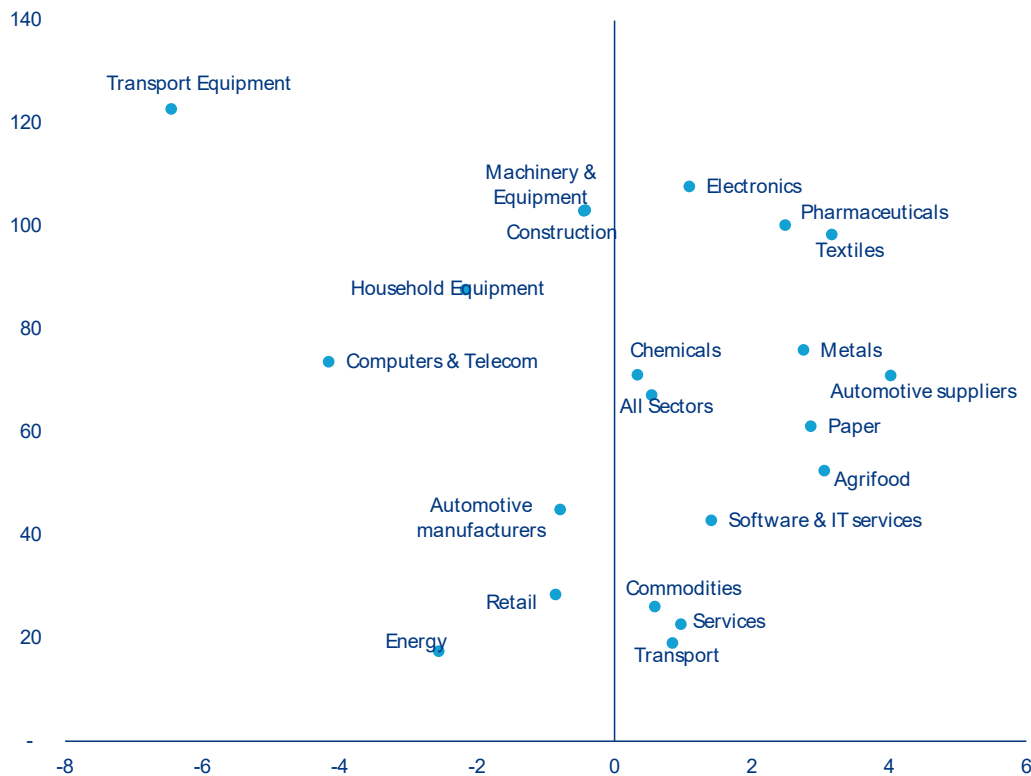
Figure 2: Regional and selected-country CCC – annual level (y-axis) and change (x-axis), in number of days of turnover, 2025



Sources: LSEG, Allianz Research

Regional aggregates mask deep country-level fragmentation. Europe is polarized: Germany and France form a high-CCC cluster (78.7d and 70.4d) but moved in opposite directions – Germany trapped cash (+1.8d) while France released it sharply (-3.1d). The UK, Italy and Spain occupied a lower band (39.6d to 54.6d) while Southern Europe extended fast (Spain +8.3d, Italy +3.9d). Asia's 70.3d baseline masks lean core markets moving in opposite directions (China 37.3d, -4.2d; Australia 33.4d, +5.8d; Japan 68.3d, +2.1d), implying a very high cash conversion cycle across the rest of the region. South America's 62.6d reflects Brazil's high and rising cycle (87.2d, +5.1d) more than the region. North America is the exception – a single consistent bloc (62.8d) releasing cash across both the US (-1.9d) and Canada (-1.8d).

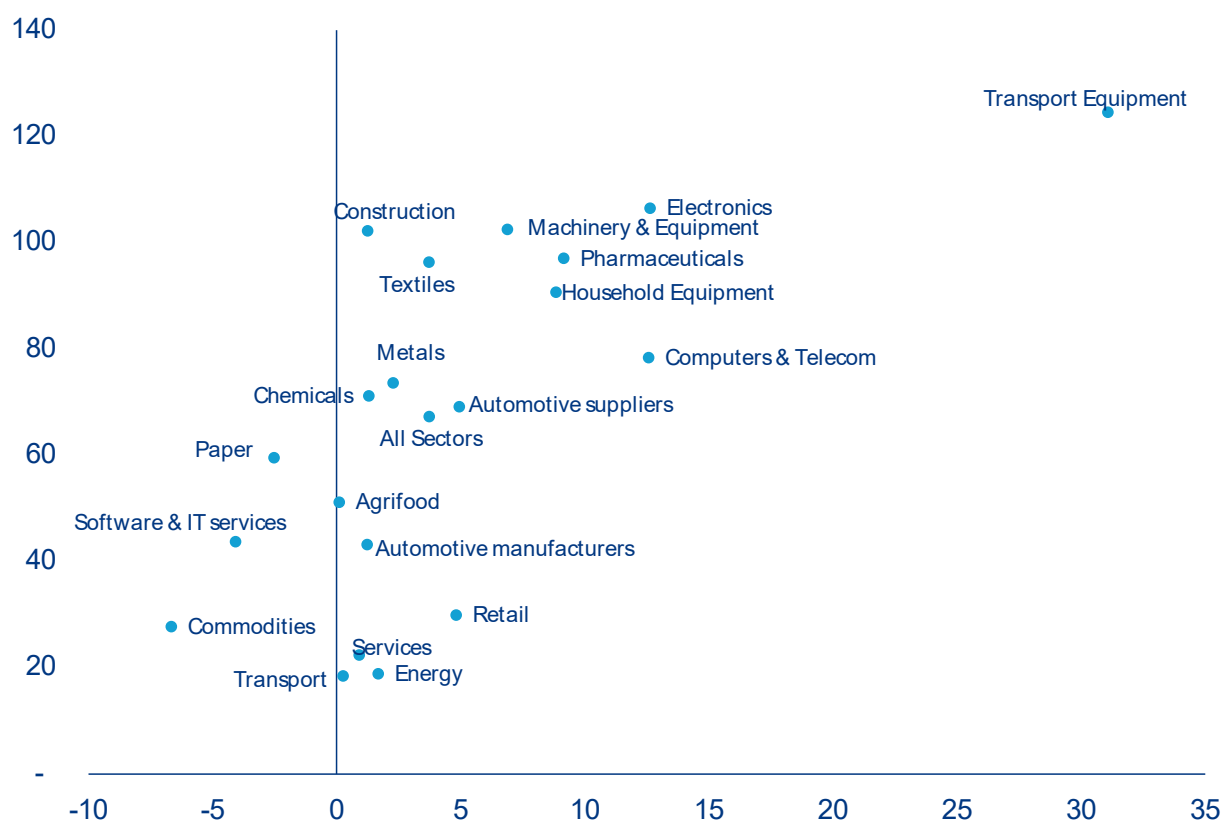
Across sectors, diverging trends prolong the usual wide discrepancies. Twelve global sectors expanded their cash conversion cycle in 2025 – with the largest increases for automotive suppliers (+4d), followed by paper, metals and textiles (+3d each) – while eight compressed, led by transport equipment (-6d), computers & telecom (-4d) and energy (-3d). The persistent gaps between sectors are expected: They follow structural differences in business models (retailers can run negative cycles, manufacturers long ones), strategic choices (upfront-paid subscriptions vs. credit extended to win customers) and supply-chain design (global sourcing lifts inventory; nearshoring trims it but raises costs).

Figure 3 Global sector CCC — annual level (y-axis) and change (x-axis), in number of days of turnover, 2025

Sources: LSEG, Allianz Research

Ranked by the length of the cash conversion cycle, the sectors fall into three financing-need clusters – and nine recognizable profiles within them. The higher the cycle, the more capital a sector must permanently fund. So the direction of the 2025 move tells us whether that burden is easing or building. At the top (CCC > 76 days) sit the 'trapped' – electronics, pharmaceuticals and textiles, heavy on working capital and trapping still more as their cycle rises — alongside 'heavy standards' construction and machinery & equipment (~103 days locked up, but frozen at -0.4d) and the 'turnarounds' transport equipment and household equipment, which cut hard (-6.5d and -2.2d). The mid-tier (57-76 days) holds 'creeping risks' (automotive suppliers, metals, paper – rising and at risk of tipping into the trapped next year), a 'balanced pivot' in chemicals (71d, +0.3d) and a 'structural optimizer' in computers & telecom (-4.2d). The efficiency zone (< 57 days) contains 'hidden drags' expanding fast off a low base (agrifood +3.1d, software & IT +1.4d), 'lean anchors' running flat (retail, commodities, services, transport, automotive manufacturers) and the 'accelerator', energy, hyper-lean at 17.5 days and cutting a further -2.6d.

Sectors are rarely in sync across regions. Across the three largest economic regions – Western Europe, North America and Asia – only four of 20 sectors saw their CCC moving in the same direction in 2025: chemicals and paper rose in all three, while computers & telecom and machinery & equipment fell. Asia is the epicenter, driving the change in 12 of 20 sectors and posting the highest structural level in nine (agrifood, auto suppliers, commodities, construction, electronics, machinery equipment, retail, textiles and transportation). Western Europe remains the efficiency benchmark, holding the lowest CCC in 12 of 20 sectors even where it drives the change (chemicals, electronics, machinery, paper, textiles, transport). North America is the contrarian swing factor, the most frequent outlier, trending against the dominant move in seven sectors and acting as a natural diversifier, but still maintaining large CCC in six sectors (auto suppliers, computer/telecoms, energy, household equipment, services and services/IT services) and lower CCC in five sectors (chemicals, construction, metals, paper and pharmaceuticals).

Figure 4: Global sector CCC — 2022-2025 mean vs 2013-2019 (x-axis) and 2025 level (y-axis), in number of days of turnover

Sources: LSEG, Allianz Research

Figure 5 Global and regional sector CCC — 2022-2025 mean vs 2013-2019 and 2025 level, in number of days of turnover

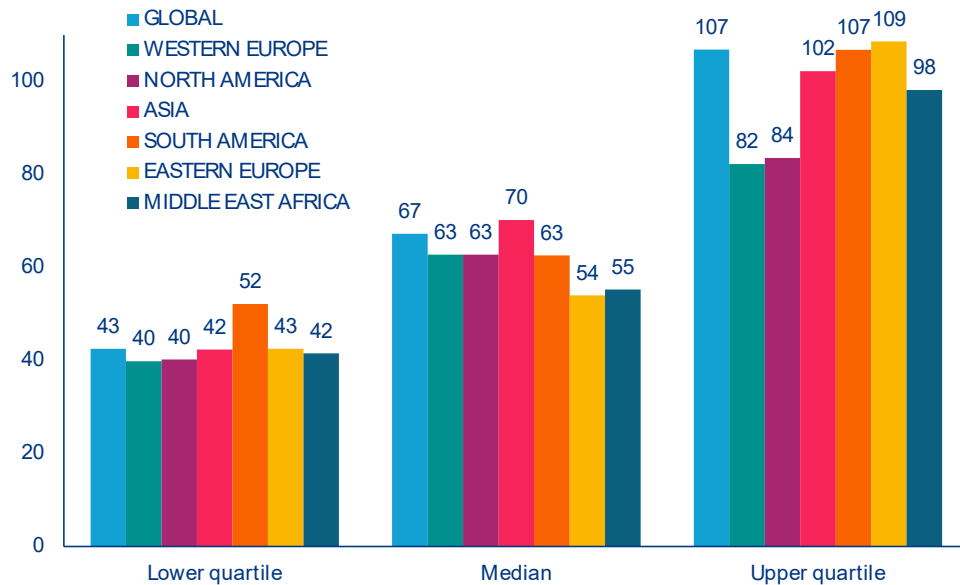
	Change					Level				
	GLOBAL	WESTERN EUROPE	NORTH AMERICA	ASIA	Other regions (avg)	GLOBAL	WESTERN EUROPE	NORTH AMERICA	ASIA	Other regions (avg)
Agrifood	3	2	-1	4	1	53	45	51	55	61
Automotive manufacturers	-1	0	-2	4	-9	45	43	48	44	60
Automotive suppliers	4	2	-2	3	11	71	64	64	74	88
Chemicals	0	2	1	2	-7	71	77	61	75	70
Commodities	1	-13	-2	-13	-4	26	-25	19	29	20
Computers & Telecom	-4	-1	-7	-5	-3	74	78	82	66	2
Construction	0	-1	2	-6	3	103	54	50	135	93
Electronics	1	-6	-4	2	4	108	97	101	110	80
Energy	-3	2	-3	-2	0	18	17	19	7	29
Household Equipment	-2	0	-3	-2	4	88	75	96	90	84
Machinery & Equipment	0	-4	-2	0	10	103	92	97	103	110
Metals	3	3	-4	6	8	76	85	58	67	91
Paper	3	5	4	4	-2	61	71	53	58	64
Pharmaceuticals	2	2	-3	5	-13	100	96	88	94	101
Retail	-1	-3	2	-3	0	29	15	29	31	48
Services	1	2	-2	3	-4	23	15	32	25	16
Software & IT services	1	-2	-1	4	3	43	34	50	43	43
Textiles	3	8	-4	3	8	98	77	85	99	126
Transport	1	-3	0	2	2	19	14	17	25	40
Transport Equipment	-6	3	-1	-9	-14	123	121	113	100	116
All Sectors	1	2	-2	1	0	67	63	63	70	57

Sources: LSEG, Allianz Research

The distribution is the ultimate fragmentation. Globally, 25% of firms run a cash conversion cycle below 43 days and 25% above 107. The worst quartile clusters in Asia (25% of firms above 102 days), South America (107), Eastern Europe (109) and Middle East & Africa (98) – a far wider spread than in North America and Western Europe, where outliers are fewer. By sector, five stand

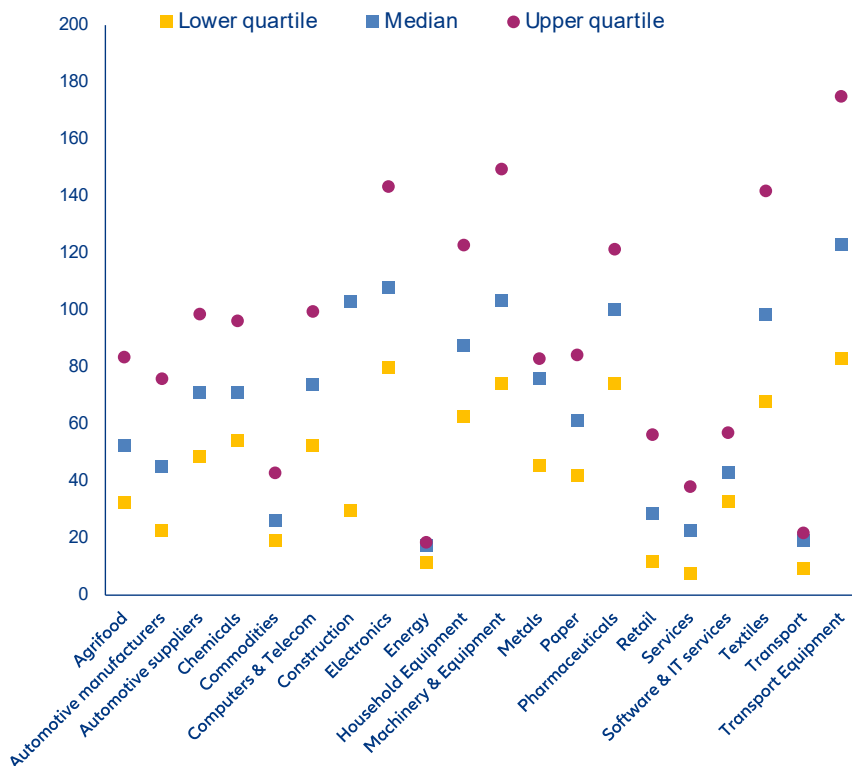
out with a quarter of firms above 140 days (construction, transport equipment, machinery & equipment, electronics and textiles), followed by pharmaceuticals and household equipment (25% above 120); at the other end, services, retail, energy and transport keep a quarter of firms below 15 days.

Figure 6: 2025 CCC by quartile, globally and by region, in number of days of turnover

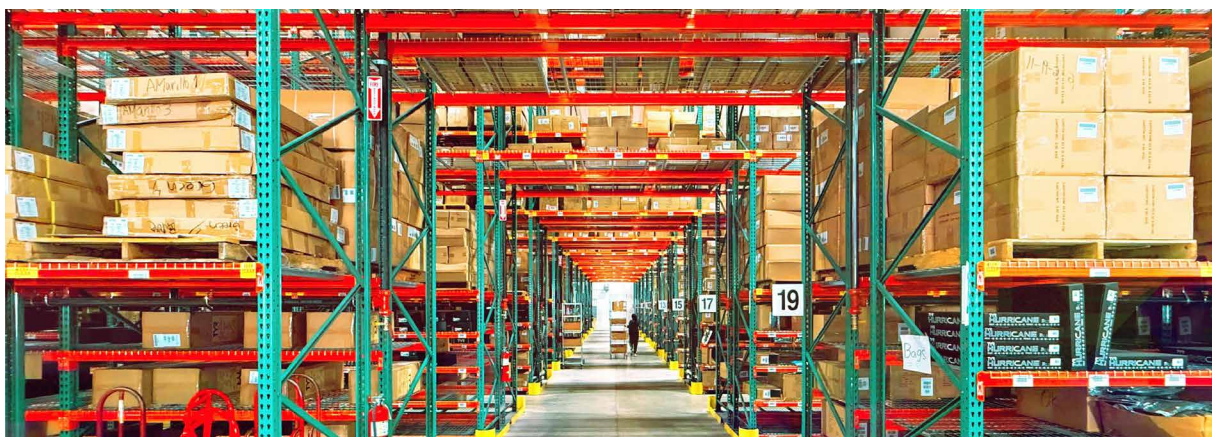


Sources: LSEG, Allianz Research

Figure 7: 2025 CCC by quartile, globally and by region, in number of days of turnover



Sources: LSEG, Allianz Research



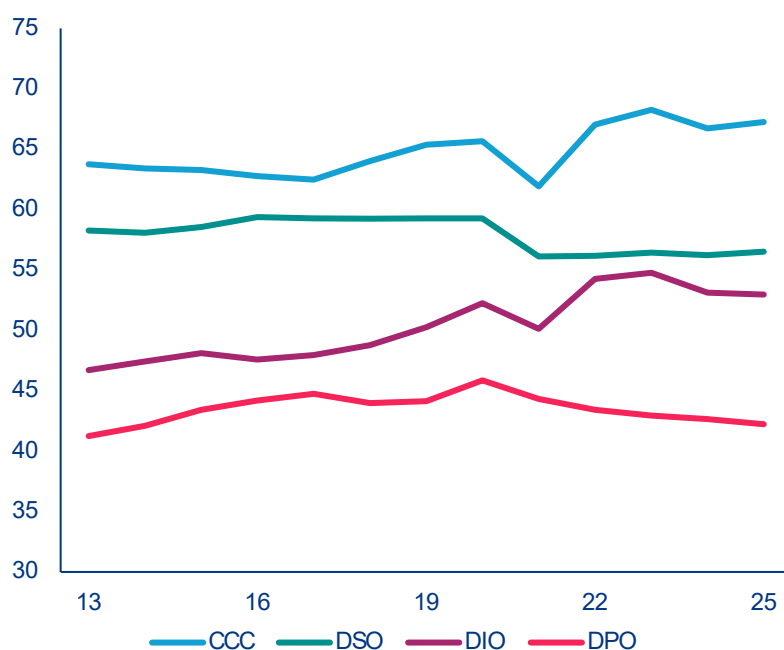
An inventory-dominated landscape

The warehouse is the balance sheet: inventories are the primary force – and risk – behind the cash conversion cycle

At the global level, net payment terms did the lifting in 2025. A rise in Days Sales Outstanding (DSO) met a fall in Days Payable Outstanding (DPO) while inventories edged down: paying suppliers faster pushed the cycle

up. DSO rose modestly (+0.3d) to a contained 56.5 days – roughly stable at a low level since 2022 and still -3d below the pre-pandemic norm. DPO fell for the fifth consecutive year since 2021, widening the net commercial balance (DSO minus DPO). Days Inventory Outstanding (DIO) slipped -0.2d to 53 but remained well above the pre-pandemic average of 48 – confirming the structural supply-chain shift from ‘just in time’ to ‘just in case’.

Figure 8: Global CCC, DSO, DPO and DIO, annual levels, in number of days of turnover



Sources: LSEG, Allianz Research

The same move can be a strategic choice or a warning signal. A +1 day increase is not good or bad in itself: it can reflect either bulk-buying for volume discounts, safety stocks against supply shocks and flexible terms to win clients or unsold product, deteriorating collections and lost supplier trust. Figure 9 sets out how to read each subcomponent move on both sides of the ledger.

But the headline again hides asymmetric regional dynamics. The upside was inventory-driven in Western Europe, where stock accumulation (+1.3d to 54) trapped cash, and the downside inventory-driven in North

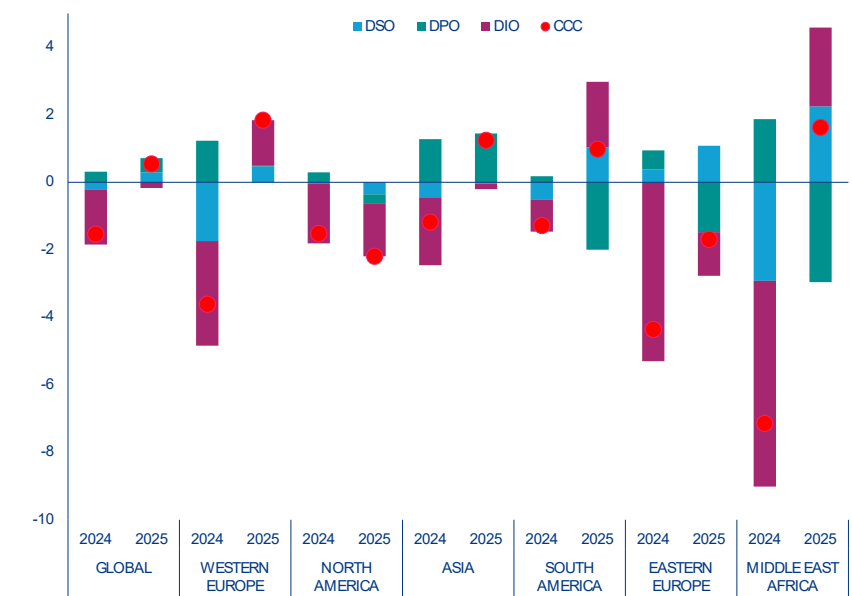
America, where a leaner book (-1.6d to 46) released it. Elsewhere the lever was payables: In Asia a sharp cut in supplier-payment windows (-1.4d to 44) drove the cycle up, while Eastern Europe, South America and the Middle East & Africa stretched suppliers (+1.5d to +3d) as a defensive reaction to inventory pressure. DSO, by contrast, was not the trigger in relative terms despite its broad-based rise from Middle East & Africa (+2d), Eastern Europe (+1d) and Western Europe (+0.5d) down to a stable Asia and a slightly lower North America (-0.4d).

Figure 9: Decoding a CCC move — strategic moves vs. warning signals

CCC move	Subcomponent drivers	Good reasons (strategy / efficiency)	Bad reasons (risk / inefficiency)
RISE IN CCC (cash locked up longer)	• DIO ↑ (more stock) • DSO ↑ (slower collections) • DPO ↓ (faster vendor pay)	• Bulk buying for volume discounts • Safety stocks against supply-chain shocks • Flexible terms to win major clients	• Unsold products / inventory obsolescence • Poor collection / rise in bad debts • Loss of supplier trust (shorter terms)
DROP IN CCC (cash generated faster)	• DIO ↓ (less stock) • DSO ↓ (faster collections) • DPO ↑ (slower vendor pay)	• Just-in-time inventory optimization • Strict credit control & automated dunning • High supplier leverage (longer terms)	• Severe stockouts causing missed sales • Damaged client relations via aggressive pressure • Stretching vendors under liquidity stress

Sources: Allianz Research

Figure 9: Annual change in CCC by subcomponent, by region, in number of days of turnover, 2024 and 2025

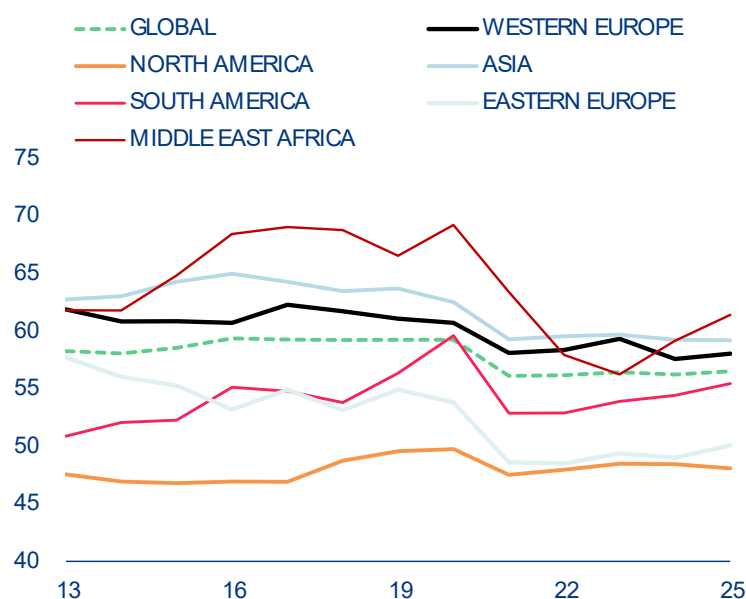


Sources: Allianz Research

Cross-referencing commercial terms, inventory exposure and the level of the cycle sorts the regions into four structural clusters. Where the cycle is long, is it the commercial gap (DSO minus DPO) or the inventory that keeps it there? The answer splits the map cleanly:

- **Structural capital trap** – Asia (payment-terms driven) – The highest cycle (70d, +3 above global). Long customer terms (59d of DSO) that a low DPO (44d) cannot offset leave about 16 days of cash permanently trapped in the commercial cycle; here a high CCC is structural.
- **Cash-lean commercials** – Eastern Europe (payment-terms driven) – The leanest footprint (54d, -13 below global): a net commercial balance of just six days (50d DSO – 44d DPO) and controlled inventory (48d) give it the smallest structural cycle of any region.
- **Balanced supply chains** – Middle East & Africa (inventory driven) – High DIO (55d) and the longest collection (61d of DSO) are matched almost exactly by supplier financing (61.5d of DPO): a near-zero commercial gap leaves inventory as the only asset requiring funding.
- **Inventory-dependent** – Western Europe, North America, South America (inventory driven) – Three near-identical 63-day cycles, about 4.5 days leaner than global and dictated by inventory (54d / 46d / 50d) rather than terms – with North America a hyper-lean, cash-is-king outlier that is the most exposed to an inventory shock.

Figure 11: Global and regional DSO, annual levels, in number of days of turnover

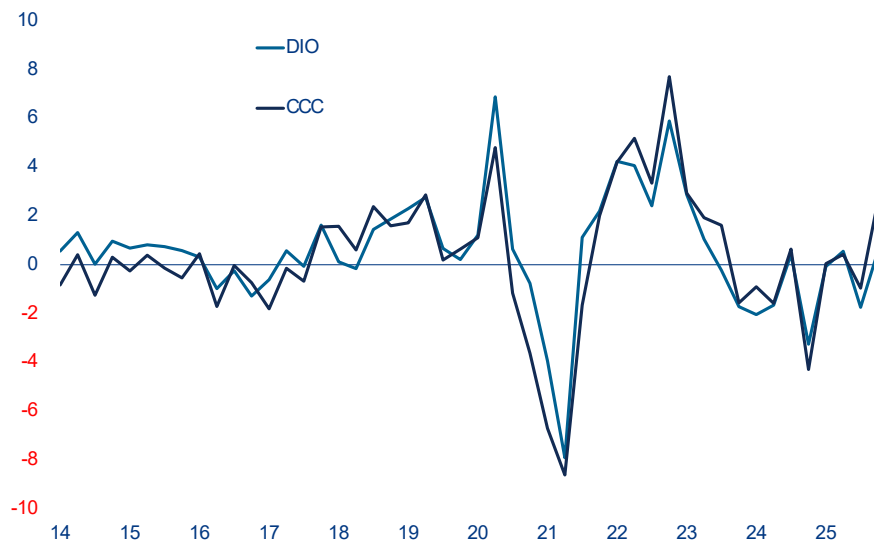


Sources: LSEG, Allianz Research

The key outcome: Inventories now explain a growing share of the cycle. DIO contributes almost 80% of the CCC level today, up from 75% a decade ago, and accounts for more than 80% of the change over 2014-2025. The intensification is recent and broad: DIO’s explanatory role in the cycle jumped from 68% before 2021 to 90% since – with Western Europe (30% to 90%) and Asia (32% to 90%) leading the shift, and North America already at 94%.

Reading a DIO move takes the same care. A rising DIO can be a deliberate geopolitical buffer or a margin play – or dead stock and overproduction; a falling one can be lean logistics – or liquidity starvation and stockouts. A low level is only efficiency if fulfilment holds up: paired with a spike in backorders, a dropping DIO destroys reputation to save cash. Figure 13 decodes the signal.

Figure 12: Global CCC and DIO, quarterly, y/y change in %



Sources: LSEG, Allianz Research

Figure 13: Decoding a DIO move – supply-chain mastery vs. structural risks (COGS = cost of goods sold)

DIO move	Underlying driver	Good reasons (strategic choice)	Bad reasons (operational drag)
RISE IN DIO (inventory sits longer)	• Average inventory ↑ • or COGS ↓	• Geopolitical buffer: safety stock against supply shocks • Margin optimization: bulk buying to lock in low input prices • Peak-season readiness: stacking ahead of promotions	• Sales misjudgment: sluggish demand leaves goods stuck • Product obsolescence: perishable, outdated or dead stock • Overproduction: output despite shrinking demand
DROP IN DIO (faster inventory turnover)	• Average inventory ↓ • or COGS ↑	• Lean operations: successful just-in-time rollout • Assortment rationalization: cutting slow, low-margin lines • High-velocity demand: strong pull clearing product	• Liquidity starvation: no cash to buy critical inputs • Frequent stockouts: empty shelves, missed sales • Supply disruptions: unable to restock on vendor failure

Sources: Allianz Research

Sector by sector, it is mostly an inventory story – sorting into five clusters (Figure 14). DIO drives the 2025 change in half the sectors (10 of 20) and underpins the structural level in 14 of 20 – most visibly in physical-goods manufacturing, where production cycles and storage lock capital into stock. The exceptions prove the rule: where receivables or payables lead, the businesses are asset-light or mid-renegotiation. The pattern is regionally polarized between Asia (highest, most volatile) and Western Europe (leanest, the benchmark), with North America counter-cyclical.

- Pure inventory (four sectors) – Agrifood, construction, household equipment, textiles – DIO drives both the 2025 change and the structural level; any CCC move traces straight back to stock.
- Inventory-shifting / dual receivables-inventory (five sectors) – Automotive suppliers, chemicals, machinery

& equipment, paper, transport equipment – DIO drives the change, but the level is shaped jointly by DSO and DIO (B2B manufacturers with long client terms and heavy stock).

- Payables under an inventory regime (five sectors) – Commodities, electronics, energy, metals, pharmaceuticals – inventory sets the level, but 2025 moves came from DPO, signaling active renegotiation of supplier terms.
- Asset-light / services (two sectors) – Services, software & IT – DSO defines the level (no meaningful inventory); the 2025 move comes from how fast these firms pay their vendors (DPO).
- DSO-driven outliers (three sectors) – Automotive manufacturers, retail, transport – the only sectors where receivables lead the change, pointing to client-payment shifts rather than supply-chain forces.

Figure 14: Global sector CCC — key change and level drivers, by component and by region, 2025

Sector	Change driver	Level driver	Regional change driver	Regional outlier	Top level region	Lowest level region
Agrifood	DIO ↑	DIO ↑	Asia ↑	North America ↓	Asia	Western Europe
Automotive mfg.	DSO-DPO ↓	DPO ↑	Asia ↑	North America ↓	North America	Western Europe
Automotive suppliers	DIO ↑	DSO/DIO	Asia ↑	North America ↓	Asia	Western Europe
Chemicals	DIO	DSO/DIO	Western Europe ↑	—	Western Europe	North America
Commodities	DPO	DIO ↓	Western Europe ↓	—	Asia	Western Europe
Computers & Telecom	DIO ↓	DSO	North America ↓	—	North America	Asia
Construction	DIO	DIO ↑	Asia ↓	North America ↑	Asia	North America
Electronics	DPO ↑	DIO ↑	Western Europe ↓	Asia ↑	Asia	Western Europe
Energy	DPO ↓	DIO ↓	North America ↓	Western Europe ↑	North America	Asia
Household Equipment	DIO ↓	DIO ↑	North America ↓	—	North America	Western Europe
Machinery & Equip.	DIO	DSO/DIO ↑	Western Europe ↓	—	Asia	Western Europe
Metals	DPO ↓	DIO	Asia ↑	North America ↓	Western Europe	North America
Paper	DIO ↓	DSO/DIO	Western Europe ↑	—	Western Europe	North America
Pharmaceuticals	DPO ↓	DIO ↑	Asia ↑	North America ↓	Western Europe	North America
Retail	DSO ↑	DPO ↓	Asia ↓	North America ↑	Asia	Western Europe
Services	DPO	DSO ↓	Asia ↑	North America ↓	North America	Western Europe
Software & IT services	DPO ↓	DSO ↓	Asia ↑	Asia ↑	North America	Western Europe
Textiles	DIO ↓	DIO ↑	Western Europe ↑	North America ↓	Asia	Western Europe
Transport	DSO	DSO ↓	Western Europe ↓	Asia ↑	Asia	Western Europe
Transport Equipment	DIO ↑	DSO/DIO ↑	Asia ↓	Western Europe ↑	Western Europe	Asia

Sources: LSEG, Allianz Research



What we expect for 2026

We expect a prolonged rise in the cash conversion cycle. Measures to reassess energy security, strategic inventories and supply-chain resilience improve robustness but raise inventories – and thus working-capital needs. As seen during the 2022 energy-price shock, we expect the disruption from the US-Iran conflict to be only faintly visible in listed firms' financials in the first half of 2026 and more tangible in the second as flows normalize unevenly and shocks take time to feed through sectors and supply chains

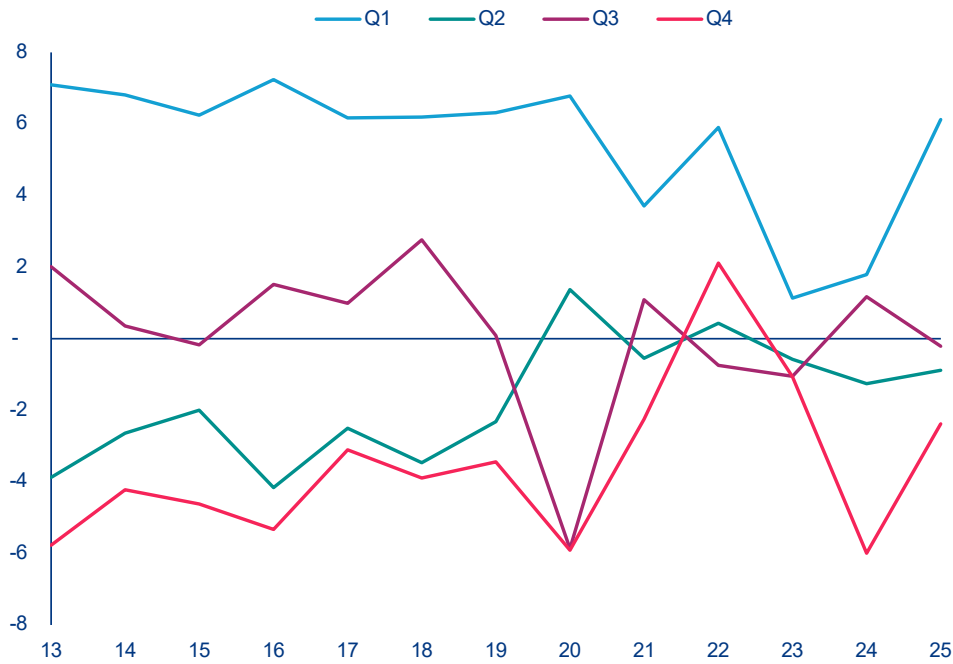
A repeat of 2022 would massively increase the cash conversion cycle – but we do not expect a full replication. Our elasticity puts one extra day of inventory at +1.16 days of CCC globally (Western Europe +1.19, North America +1.17, South America +1.34, Eastern Europe +1.25, Asia and Middle East & Africa +1.0). In 2022, the cycle surged +5 days over the year, roughly +4 of it (81%) from inventories – 95% in Western Europe,

110% in North America, 78% in Asia and 76% on average for other regions. This time the shock looks strong but shorter, and should be partly offset by the AI boom, with data centers and large tech platforms driving private-sector spending.

The net result is expected at around +2.4 days of CCC, alongside higher financing costs. We expect global DIO to rise by about +2 days, translating into +2.4 days of CCC to 70 days (annual average, +6d above the 10-year average). Ultimately this means higher short-term financing costs: with the credit and macro effects of the US-Iran disruption likely to persist, central-bank easing may proceed more slowly than anticipated, weighing on demand and refinancing conditions.

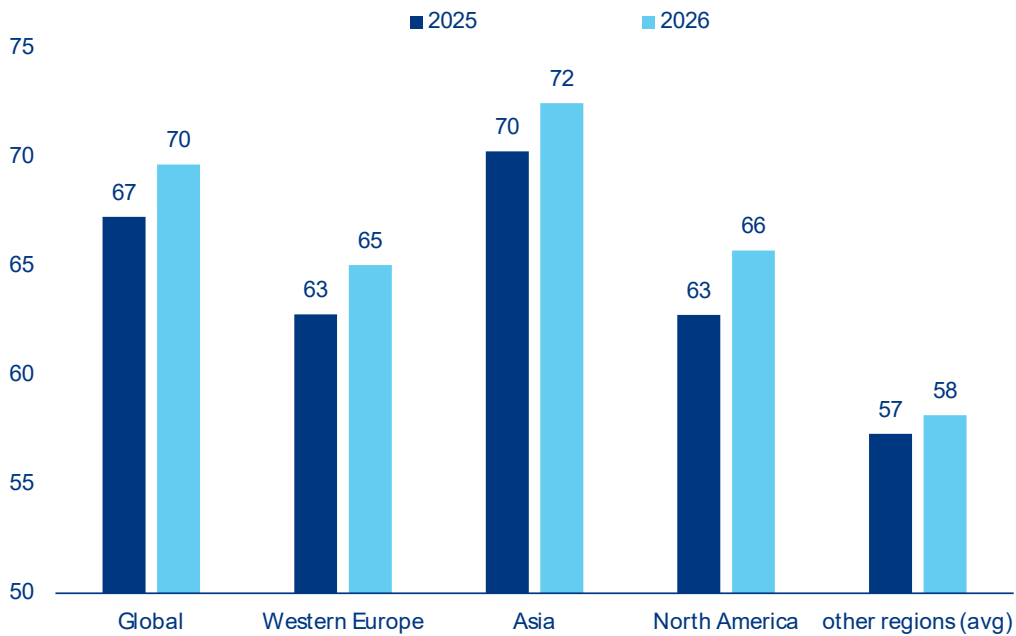
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Figure 15: Global and regional CCC, q/q changes by quarters



Sources: LSEG, Allianz Research

Figure 16: Global and regional CCC 2026 forecasts



Sources: LSEG, Allianz Research

Late payments: Who blinks first, buyer or supplier?

Few regulatory questions bear as directly on Days Sales Outstanding as the one Europe has spent three years failing to answer: how fast must a business be paid? Payment terms are the raw material of working capital. Stretch them and DSO climbs, liquidity tightens and the weakest suppliers edge towards insolvency; compress them and the working-capital gap begins to close. That is why the fate of Europe's late-payment rules matters well beyond Brussels – and why the answer now emerging is not a single European rule but a patchwork of national ones, moving at different speeds and in the same direction: towards faster payment.

The story is best read as a suspended center and an active periphery. The EU-level file has stalled, while France and the UK legislate at home and a cluster of adjacent files – on insolvency, on the new EU company form and on e-invoicing – quietly reshape the same terrain. Below, we take each in turn.

The EU file has stalled. The European Commission's September 2023 proposal for a Regulation on combating late payments – intended to give teeth to the 2011 Late Payment Directive (2011/7/EU) through a hard 30-day cap on B2B terms – has been effectively shelved. Blocked from the outset by a German-led coalition, the file was formally paused when an internal Council document confirmed in February 2026 that, with 21 member states opposed, no further meetings would be convened until at least 2027. The compromise reached before the pause had already softened the text: a deviation to 60 days (120 for slow-moving or seasonal goods), financial services and insurance-claim payments taken out of scope, eased default-interest charges and a possible reclassification from Regulation to Directive.

The center of gravity has moved to national capitals. Payment delays keep rising and SME associations remain vocal, so the debate is migrating to domestic initiatives. In France, a Senate bill authored by Senator Olivier Rietmann and adopted unanimously on 19 February 2026 tightens the national regime – citing a 25% increase in business-failure risk attributable to payment delays in 2024 (rising to 42% beyond 60 days) and EUR17bn of receivables weighing on SME liquidity. It raises the sanctions ceiling to 1% of global turnover with a three-year recurrence window for doubled fines, prohibits waiving late-payment penalties and creates a public factoring fund for micro-enterprises and SMEs exposed to a public debtor. A cap on payment terms is, so far, out of scope; but the bill now sits with the National Assembly, where a fragmented chamber could table more restrictive amendments – even if the Government's accelerated procedure limits the room for a wholesale rewrite.

The UK has moved from announcement to legislation. The Commercial Payments Bill (Small Business Protections (Late Payments) Bill) was confirmed in the King's Speech on 13 May 2026 and introduced to the House of Lords on 19 May. It grants the Small Business Commissioner new powers of investigation, adjudication and fines reaching tens of millions of pounds for persistent offenders, sets a statutory 60-day maximum term for large companies dealing with smaller suppliers and mandates interest on overdue invoices at 8% above the Bank of England base rate. Contractual flexibility is preserved between large firms and in international trade, with phased entry into force from H2 2026 and full application not expected before 2027. Early business reaction, including from the CBI, has been supportive.

The surrounding regulatory context. Late payment is the near end of a spectrum whose far end is insolvency, and three adjacent files shape the same terrain. On insolvency, the direction favors creditors: EU co-legislators reached provisional agreement on the Insolvency Directive on 19 November 2025, harmonizing transaction-avoidance rules, asset-tracing, creditors' committees and pre-pack proceedings – measures that tend to lift cross-border recovery rates, with adoption and transposition still to come. Running alongside, the 28th Regime (‘EU Inc.’, COM(2026)321) carries its own simplified insolvency chapter; at the 28 May Competitiveness Council its Chapter X emerged as the live battleground, with several member states resisting a zero-minimum-capital company form on creditor-protection grounds. The unresolved questions – subrogation rights, retention of title and creditor ranking – are precisely those that determine how much an unpaid creditor recovers when a buyer fails.

E-invoicing: the quiet accelerant. The VAT in the Digital Age (ViDA) package, adopted on 11 March 2026, will make structured electronic invoicing the default and mandatory for intra-Community B2B transactions by 1 July 2030, with national rollouts already running ahead – France for large companies and ETIs from September 2026, Germany for B2B from 1 January 2027. E-invoicing has no direct bearing on statutory payment terms, but by shortening invoice-processing cycles and improving traceability it is likely, over time, to compress effective payment behavior. The first French data after September 2026 will be the earliest real-world signal of whether that compression is material.

Figure 17: 2025 DSO Heat map by sector, region and selected countries, in number of days of turnover

	GLOBAL	WESTERN EUROPE	Germany	France	UK	Italy	Spain	NORTH AMERICA	US	Canada	ASIA	China	Japan	Australia	SOUTH AMERICA	Brazil	EASTERN EUROPE	MIDDLE EAST AFRICA
Agrifood	41	46	44	53	61	51	45	29	29	34	40	22	50	37	42	34	47	57
Automotive manufacturers	33	40	41	81	43			20	21	16	32	17	30	30	34	34	33	59
Automotive suppliers	61	67	68	60		63	21	56	56	67	61	101	65	53	59	59	62	63
Chemicals	61	59	48	55	69		60	52	48	61	62	60	91	57	66	44	54	73
Commodities	57	74		79				40	40	39	70	158	46	56	62	61		77
Computers & Telecom	64	63	57	74	72	62		58	57	65	65	60	75	61	68	78	60	59
Construction	65	58	32	91	60	68	82	45	47	51	78	138	83	56	72	116	56	71
Electronics	69	71	87	80	74	78		61	61	90	73	103	91	68	44		84	48
Energy	48	52	52	40	54	72	52	45	43	48	48	58	42	60	53	54	43	51
Household Equipment	58	58	64	70	50	27		50	51	48	62	108	72	59	59	76	52	93
Machinery & Equipment	72	67	58	68	76	73	94	58	59	57	80	103	85	53	69	81	83	55
Metals	58	55	54	59	61	48	50	42	46	24	66	21	85	125	46	47	45	92
Paper	58	57	73		64		68	47	50	40	60	50	64		50	48	47	90
Pharmaceuticals	66	69	71	69	83	82	63	59	59	52	68	77	78	68	60	60	93	25
Retail	14	17	10	21	22	36	10	7	6	17	16	20	19	10	44	59	12	55
Services	45	50	53	79	34	68	75	46	47	43	42	46	26	51	46	66	24	87
Software & IT services	62	74	65	98	74	80		56	55	57	59	188	57	53	61	74	68	61
Textiles	50	42	49	18	54		19	47	46	49	51	38	62	56	78	87	37	49
Transport	42	39	44	52	44			35	35	41	48	39	42	44	26	25	37	137
Transport Equipment	79	99		84	92		205	64	69	21	99	208	100	132	51	51	58	
All Sectors	57	58	55	70	58	65	67	48	49	46	59	59	61	59	55	62	50	61

Sources: LSEG, Allianz Research

Statistical Appendix

GLOBAL	2025 changes				2025 levels			
	DSO	DPO	DIO	CCC	DSO	DPO	DI	CCC
Agrifood	1	-1	2	3	41	39	51	53
Automotive manufacturers	-1	1	1	-1	33	43	55	45
Automotive suppliers	0	-2	2	4	61	48	58	71
Chemicals	0	0	1	0	61	48	58	71
Commodities	0	-1	-1	1	57	43	12	26
Computers & Telecom	-1	1	-2	-4	64	47	56	74
Construction	2	0	-3	0	65	57	95	103
Electronics	-2	-4	-1	1	69	34	73	108
Energy	0	3	0	-3	48	49	19	18
Household Equipment	-1	-1	-2	-2	58	40	69	88
Machinery & Equipment	0	-2	-2	0	72	42	74	103
Metals	0	-4	-1	3	58	50	67	76
Paper	2	2	3	3	58	50	53	61
Pharmaceuticals	0	-1	1	2	66	36	71	100
Retail	-1	0	0	-1	14	38	52	29
Services	0	-1	0	1	45	32	9	23
Software & IT services	0	-1	0	1	62	27	8	43
Textiles	0	0	3	3	50	36	84	98
Transport	1	0	0	1	42	30	7	19
Transport Equipment	-2	2	-3	-6	79	50	94	123
All Sectors	0.3	-0.4	-0.2	0.5	57	42	53	67

WESTERN EUROPE	2025 changes				2025 levels			
	DSO	DPO	DI	CCC	DSO	DPO	DIO	CCC
Agrifood	1	3	4	2	46	59	57	45
Automotive manufacturers	1	0	-1	0	40	57	60	43
Automotive suppliers	-2	-3	1	2	67	57	54	64
Chemicals	1	0	1	2	59	45	62	77
Commodities	-2	13	1	-13	74	110	11	-25
Computers & Telecom	-1	1	0	-1	63	50	66	78
Construction	1	2	0	-1	58	65	61	54
Electronics	-5	-5	-6	-6	71	43	68	97
Energy	-1	-5	-2	2	52	55	20	17
Household Equipment	0	0	0	0	58	44	62	75
Machinery & Equipment	1	-1	-5	-4	67	45	69	92
Metals	2	-1	1	3	55	48	78	85
Paper	4	4	5	5	57	56	70	71
Pharmaceuticals	5	3	0	2	69	43	69	96
Retail	-1	1	0	-3	17	54	52	15
Services	-1	-2	2	2	50	44	9	15
Software & IT services	0	1	-1	-2	74	45	5	34
Textiles	-1	-5	3	8	42	48	83	77
Transport	-1	1	0	-3	39	32	7	14
Transport Equipment	13	-4	-14	3	99	75	96	121
All Sectors	0.5	0.0	1.3	1.8	58	49	54	63

ASIA	2025 changes				2025 levels			
	DSO	DPO	DIO	CCC	DSO	DPO	DI	CCC
Agrifood	0	-2	2	4	40	33	48	55
Automotive manufacturers	-2	-3	2	4	32	40	53	44
Automotive suppliers	-1	-4	1	3	61	42	55	74
Chemicals	-1	-1	1	2	62	46	59	75
Commodities	-1	8	-3	-13	70	56	15	29
Computers & Telecom	-2	2	-1	-5	65	56	58	66
Construction	1	-4	-12	-6	78	60	117	135
Electronics	-6	-9	-1	2	73	38	75	110
Energy	0	2	1	-2	48	60	20	7
Household Equipment	-2	-2	-3	-2	62	42	71	90
Machinery & Equipment	-1	-3	-2	0	80	51	74	103
Metals	-2	-8	0	6	66	66	68	67
Paper	3	-1	0	4	60	45	43	58
Pharmaceuticals	0	-2	3	5	68	48	74	94
Retail	-1	1	-1	-3	16	33	48	31
Services	1	-2	0	3	42	27	9	25
Software & IT services	2	-1	0	4	59	25	9	43
Textiles	1	1	3	3	51	35	83	99
Transport	2	0	0	2	48	30	7	25
Transport Equipment	-11	-2	0	-9	99	115	116	100
All Sectors	0.0	-1.4	-0.2	1.2	59	44	55	70

NORTH AMERICA	2025 changes				2025 levels			
	DSO	DPO	DI	CCC	DSO	DPO	DIO	CCC
Agrifood	-2	1	2	-1	29	31	53	51
Automotive manufacturers	-2	1	0	-2	20	28	56	48
Automotive suppliers	-1	-1	-2	-2	56	49	58	64
Chemicals	2	1	0	1	52	41	50	61
Commodities	2	2	-1	-2	40	34	13	19
Computers & Telecom	-2	-1	-6	-7	58	32	56	82
Construction	2	0	0	2	45	37	41	50
Electronics	1	1	-3	-4	61	28	67	101
Energy	-1	3	0	-3	45	45	20	19
Household Equipment	0	0	-3	-3	50	27	72	96
Machinery & Equipment	1	0	-3	-2	58	31	70	97
Metals	0	-3	-7	-4	42	35	51	58
Paper	1	-2	2	4	47	44	51	53
Pharmaceuticals	-2	-2	-4	-3	59	24	53	88
Retail	0	0	2	2	7	31	53	29
Services	-1	0	-1	-2	46	23	8	32
Software & IT services	-1	0	0	-1	56	20	14	50
Textiles	-2	0	-2	-4	47	32	70	85
Transport	0	0	-1	0	35	25	7	17
Transport Equipment	2	1	-2	-1	64	31	80	113
All Sectors	-0.4	0.3	-1.6	-2.2	48	32	46	63

EASTERN EUROPE	2025 changes				2025 levels			
	DSO	DPO	DIO	CCC	DSO	DPO	DI	CCC
Agrifood	6	1	2	7	47	38	55	64
Automotive manufacturers	2	0	-6	-4	33	44	65	54
Automotive suppliers								
Chemicals	3	0	-9	-6	54	43	55	66
Commodities								
Computers & Telecom								
Construction	6	7	4	3	56	66	63	53
Electronics	26	-2	-17	11	84	42	64	106
Energy	4	7	-3	-6	43	38	21	26
Household Equipment	-4	-7	11	13	52	47	81	85
Machinery & Equipment	16	13	2	5	83	63	92	113
Metals	6	-3	3	11	45	49	85	81
Paper	1	-2	-7	-4	47	44	33	37
Pharmaceuticals	-11	-2	5	-4	93	39	79	132
Retail	-1	-3	-3	-1	12	39	62	35
Services	-3	-2	-1	-1	24	33	8	-1
Software & IT services	1	4	0	-3	68	39	6	35
Textiles	-3	-6	-2	0	37	33	116	120
Transport	-1	-1	6	6	37	41	11	6
Transport Equipment	-7	7	-1	-16	58	38	135	155
All Sectors	1	1	-1	-2	50	44	48	54

MIDDLE EAST AFRICA	2025 changes				2025 levels			
	DSO	DPO	DIO	CCC	DSO	DPO	DI	CCC
Agrifood	1	-2	-6	-3	57	48	58	67
Automotive manufacturers	-1	1	-8	-10	59	52	43	50
Automotive suppliers	10	1	8	17	63	62	59	60
Chemicals	-4	4	-1	-9	73	71	55	56
Commodities								
Computers & Telecom								
Construction	6	8	7	6	71	86	100	85
Electronics	-3	-10	10	18	48	57	91	81
Energy	3	4	4	3	51	42	26	35
Household Equipment	15	6	4	12	93	53	84	123
Machinery & Equipment	3	-6	5	14	55	57	95	92
Metals	14	2	4	16	92	45	83	130
Paper	2	7	5	0	90	67	62	84
Pharmaceuticals	4	10	5	0	25	75	102	52
Retail	-1	2	-2	-4	55	60	54	49
Services	4	7	1	-2	87	61	16	41
Software & IT services	9	0	1	9	61	48	10	22
Textiles	-6	0	22	16	49	46	99	102
Transport	-12	-7	1	-4	137	41	8	104
Transport Equipment								
All Sectors	2.2	3.0	2.3	1.6	61	62	55	55

A photograph showing a variety of hands of different skin tones stacked on top of each other, resting on a thick, textured tree branch. The background is a lush green forest with sunlight filtering through the leaves. The text 'Our team' is overlaid in the center, with 'Our' in white and 'team' in orange.

Our team

Chief Investment Officer
& Chief Economist
Allianz Investment Management SE



Ludovic Subran
ludovic.subran@allianz.com

Head of Economic Research
Allianz Trade



Ana Boata
ana.boata@allianz-trade.com

Head of Macroeconomic & Capital
Markets Research
Allianz Investment Management SE



Bjoern Griesbach
bjoern.griesbach@allianz.com

Head of Outreach
Allianz Investment Management SE



Arne Holzhausen
arne.holzhausen@allianz.com

Head of Corporate Research
Allianz Trade



Ano Kuhanathan
ano.kuhanathan@allianz-trade.com

Head of Thematic & Policy Research
Allianz Investment Management SE



Katharina Utermoehl
katharina.utermaehl@allianz.com

Macroeconomic Research



Lluís Dalmau Taules
Economist for Africa & Middle East
lluis.dalmau@allianz-trade.com



Maxime Darmet Cucchiari
Senior Economist for UK, US & France
maxime.darmet@allianz-trade.com



Jasmin Gröschl
Senior Economist for Europe
jasmin.groeschl@allianz.com



Françoise Huang
Senior Economist for Asia Pacific
francoise.huang@allianz-trade.com



Maddalena Martini
Senior Economist for Southern Europe & Benelux
maddalena.martini@allianz.com

Outreach



Luca Moneta
Senior Economist for Emerging Markets
luca.moneta@allianz-trade.com



Giovanni Scarpato
Economist for Central & Eastern Europe
giovanni.scarpato@allianz.com



Heike Baehr
Content Manager
heike.baehr@allianz.com



Melissa Eddy
Content Manager
melissa.eddy@allianz.com



Maria Thomas
Content Manager and Editor
maria.thomas@allianz-trade.com

Corporate Research



Guillaume Dejean
Senior Sector Advisor
guillaume.dejean@allianz-trade.com



Maria Latorre
Sector Advisor, B2B
maria.latorre@allianz-trade.com



Maxime Lemerle
Lead Advisor, Insolvency Research
maxime.lemerle@allianz-trade.com



Pierre Lebard
Public Affairs Officer
pierre.lebard@allianz-trade.com

Thematic and Policy Research



Michaela Grimm
Senior Economist,
Demography & Social Protection
michaela.grimm@allianz.com



Patrick Hoffmann
Economist, ESG & AI
patrick.hoffmann@allianz.com



Simon Krause
Economist, ESG & Insurance
simon.krause@allianz.com



Hazem Krichene
Senior Economist, Climate
hazem.krichene@allianz.com



Kathrin Stoffel
Economist, Insurance & Wealth
kathrin.stoffel@allianz.com



Markus Zimmer
Senior Economist, ESG
markus.zimmer@allianz.com

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Director of Publications
Ludovic Subran, Chief Investment Officer & Chief Economist
Allianz Research
Phone +49 89 3800 7859

Allianz Group Economic Research
https://www.allianz.com/en/economic_research
<http://www.allianz-trade.com/economic-research>
Königinstraße 28 | 80802 Munich | Germany
allianz.research@allianz.com

 @allianz

 allianz

Allianz Trade Economic Research
<http://www.allianz-trade.com/economic-research>
1 Place des Saisons | 92048 Paris-La-Défense Cedex | France

 @allianz-trade

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Allianz Research | 15 July 2026

Super El Niño risk: more inflation in Asia, oversupply in Latin America yet little disruption in financial markets

Hazem Krichene
Senior Climate Economist
[hazem.krichene@allianz.com](mailto: hazem.krichene@allianz.com)

Ano Kuhanathan
Head of Corporate Research
[ano.kuhanathan@allianz-trade.com](mailto: ano.kuhanathan@allianz-trade.com)

Garance Tallon
Economist for APAC
[garance.tallon1@allianz-trade.com](mailto: garance.tallon1@allianz-trade.com)

Lina Manthey
Equity Strategist
[lina.manthey@allianz.com](mailto: lina.manthey@allianz.com)

In Summary

- **The emerging 2026-27 El Niño has the potential to become the strongest climate event in more than a decade.** While weather events are often viewed as operational risks, this one is more appropriately understood as a macroeconomic event with implications for inflation, trade flows, supply chains and corporate earnings. The IMF estimates that a typical El Niño raises global food prices by around 5% within a year, and the 1982-83 and 1997-98 events led to USD4.1trn and USD5.7trn in cumulative global income losses over the following five years.
- **El Niño creates an uneven commodity shock, with the greatest disruption concentrated in commodities produced in Asia while grains in Latin America – mainly Brazil and Argentina – could see oversupply from improved growing conditions.** Southeast Asia, India and parts of West Africa face heightened risks of drought and agricultural disruption. At the same time, large parts of South America are likely to benefit from improved growing conditions, supporting grain and oilseed production. The result is a divergence across commodity markets, countries and sectors rather than a broad-based inflationary surge. The highest risks are in sugar, palm oil and rice, where drought, low inventories and potential export restrictions could drive price volatility. Cocoa and robusta coffee also face supply pressures, while soybeans, corn and arabica coffee are likely to face price declines on the back of favorable conditions.
- **For corporates, the implications extend well beyond agriculture.** Food manufacturers may face renewed input-cost pressure from sugar, palm oil, rice and cocoa but could pass on higher prices. Consumer-goods companies operating in emerging markets may encounter renewed inflation sensitivity among lower-income consumers. For corporates, involved in the trade of crops that will see a supply surge, especially in Latin America, higher production is not necessarily positive news as selling prices fall, storage capacity could overflow and demand is inelastic (i.e. lower prices do not drive more volume). Governments in food-importing economies could respond with export restrictions, price controls or strategic stockpiling measures that amplify supply-chain disruption.
- **A super El Niño would transmit inflationary pressure in Asia primarily through food prices, with Indonesia the most affected (+2.3pp) and Malaysia and the Philippines the most insulated (+0.7pp). In Latin America this picture is mixed, with Colombia, Peru and Brazil facing potential pressures.** Central banks in the Philippines, Indonesia and India could be forced to hike further or delay easing into 2027, while Malaysia and Thailand would remain on hold. Export bans by major agricultural producers (India, Thailand, Vietnam) would ease domestic pressures but amplify inflation risks for importers, notably the Philippines and Indonesia. In Latin America, drought conditions in Northern Brazil and Colombia could push up inflation while Peru could face disruptions from coastal El Niño.
- **Super El Niño events normally do not matter much for broader financial markets.** The S&P 500 has rallied strongly in every such episode, driven by the concurrent macro backdrop rather than the weather event itself. For emerging markets, food-importing economies like the Philippines and Indonesia show a tendency to underperform global benchmarks in the 12-18 months following an ONI peak (with the exception of 2015-2016). However, each event occurred in a radically different macro backdrop (i.e. Asian crisis, China slowdown, post-Covid inflation) so the signal is too weak and too confounded to be attributed to El Niño alone.

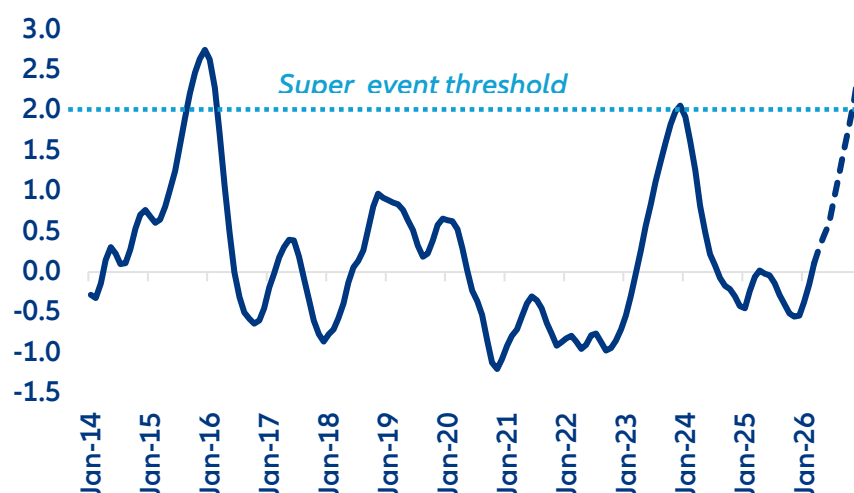
The world is facing yet another major climate shock with the incoming El Niño

Most climate events are operational concerns and this one has the potential to become a macroeconomic event.

The IMF estimates that past El Niño episodes raised global food prices by roughly 5% over a year and lifted headline inflation by between 0.1-1pp across affected economies, with the effect rising in line with the weight of food in the consumer basket¹. Going further, Callahan & Mankin (2023)² attribute some USD4.1trn and USD5.7trn in cumulative global income losses to the 1982-83 and 1997-98 events – concentrated in the tropics, where economies such as Peru and Indonesia ran more than 10% below trend several years later – and put a possible USD3trn drag on global output through 2029 on the 2023 event alone. These are upper-bound, still-debated estimates, but they frame the order of magnitude: this is an inflation and income event, not merely an agricultural one.

Three factors make the current outlook particularly important. First, the probability of a major event is now high and rising. NOAA's Climate Prediction Center upgraded its outlook to an El Niño Advisory in June 2026 and assigns around a 63% chance of a very strong event by November-January – one it judges would rank among the largest in the record since 1950; the WMO puts the probability near 80% and the World Bank cited a 61-87% chance of emergence by mid-2026, persisting into 2027. Second, several agricultural markets enter the event with limited buffers. Inventories in commodities such as cocoa, palm oil and rice are already tighter than historical norms, reducing the system's ability to absorb supply disruptions. Third, governments have become more interventionist since previous El Niño cycles. Export restrictions, strategic stockpiling and domestic price controls are now common policy tools. These interventions can amplify the economic impact of weather shocks far beyond the initial production losses. The result is that relatively modest supply disruptions can rapidly become significant economic events.

Figure 1: Oceanic Niño Index



Sources: NOAA/IRI, Allianz Research

What past El Niños did to global catastrophe patterns

Using a record of some 9,400 events since 2000 we explore how El Niño affects NatCat events distribution globally. Each event was tagged to an ENSO phase using NOAA's Oceanic Niño Index episode record, and we measured how the frequency of each hazard and each region shifts in El Niño years relative to neutral ones³. The exercise is validated by a placebo: earthquakes and volcanoes, which have no link to ocean temperature, show no El Niño signal (1.0x), which is what allows the elevated hazard ratios to be read as signal rather than an artefact of rising reporting over time.

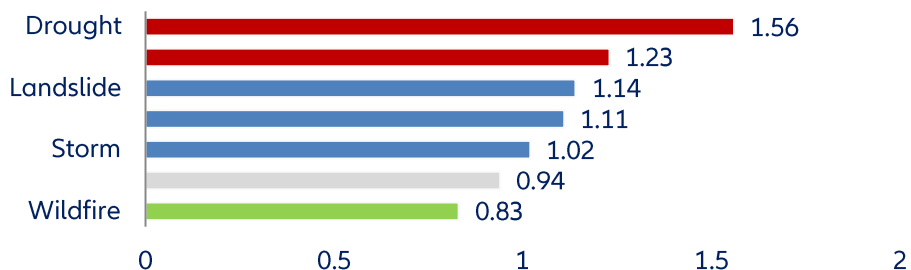
¹ [Fall 2023 Global Food Crisis Update—Recent Developments, Outlook, and IMF Engagement](#)

² [Persistent effect of El Niño on global economic growth | Science](#)

³ ENSO-neutral months are those in which the Oceanic Niño Index sits between -0.5°C and $+0.5^{\circ}\text{C}$; neither El Niño (warm) nor La Niña (cool), with equatorial-Pacific sea-surface temperatures close to their long-run average. We benchmark against neutral rather than La Niña so the ratios isolate the El Niño signal itself.

El Niño rewires the mix of catastrophes rather than lifting them across the board (Figure 2). Drought (1.6×) and extreme heat (1.2×) rise clearly, floods more modestly (1.1×), while storm frequency is essentially unchanged globally (1.0×). That last point is the important one: El Niño redistributes cyclone activity rather than adding to it. Wildfire actually sits below its neutral rate worldwide, because its signal is regional and washes out in the global total. In the two strongest past events (the closest analogues to an expected strong 2026-27) drought and heat intensify further still.

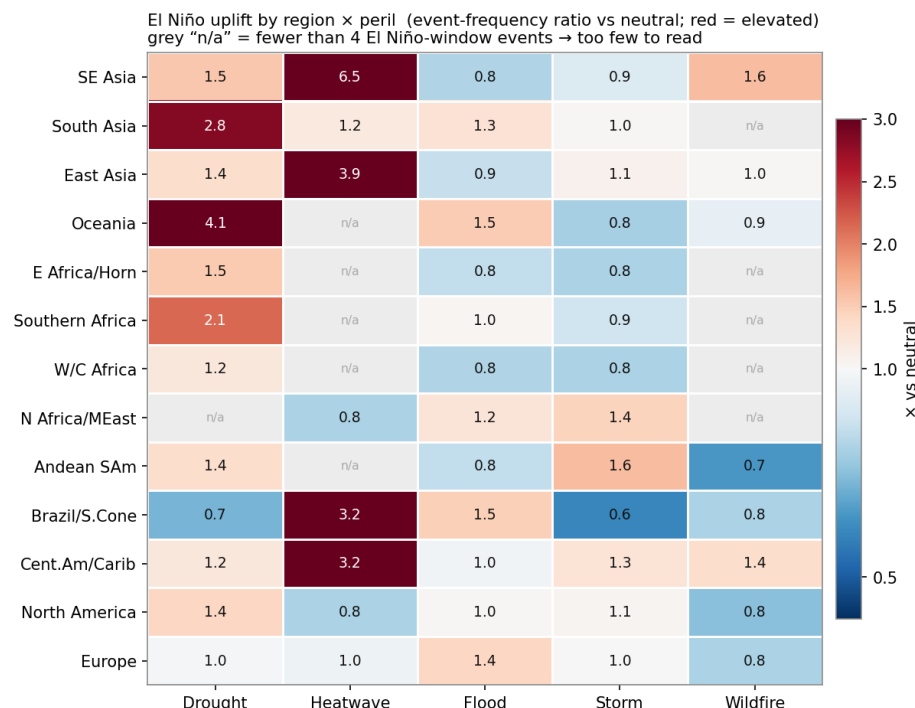
Figure 2: El Niño vs neutral : global event-frequency ratio by hazard



Sources: Allianz Research; global catastrophe-alert dataset (2000-2026), NOAA/CPC. Earthquake (grey) is the placebo control.

There is a clear El Niño drought belt: Australia (4.1×), India and the wider South Asia (2.8×), Southern Africa (2.1×), Southeast Asia and the Horn of Africa. Heatwaves cluster in Southeast and East Asia and floods lift across the Southern Cone. Australian wheat, Indian rice and sugar, Southeast Asian palm oil and Southern African maize all sit inside the drought zone. Storms again confirm redistribution rather than amplification, where Australian-basin cyclones fall to 0.8× while eastern-Pacific and Central-American activity rises.

Figure 3: El Niño uplift by region and hazard (ratio vs neutral; red = elevated)

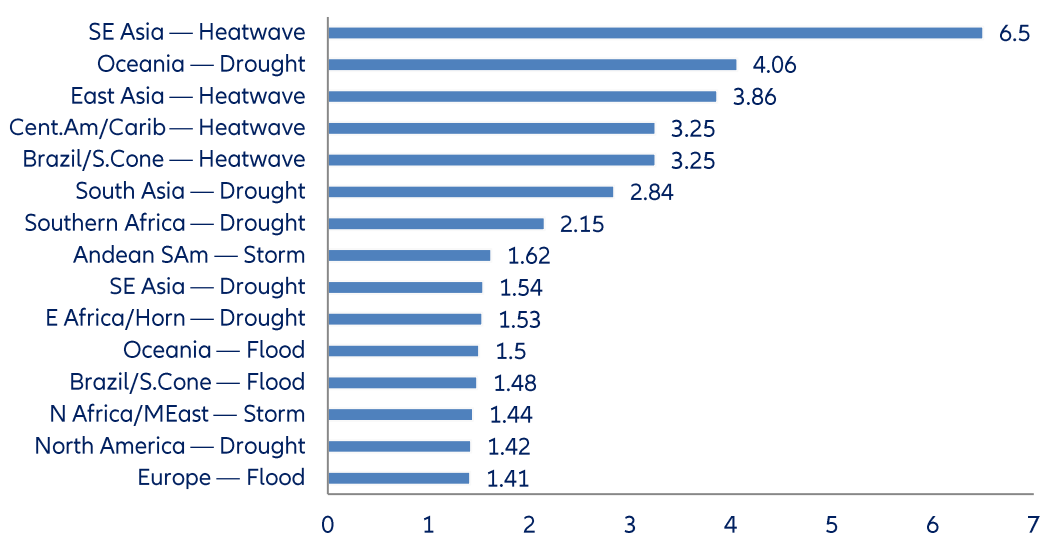


Sources: Allianz Research; NOAA/CPC. Cells with fewer than four El Niño-window events are greyed out as too thin to read.

Translating the record into the 2026-27 outlook (Figure 4). This is an analogue rather than a probabilistic forecast. For each region and hazard we take its behavior across the eight past El Niños and score it on three things: the size of the uplift, whether the sample is large enough to trust, and whether the signal also holds up in the two strong episodes, which are the closest match to a potentially strong 2026-27. On that basis the highest-conviction elevated risks are drought in Australia, Southern Africa and Southeast Asia, heatwaves in Southeast and East Asia, and

flooding across the Southern Cone. The multipliers are frequency ratios, not probabilities. Southeast Asia's 6.5 for heatwaves means such events occurred about six and a half times as often under El Niño as in neutral conditions, once you adjust for how long each phase lasted. In the raw record that is eight heat events across the El Niño windows against one in the neutral windows, and seven of the eight fell in the two strong episodes of 2015-16 and 2023-24, six of them in 2024. Drought shows the same clustering at country level, with Indonesia logging two droughts during those two strong El Niños, in 2015 and 2023, against one across the entire run of neutral months. A large ratio on a small base therefore signals a sharp rise in relative risk rather than a large number of separate disasters, which is why conviction rests on recurrence and strong-episode consistency more than on the headline figure. India's drought risk is the clearest example of that distinction: it carries the largest human and agricultural exposure of any cell but the weakest strong-episode signal, so it belongs in a "materially elevated but uncertain" tier rather than the central case.

Figure 4: Analog risk elevation for a strong 2026-27 El Niño



Sources: Allianz Research; NOAA/CPC. Green = signal also strengthens in the two strong past episodes (higher conviction); amber = elevated but the strong-episode signal is noisy.

Where could we see most disruptions?

The strongest impacts are expected across Southeast Asia, India and parts of West Africa. These regions sit at the center of global production for several critical food commodities. Any further reduction in output could renew volatility in commodity markets and increase pressure on food product manufacturers. India is the world's largest rice exporter and a major producer of sugar. A weaker-than-normal monsoon could reduce agricultural output and place renewed pressure on domestic food prices. Historically, the government's response has often been to restrict exports in order to protect domestic consumers. Such decisions have an outsized impact: India alone accounts for roughly 40% of global rice trade. The 2023 episode is instructive. As El Niño concerns built, India banned non-basmati white rice exports in July 2023, and benchmark Thai prices jumped more than 20% within weeks; the FAO rice price index rose almost 10% month-on-month in August, a 15-year high⁴. For multinational businesses, Indian policy decisions may prove more important than the weather itself.

Indonesia, Malaysia, Thailand and Vietnam are exposed to drought risk during major El Niño events. The region is critical for palm oil, sugar, rice and coffee. Disruptions in any of these markets can quickly affect food manufacturers, consumer-goods companies and global retailers. Palm oil deserves particular attention because of its importance across processed foods, personal care products and industrial supply chains.

Fallout in Côte d'Ivoire and Ghana could disrupt global cocoa production. The cocoa market enters this event with already constrained inventories and structural production challenges. The most recent El Niño shows how

⁴ [Rice Export Prices Highest in More Than a Decade as India Restricts Trade | USDA Foreign Agricultural Service](#)

violent that disruption can be: cocoa roughly tripled in 2024, from about USD 2,500 to over USD 10,000 per ton, as Ivorian output fell around 21% to an eight-year low. But the episode is also a warning against single-cause attribution – the collapse combined El Niño dryness with excessive-rain disease outbreaks (black pod), swollen-shoot virus and aging trees. With West Africa supplying roughly 70% of the world's cocoa and structural decline already underway, El Niño acts as an accelerant on a fragile crop rather than the sole driver⁵.

Higher crops expected in Latin America, not necessarily good news

One of the most misunderstood aspects of El Niño is that it does not create uniform disruption. In fact, some regions are likely to benefit. IMF research confirms this asymmetry directly: a typical El Niño shaves activity in Australia, Chile, India, Indonesia, Japan, New Zealand and South Africa, while the US tends to gain directly and Europe indirectly, through positive trade spillovers.⁶

Brazil and Argentina may emerge as beneficiaries in yields but not necessarily in economic terms. Historically, El Niño improves growing conditions across some agricultural regions, supporting: soybean, corn, and sugar output in parts of Latin America. Stronger harvests have the potential to improve trade balances, support economic activity and provide a natural offset to inflationary pressures. Argentina's grain sector also typically benefits from the wetter conditions associated with El Niño. El Niño's effect on Brazil is regionally split – supportive for southern crops, but historically associated with drought in the north and the Amazon basin – and much of Argentina's upside reflects a rebound from the severe La Niña drought that preceded it, rather than an absolute windfall. However, higher production could also put downward pressure on prices while demand is rather inelastic (i.e. consumption of food in volume do not increase significantly when prices increase). Lastly, when there is a bit of oversupply, inventory facilities fills up rapidly and prices increase steeply – another headwind for farmers and corporates looking for storage.

Parts of the US often experience favorable agricultural conditions during major El Niño events. Corn and soybean production have historically performed well, providing an important counterweight to disruptions elsewhere. The implication is important, the world is unlikely to experience a broad-based food shortage. Instead, supply will become increasingly uneven across regions and commodities.

The Northern Hemisphere is mostly a beneficiary. Strong El Niños raise wind shear over the Atlantic and hold hurricane activity down. NOAA's 2026 outlook puts the odds of a below-normal Atlantic season at 55% and forecasts 8 to 14 named storms versus a norm of 14; CSU now sees 11 named storms, 5 hurricanes and 2 major.⁷ Fewer Atlantic landfalls means lower insured catastrophe losses and fewer Gulf oil, gas and LNG outages. Northern US winters also tend to run mild because the cold Canadian air stays bottled up, which trims heating demand: the difference between a cold winter and a mild El Niño one is roughly 4 Bcf/d⁸ of US gas demand, about 5%, and the last episode pushed Henry Hub to multi-year lows. So for the North the price signal leans disinflationary, the opposite of the food story in Asia.

Figure 5: Impact of previous strong El Niño (% deviation vs 5y trend)

Region	Country	Commodity	1982	1997	2015	2023
Africa	Côte d'Ivoire	Cocoa	-5	-8	-11	-7
Africa	Ghana	Cocoa	-4	-7	-9	-6
Asia	India	Rice	-3	-5	-7	-4
Asia	India	Sugar	-8	4	-14	-7
Asia	Indonesia	Palm Oil	-4	-8	-6	-3
Asia	Malaysia	Palm Oil	-5	-6	-5	-2
Asia	Thailand	Rice	-4	-6	-8	-5
Asia	Thailand	Sugar	-5	-3	-9	-5
Asia	Viet Nam	Robusta	-2	-8	-5	-3
LatAm	Argentina	Soybean	4	6	3	5

⁵ [Chocolate's climate crisis - The Salata Institute](#)

⁶ [El Niño Good Boy or Bad? -- Finance & Development, March 2016](#)

⁷ [WATCH: NOAA forecasts milder Atlantic hurricane season thanks to El Nino | PBS News](#)

⁸ Billion cubic feet per day

LatAm	Brazil	Arabica	2	3	5	3
LatAm	Brazil	Soybean	2	3	4	3
N.America	US	Corn	1	2	3	2
Oceania	Australia	Wheat	-15	-21	-11	-9

Sources: USDA, CONAB, ICO, ICCO, FOA and Allianz Research

Figure 6: Summary of potential impact on commodity prices

Commodity	Expected Impact	Key Trigger
Sugar	▲ Significant Upside Risk	Weak Indian monsoon and lower kharif harvests
Palm Oil	▲ Significant Upside Risk	Falling Indonesian and Malaysian production estimates
Rice	▲ Significant Upside Risk	Indian harvest disappointment
Robusta Coffee	▲ Moderate Upside Risk	Weak Southwest Monsoon in Vietnam
Cocoa	▲ Moderate-to-High Upside Risk	Poor Côte d'Ivoire and Ghana harvests
Wheat	▲ Conditional Upside Risk	Australian production vulnerable if El Niño coincides with a positive Indian Ocean Dipole.
Arabica Coffee	► Stable to Slightly Negative	Strong Brazilian crop forecasts
Soybeans	▼ Downside Risk	Strong South American harvest outlook
Corn	▼ Downside Risk	Above-trend US crop conditions

Source: Allianz Research

Energy, power and freight could also be disrupted, and governments take action

El Niño does not only affect food. The droughts it brings to Latin America and parts of Asia curtail hydropower and, as the IMF notes, raise demand for coal and crude oil to fill the gap in thermal generation; offsetting this, milder Northern-Hemisphere winters can soften heating-gas demand. Net energy effects are therefore regional and partly self-cancelling, but the power-generation channel is a real source of price and fiscal pressure in hydro-dependent economies.

The supply-chain impact also runs through shipping chokepoints. In 2023-24, an El Niño-driven drought forced the Panama Canal, a route for around 6% of global trade, to cut daily transits from a normal 36 to as few as 24 and impose draft limits that reduced cargo loads; queues swelled to roughly 160 vessels and full-year transits fell 29%, with LNG shipments down by two-thirds as carriers rerouted around Africa. With NOAA already flagging the canal as a watch-point for 2026-27, freight and routing risk belongs on the corporate exposure map alongside commodity prices.

History suggests we should pay attention to policymakers. During previous food-price shocks, government intervention frequently became the dominant market driver. Potential policy responses include export restrictions, strategic stockpile accumulation, price controls, import subsidies and tariff adjustments. India's interventions on rice markets is a textbook illustration. A modest production shortfall can trigger export restrictions. Those restrictions then affect importing countries across Asia, Africa and the Middle East. Importing governments respond with emergency purchases, amplifying price movements further. In Asia, India, Thailand and Vietnam, as major rice and agricultural exporters, are the most likely to do so. While export restrictions could ease domestic supply pressures and give local central banks more flexibility, they would raise inflation risks for regional food importers, especially the Philippines and Indonesia.

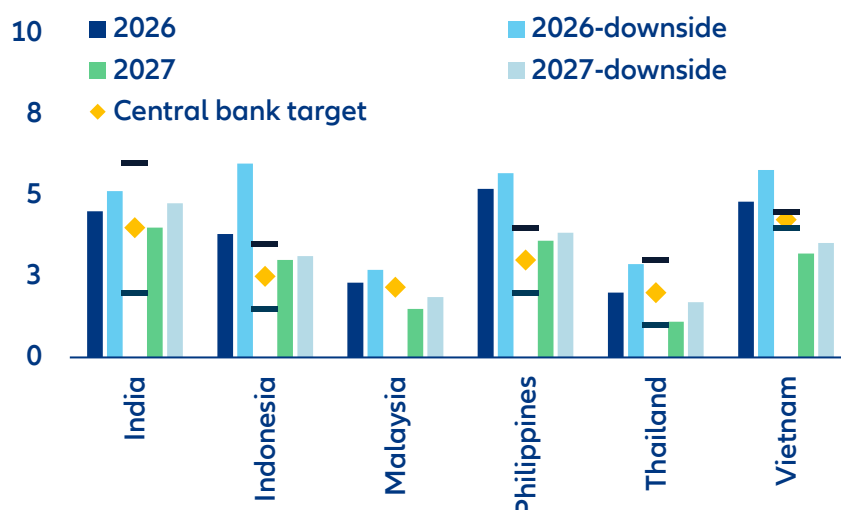
The economic impact becomes far larger than the original weather event. For global businesses, this policy feedback loop often creates greater uncertainty than the agricultural disruption itself.

Southeast Asia and some parts of Latin America may face inflationary pressure

Inflation has already hit consumers across ASEAN and India following the Iran war. With the Philippines most affected initially as headline inflation reached +7.2% y/y in April. While second-round effects are expected to lift inflation modestly across the region, Thailand remains an exception, with inflation still below target due to deflationary pressures. Energy-price pressure has eased after the gradual reopening of the Strait of Hormuz, though prices remain above pre-crisis levels. Current 2026 inflation forecasts are 4.5% for India, 3.8% for Indonesia, 2.3% for Malaysia, 5.2% for the Philippines, 2.0% for Thailand and 4.8% for Vietnam. In 2027, inflation is expected to normalize to 4.0%, 3.0%, 1.5%, 3.6%, 1.1% and 3.2%, respectively.

A new inflation risk for 2027 now comes from a potential super El Niño, mainly through food prices. Exposure depends on food's weight in CPI baskets: Malaysia is least exposed at around 30%, followed by Indonesia and Vietnam at around 35%, Thailand and India at around 40%, and the Philippines above 45%. Given these exposures, a strong El Niño would largely undo the disinflationary gains from the reopening of Hormuz for 2026 and push inflation higher into 2027. Drawing on the 2015–2016 El Niño precedent, we expect Indonesia to be the most affected, with a cumulated impact of +2.3pp, followed by Thailand, Vietnam and India at approximately +1.3pp each, while Malaysia and the Philippines appear more insulated at around +0.7pp.

Figure 5: Inflation across ASEAN and India in 2026-2027



Sources: IMF, Allianz Research

Policy responses across ASEAN and India are likely to diverge, as central banks start from different positions. Indonesia and the Philippines have already tightened policy, raising rates by 100bps and 50bps, respectively. The RBI is expected to begin hiking in H2 with two rate increases, while Malaysia, Thailand and, to a lesser extent, Vietnam have remained on hold.

If El Niño materializes at full strength, the Philippines, Indonesia and India may need to hike again or delay easing until 2027. Malaysia and Thailand should stay on hold but remain ready to act. For Thailand, a severe El Niño would likely rule out rate cuts to support growth, which had already been largely postponed due to the Iran war.

In Latin America, based on the last two precedents episodes, Brazil, Colombia and Peru face the largest risks. Drought conditions in Northern Brazil and Colombia could push up inflation, through food prices, logistics and electricity prices (especially for Colombia). While Peru could face disruptions from coastal El Niño. However in previous inflationary shocks, these countries were also facing forex depreciations.

East-Asian supply-chains to be tested but markets unlikely to be fazed

Companies with exposure to food ingredients should prepare for renewed input-cost volatility. The highest-risk commodities include sugar, palm oil, cocoa and rice. Margin management may become more challenging if inflation-sensitive consumers resist further price increases.

Companies with sourcing exposure to Southeast Asia should review supply-chain resilience and inventory strategies. The risk is less about physical shortages and more about cost volatility and policy intervention.

Credit stress could emerge among food processors, agricultural exporters and businesses operating in vulnerable emerging markets. Conversely, agricultural exporters in Brazil and other beneficiary regions may experience stronger operating conditions.

Super El Niño events should not be considered as a potential driver for major market turbulences. For developed markets, the signal is essentially absent at the index level: the S&P 500 has delivered strong positive raw returns in every super El Niño window in our sample, averaging +30% over the 18 months following an ONI peak - not because El Niño helps the US economy, but because these episodes have historically coincided with US mid-cycle expansions or early Fed easing cycles, which dominate any weather-related noise. The global business cycle, monetary policy, and earnings revisions are orders of magnitude more important for US markets than El Niño. For emerging markets, with the exception of 2015-2016, there seems to be a pattern of equity underperformance relative to global benchmarks, concentrated in food-importing, drought-exposed economies (the Philippines and Indonesia most clearly), and it tends to peak 12-18 months after the ONI peak rather than coinciding with it. But we must underline that history only counts three to four observations per country, each episode unfolded against a radically different macroeconomic backdrop (i.e. the Asian financial crisis in 1997-98, the China slowdown in 2015-16, and the post-COVID inflation cycle in 2023-24) and in several cases these confounding factors dwarf anything El Niño could plausibly explain independently. What our event study identifies is more accurately described as a tendency for ENSO-exposed EM markets to underperform in an environment that also tends to be globally risk-off, rather than a clean causal impact of El Niño on stock prices. The commodity and food-inflation transmission are real and well-identified, their translation into markets underperformance is plausible and directionally consistent, but the signal is weak, heterogeneous, and not separable from global macro with only a few events.

What we need to monitor going forward

1. Indian monsoon performance. The June-September 2026 monsoon season will provide the clearest indication of risks to rice and sugar markets.

2. Indonesian and Malaysian production data. Palm oil inventories will offer one of the earliest indicators of whether supply conditions are tightening.

3. West African cocoa harvests. Evidence of continued production weakness would reinforce concerns about already-tight cocoa markets.

4. Government intervention. Export restrictions, stockpiling programs and food-security measures should be treated as leading indicators of broader market disruption.

5. Inflation expectations. Food-price inflation often appears in consumer expectations before it is fully visible in official prints. Monitoring inflation expectations may therefore provide earlier warning than headline CPI data.

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed or implied in such forward-looking statements.

Such deviations may arise due to, without limitation, (i) changes of the general economic conditions and competitive situation, particularly in the Allianz Group's core business and core markets, (ii) performance of financial markets (particularly market volatility, liquidity and credit events), (iii) frequency and severity of insured loss events, including from natural catastrophes, and the development of loss expenses, (iv) mortality and morbidity levels and trends, (v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures, and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

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Europe under water: The macroeconomic cost of flooding and the economic case for adaptation

09 July 2026

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Appendix

Executive Summary



Hazem Krichene
Senior Economist, Climate
hazem.krichene@allianz.com



Markus Aichinger
Senior Expert
Allianz Reinsurance
markus.aichinger@allianz.com



Katherine Wenigmann
Flood Lead - Natural Catastrophe R&D
Allianz Reinsurance
katherine.wenigmann@allianz.com



Rozalia Papaioannou
Research Assistant
rozalia.papaioannou@allianz.com



Katharina Utermoehl
Head of Thematic and Policy Research
katharina.uterhoehl@allianz.com

Flood losses are rising rapidly as economic activity continues to concentrate in high-risk areas. Floods are Europe's most frequent and costliest natural hazard, with total economic losses climbing steadily, to EUR226bn in the first quarter of this century. While flood frequency in Europe has remained broadly stable at 46 per year since 2000, their cost has risen sharply. Losses from floods climbed 17.8% from EUR63.1bn between 2000-2009 to EUR 74.3bn from 2010-2019. From 2020-2025 alone, the cost reached EUR88.6bn, a 40% increase from the first decade of the century. Insurance covers only a fraction of the damage: The July 2021 floods alone caused a record EUR38bn in damage, of which only EUR9bn was insured, leaving households, businesses and governments to bear the remainder. Looking ahead, annual river-flood losses across the EU and UK could rise more than six-fold, from EUR7.8bn today to nearly EUR50bn by 2100 in a 3°C warming scenario, driven by more intense rainfall and continued development in exposed areas.

Floods are local events in their physical origin, but their economic consequences ripple through the entire economy. To quantify these economic effects, we simulate a one-off flood in 2027, calibrated to each country's average historical maximum flood depth over 2015-2024, and compare the economic trajectory with a no-flood baseline through 2030. The analysis links observed flood depth to real gross fixed capital formation and real household net disposable income, tracing how flood damage spreads through investment, consumption and public finances. Investment is hit hardest: cumulative gross fixed capital formation losses between 2027 and 2030 range from 10.5% in Norway to 14.6% in the Netherlands, with Germany at 12% recording the largest absolute loss at around EUR84bn. Real household net disposable income meanwhile falls by 3.9-5.4% in 2027-30 as reconstruction costs and labour-market effects accumulate, in turn reducing households' capacity to absorb uninsured losses and rising insurance costs.

Floods create a stagflationary shock, raising prices, slowing growth and eroding fiscal space. Consumer prices rise as damaged infrastructure, disrupted logistics and lower local availability of goods and services create supply bottlenecks. The cumulative price-level impact remains contained, from +0.2% in the United Kingdom to +1.0% in Greece, but it still adds pressure on households already facing an income shock. Private consumption declines in all countries, with cumulative losses ranging from -0.3% in Norway to -0.9% in Czechia; in absolute terms, the United Kingdom records the largest loss at about EUR61bn, ahead of Germany and France. GDP losses then range from around -0.4% in Norway to -1.0% in Spain, with Germany and France losing approximately EUR108bn and EUR79bn respectively. The impact on public finances is also visible, with cumulative 2027-2030 government deficits widening by 1.3pp of GDP on average, from -0.3pp in Norway to -2.4pp in Spain.

The economic case for flood prevention is overwhelming: well-targeted adaptation pays for itself many times over. Adapting to flooding requires more public capital than any other climate hazard, accounting for around 65% of all predominantly public measures in our adaptation taxonomy. Yet target adaptation investments in natural flood retention, resilient infrastructure and risk-sensitive land-use planning can largely offset future losses. Concretely, flood adaptation means keeping people and assets away from high-risk floodplains where possible, restoring natural retention areas, upgrading dikes, drainage, sewers and stormwater systems, and flood-proofing buildings and critical infrastructure. These measures should be combined according to the type of risk: basin-level retention and land-use planning for fluvial floods, local drainage and blue-green urban infrastructure for pluvial floods, supported by early-warning systems, emergency planning and insurance incentives. Flood resilience investments yield roughly four times their cost in avoided damages. Land-use planning offers similarly high returns: prohibiting development in floodplains and integrating green, water-retaining infrastructure into cities offer similarly high returns. This shows that more than climate policy, adaptation is preventive fiscal policy.

Europe's challenge is to deliver effective flood adaptation solutions at the speed and scale required. The measures with the highest returns, keeping development out of floodplains and restoring natural flood retention, are also the most politically difficult to implement, while major flood protection infrastructure requires lengthy planning and construction. Germany illustrates this implementation gap: despite the EUR38bn flood disaster in 2021, only around EUR500m of the EUR6–7bn planned under the National Flood Protection Programme launched in 2013 has been spent. The main barriers are institutional fragmentation and lengthy approval procedures rather than limited fiscal capacity.

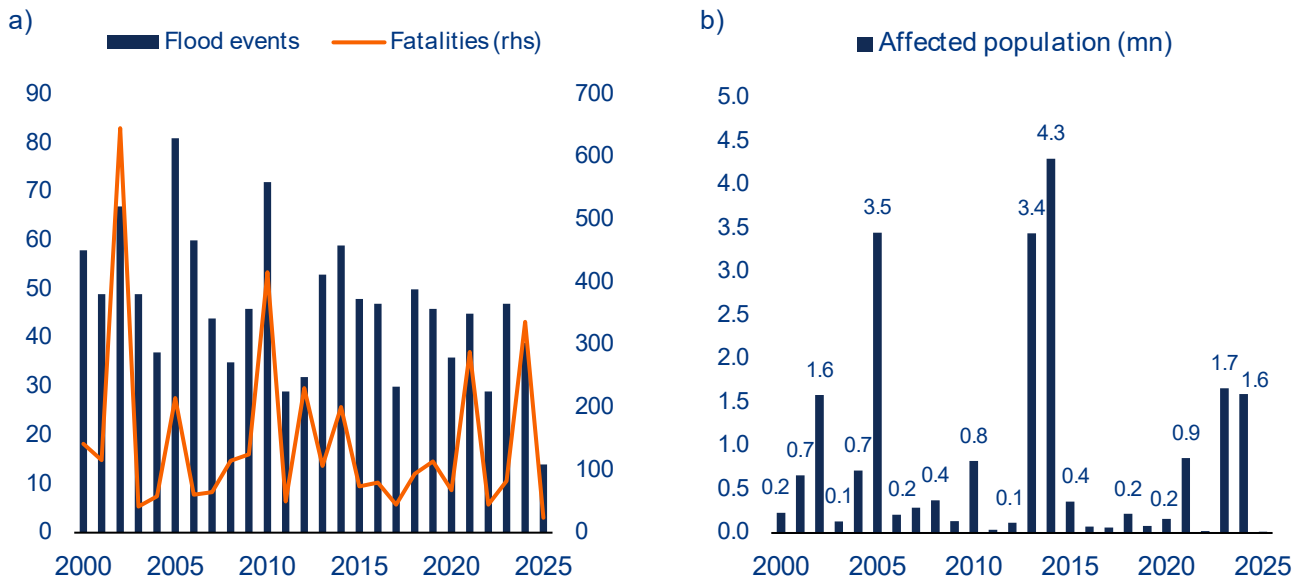
Long-term flood resilience requires integrating prevention, adaptation and insurance coverage into a single risk-management strategy. Structural flood defences, resilient building standards, property-level adaptation and greater public awareness must be complemented by sustainable public-private insurance partnerships. Experience from some European countries shows that risk transfer remains sustainable only when accompanied by meaningful risk reduction. Accelerating proven adaptation measures would strengthen public finances, reduce uninsured losses and safeguard the long-term insurability of flood risk.



The scale and cost of flooding in Europe

Europe has recorded an average of 46 flood events per year since 2000 - with close to two events per week in the most active years - but the number of floods has not shown a clear upward trend. Figure 1 plots annual flood events alongside fatalities between 2000 and 2025. Annual flood frequency fluctuates widely, from fewer than 30 events in quieter years to more than 80 in the most active, yet recent years are no more flood-prone than the early 2000s. What has changed is the severity of the worst events. Fatalities and affected populations are concentrated in a handful of catastrophe years, notably the Central European floods of 2002, the devastating floods in western Germany

and Belgium in July 2021, and the three major flood events of 2024 - the October storm in eastern Spain and the June and September floods across Central Europe. Between these episodes, flood-related deaths remain comparatively low, giving the series a pattern of intermittent but highly destructive shocks rather than a steady upward trend. The affected population follows a similar pattern, with additional peaks in 2005, 2013, 2014 and 2023, reflecting years in which floods displaced or affected large numbers of people despite relatively low mortality.

Figure 1: Flood events and associated fatalities (a), and affected population in million (b) in Europe, 2000 to 2025

Sources: ClimRad¹, EarthianAI, Allianz Research

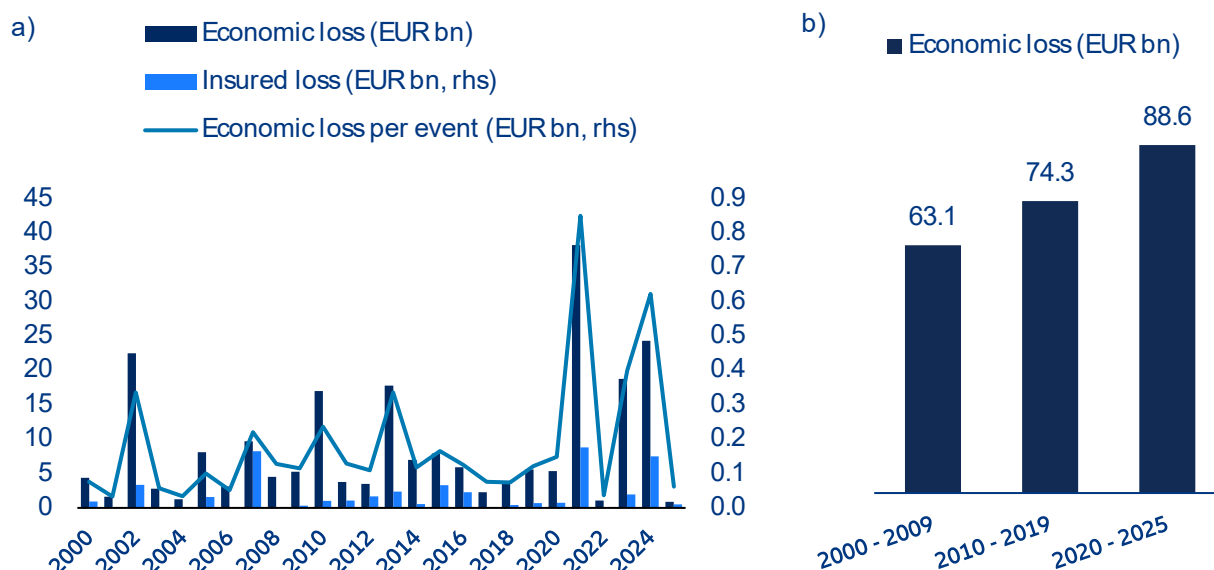
Economic losses have risen sharply as more people and assets are exposed to flood risk. The July 2021 floods remain Europe's costliest flood disaster on record, causing around EUR38bn in economic losses, followed by the Europe-wide floods of 2024 (EUR24bn) and the Central European floods of 2002 (EUR22bn; Figure 2a, 2025 prices). As with fatalities, flood losses are concentrated in a handful of catastrophe years rather than accumulating steadily over time. There is also no sign of moderation: the two most expensive years in the record - 2021 and 2024 - occurred within the past five years. The economic cost per event has risen sharply over this period. Between 2000 and 2019, a flood event caused on average around EUR0.1bn in economic losses; between 2020 and 2025 that figure climbed to approximately EUR0.4bn, with the 2021 floods alone averaging EUR0.8bn per recorded event. The rising economic cost of floods is clear when we look at the trend since 2000. Total economic losses grew 17.8% between the 2000–2009 and 2010–2019 decades, rising from EUR 63.1bn to EUR 74.3bn. The 2020–2025 half-decade alone has already reached EUR 88.6bn in losses, driven largely by the disastrous events of 2021, 2023, and 2024 – an increase of 19.1% over 2010–2019 and 40% over the first decade of the century (Figure 2b). This increase reflects not only more damaging weather but, above all, growing exposure. As urbanisation and economic activity continue to concentrate in flood-

prone areas, similar flood events destroy substantially more capital than they did two decades ago. Because exposure is shaped by land-use decisions, building standards and flood protection, a large share of these future losses remains within policy control.

Most flood losses remain uninsured, leaving households, businesses and governments to absorb the economic cost. In 2021, insured losses of around EUR9bn covered less than one-quarter of total damage; in 2002 the insured share was around 15%, and in 2010 below 10%. These swings reflect which countries flood in a given year, not a steady trend, because flood cover differs sharply across Europe. It is bundled into standard home insurance in some countries, such as the UK, but remains optional in others, including Germany, so a year dominated by losses in low-coverage markets pulls the European insured share down. Moreover, in several European markets, protection and modelling have historically been stronger for river flooding than for localised heavy-rain and surface-water flooding. Where losses are uninsured, they fall directly on households and businesses or are ultimately borne by the public sector through emergency relief and reconstruction. This protection gap amplifies the macroeconomic consequences of flooding, weakening household balance sheets, constraining investment and increasing pressure on public finances.

¹ ClimRad is a climate data tool developed by Allianz Research through a collaboration with EarthianAI

Figure 2: (a) Economic and insured flood losses in Europe, 2000 to 2025 (EUR billion); (b) Evolution of economic losses in Europe per decade



Sources: ClimRad, EarthianAI, Allianz Research

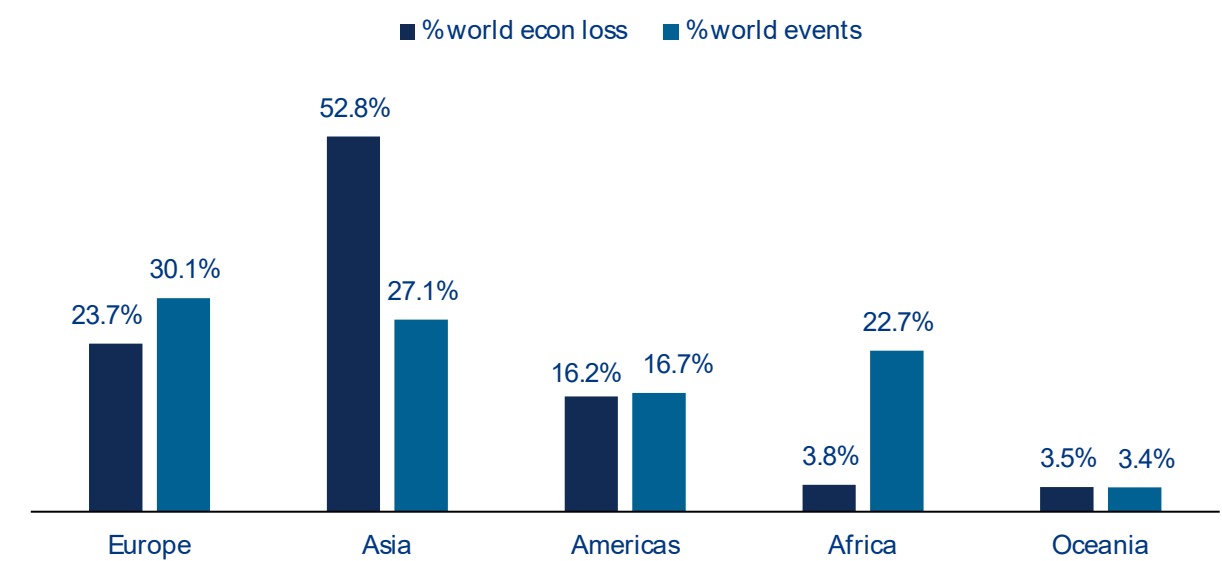
Floods impose a disproportionately large economic burden on Europe. Although Europe accounts for around 30% of recorded global flood events, it represents just under one-quarter of global flood losses (Figure 3). By contrast, Asia generates more than half of global flood losses from only around one-quarter of recorded events, reflecting the concentration of people and economic assets in highly exposed flood-prone regions.²

Within Europe, floods are by far the most costly natural hazard. Between 2000 and 2025, they accounted for around 55% of all natural-catastrophe losses while representing roughly 45% of recorded events (Figure 4). No other hazard comes close. Storms, both severe convective storms and extratropical cyclones, rank a distant second, while earthquakes account for around one-tenth of losses despite occurring only rarely, illustrating their low-frequency, high-severity nature. Floods combine both high frequency and substantial damage, making them Europe's largest source of cumulative natural-catastrophe losses.

Flood frequency alone is a poor guide to economic risk. Italy, Spain and France recorded the largest number of flood events between 2000 and 2025, but countries with fewer floods often experienced much larger impacts on their populations (Figure 5a-b). Bulgaria, Bosnia and Herzegovina and Czechia stand out as having particularly high numbers of people affected relative to the frequency of flooding, reflecting repeated exposure to major flood events. Ranking countries by economic losses reshuffles the picture again: Germany records by far the largest cumulative losses (around USD78bn), followed by Italy (USD42bn) and Spain (USD25bn) (Figure 5c). The ranking reflects not only flood frequency but also the concentration of economic assets in exposed areas and the occurrence of exceptionally destructive events such as the 2021 floods.

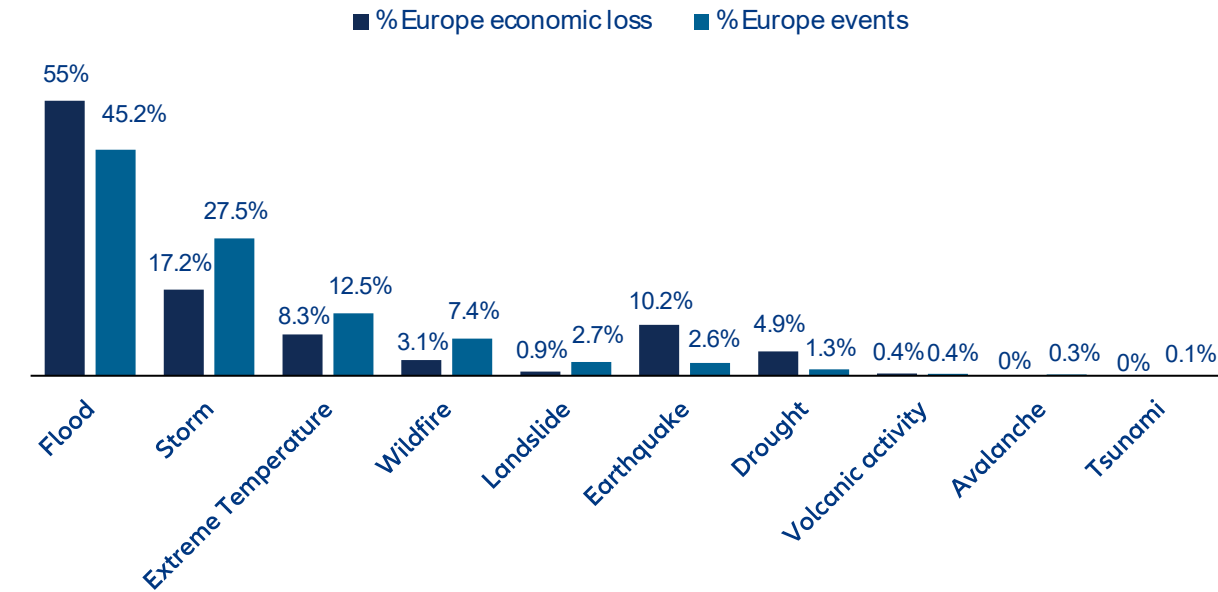
² Because flood events are captured through national reporting and emergency-mapping activations, coverage is not fully uniform across countries; this may affect individual figures but not the overall conclusions.

Figure 3: Regional shares of global flood economic losses and of recorded flood events



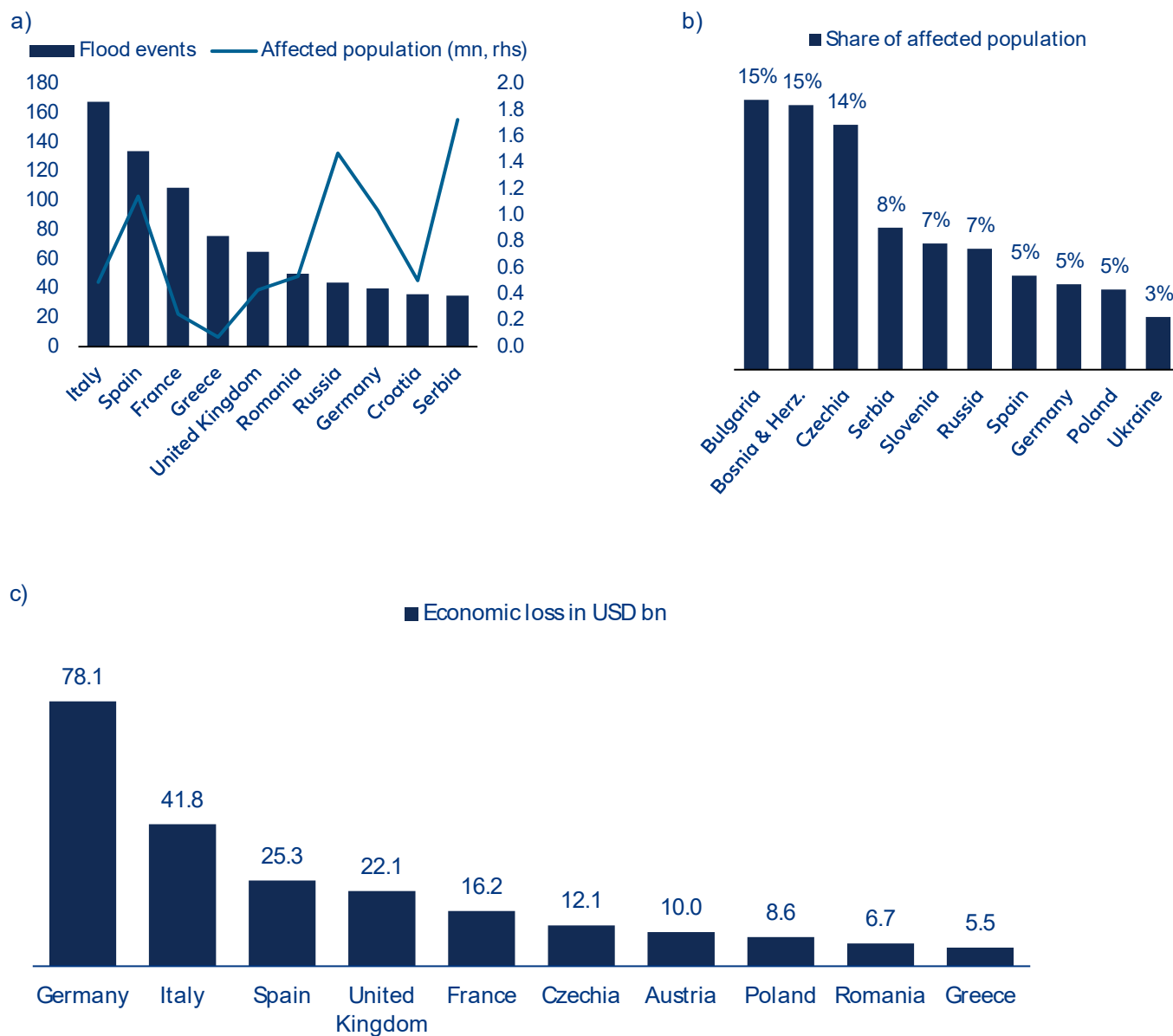
Sources: ClimRad, EarthianAI, Allianz Research

Figure 4: Shares of European natural-catastrophe losses and events, by hazard type



Sources: ClimRad, EarthianAI, Allianz Research

Figure 5: (a) Top 10 European countries ranked by number of flood events and by affected population from 2000 to 2025 (b) Top 10 European countries ranked by affected population from 2000 to 2025 (c) Top 10 countries ranked by total economic losses from 2000 to 2025



Sources: ClimRad, EarthianAI, Allianz Research



The macroeconomic impact of flood events in Europe

Floods become macroeconomic shocks that spread through the whole economy reducing investment, household incomes and fiscal space. Floods affect the macroeconomy through three channels. For households, the shock reduces real disposable income through repair costs, temporary displacement, lost working time and higher prices for essential goods and services. For firms, it reduces productive capacity through damage to buildings, machinery and inventories, while also raising short-term costs through emergency repairs, business interruption and supply-chain disruptions. These effects weaken investment. For the public sector, floods increase emergency and reconstruction spending while reducing tax intake as activity slows. We capture these channels in two steps. First, we estimate how observed flood depth³ affects real gross capital formation and real net disposable household income. Second, we feed these estimated shocks into a macro-econometric model to assess the wider macroeconomic effects. The exercise is calibrated as a one-off flood shock in 2027, based on the recent historical average of maximum flood depth over 2015 to 2024, and all results are measured relative to a no-flood baseline. This analysis is based on recently observed flood events; however, more extreme floods could have even greater economic impacts across European countries than indicated here. For example, more than 18% of the populations of Austria, Hungary,

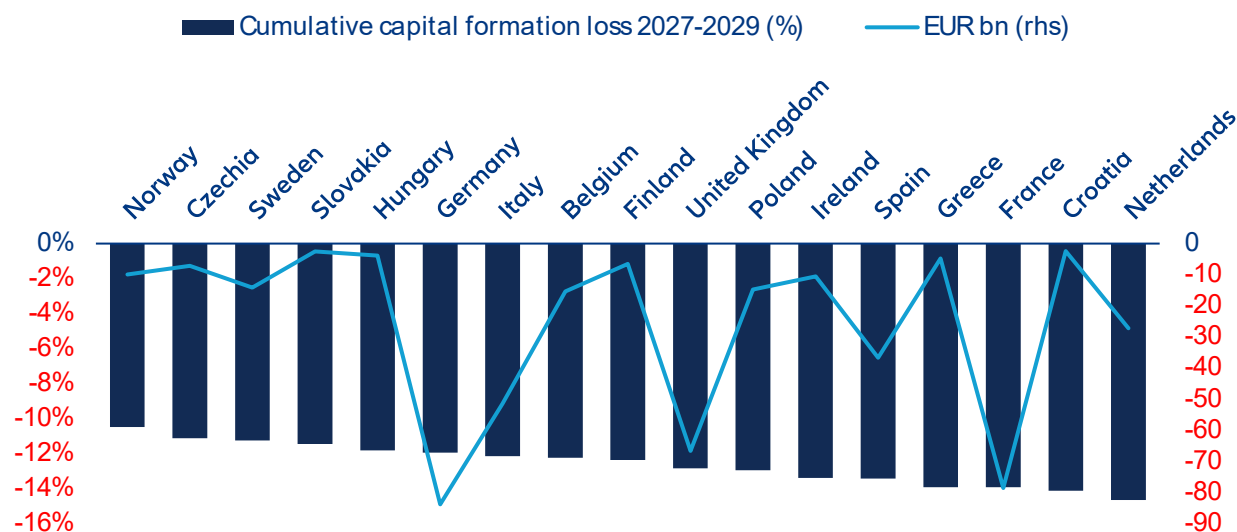
Netherlands; Liechtenstein and Slovakia live in floodplains.⁴ The numbers should therefore be read as a standardized stress test rather than a macroeconomic forecast under a tail risk event.

Investment is hit the hardest. A flood destroys part of the existing capital stock, but it also changes the conditions under which firms decide whether to commit to new capital. Since the dependent variable is real gross fixed capital formation for the total economy, the estimate already includes any reconstruction investment recorded as fixed capital formation. The reported loss is therefore a net shortfall relative to the no-flood baseline, after rebuilding spending has partly offset the initial capital damage. (See Appendix A1 for details on our empirical methodology). Estimation results show that the cumulative loss in capital formation — over the period 2027 – 2030 — is large across all countries, ranging from around -10.5% in Norway to -14.6% in the Netherlands from 2027 to 2029 (Figure 6). The Netherlands records the largest relative loss, followed by Croatia, France and Greece at about -14%, while Spain and Ireland are close to -13.4%. Germany is less affected in percentage terms, at 11.9%, but records the largest absolute loss at about EUR83.6bn. France follows with EUR78.4bn, the United Kingdom with EUR66.5bn, Italy with EUR51.3bn and Spain with EUR36.6bn.

³ See Appendix A3 on how we calculate flood depth in Europe from 2015 to 2024

⁴ [Population characteristics in potential flood-prone areas | Maps and charts | European Climate and Health Observatory Climate-ADAPT](#)

Figure 6: Cumulative capital formation loss (2027–2029) from a one-off 2027 flood shock, calibrated to the 2015–2024 average maximum depth, relative to a no-flood baseline



Source: Allianz Research

Floods leave a lasting mark on household incomes.

Cumulative real net disposable income losses for the period 2027 – 2030 range from -3.9% in Slovakia to -5.4% in Croatia (Figure 7). Sweden, Italy and Germany sit just above -4%, while Belgium is at -4.3% and Finland at -4.5%. The largest losses are recorded in Spain (-4.7%), France (-4.9%), Greece (-5.0%), Ireland (-5.1%), the Netherlands (-5.3%) and Croatia (-5.4%). Household income however does not collapse in the year of the flood rather the impact is lagged, which is consistent with the way disaster losses are absorbed in annual data, as reconstruction costs, business interruption and labor-market effects accumulate over the following years. Here the macro analysis begins to connect with insurance affordability. Under risk-based pricing, the same homeowners in floodplains often face higher than average flood premiums and more frequent losses. A flood therefore creates a double pressure: it weakens the income base from which protection is paid, while insurers might re-assess the risk, which could lead to higher prices.

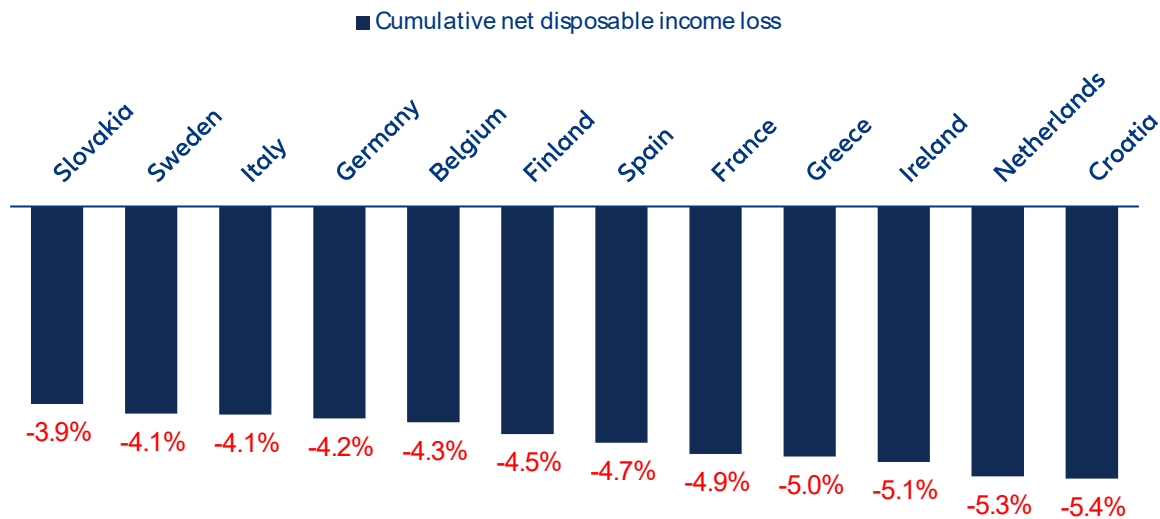
Flood shocks lead to significant GDP losses, with the strongest relative impacts in Spain, Croatia, Greece and the United Kingdom.

The largest cumulative GDP loss over 2027 to 2030 are in Spain (-1.0%), Croatia (-0.8%), Greece (-0.8%), the United Kingdom (-0.8%), France (-0.7%) and Germany (-0.7%), mainly reflecting the severity of flood events observed over the last decade. In monetary terms, Germany loses around EUR108bn relative to the baseline, followed by the United Kingdom (EUR106.3bn), France (EUR79.2bn), Spain (EUR59.2bn) and Italy (EUR56.5bn).

The key insight is that a flood is a local physical shock, but its macroeconomic effects spread beyond the flooded area.

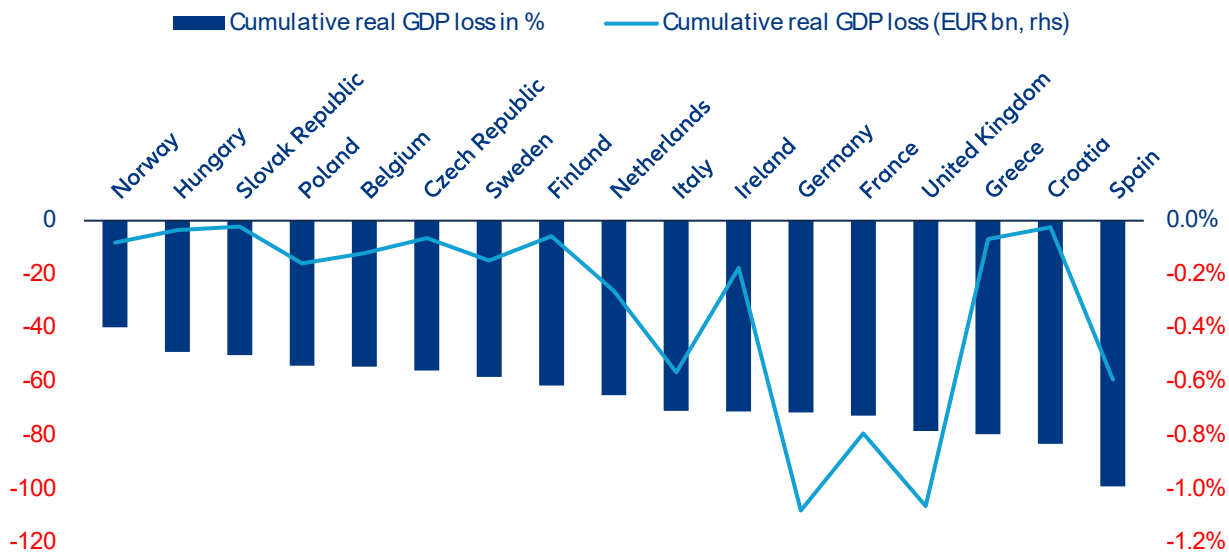
A flood lowers GDP not only through the immediate destruction of assets, but also through delayed investment, weaker household income, lower consumption and higher reconstruction needs. Reconstruction spending offsets part of the loss, but it does not erase it. The model result is consistent with the broader disaster literature: the immediate destruction is only the first-round effect; the more persistent cost comes from disrupted accumulation and weaker income and demand over the following years.

Figure 7: Cumulative net disposable income loss, constant 2013 price, (2027–2029) from a one-off 2027 flood shock, calibrated to the 2015–2024 average maximum depth, relative to a no-flood baseline



Sources: Allianz Research

Figure 8: Cumulative GDP loss (2027–2030) from a one-off 2027 flood shock, calibrated to the 2015–2024 average maximum depth, relative to a no-flood baseline

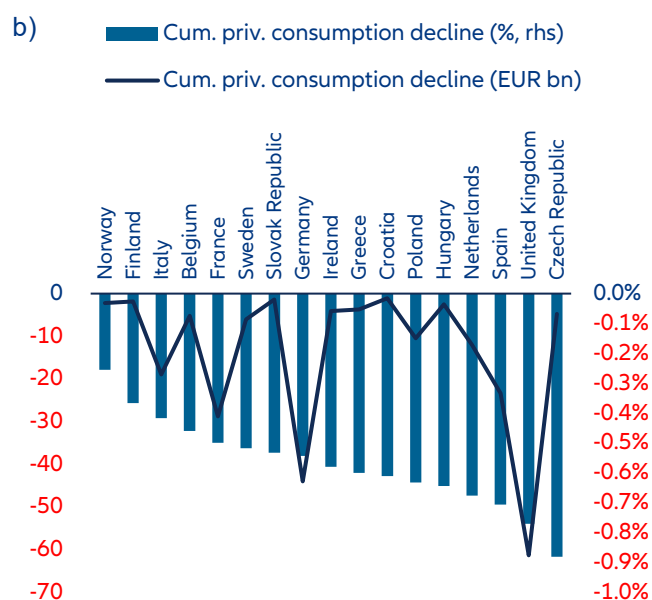
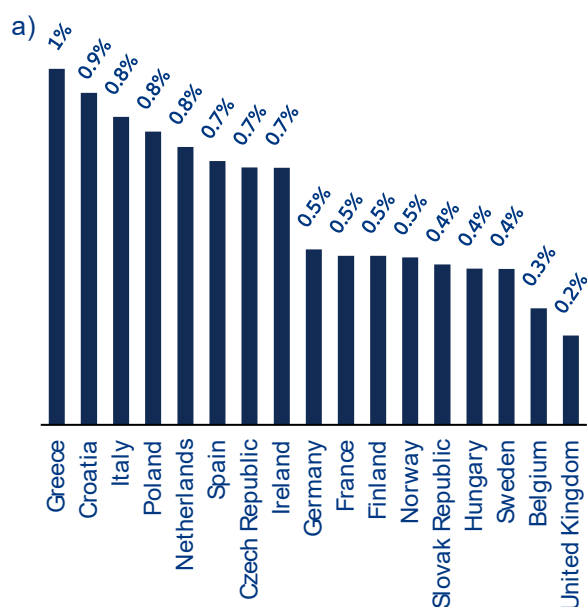


Sources: Allianz Research

Floods create a stagflationary shock by raising prices even as economic activity weakens. Damaged infrastructure, disrupted logistics and lower local supply temporarily increase prices across countries (Figure 9a). The cumulative (2027 – 2030) price-level impact is highest in Greece, at around +1.0%, followed by Croatia (+0.9%), Italy (+0.8%), Poland (+0.8%), the Netherlands (+0.8%) and Spain (+0.7%). At the lower end, the impact is more contained in Belgium (+0.3%) and the United Kingdom (+0.2%). While the inflation effect is moderate, it compounds the loss of household purchasing power.

Floods reduce household consumption by eroding real incomes and purchasing power (Figure 9b). Private consumption falls in every country, with cumulative losses ranging from around -0.3% in Norway to -0.9% in Czechia. The largest relative declines are recorded in Czechia (-0.9%), the United Kingdom (-0.8%), Spain (-0.7%), the Netherlands (-0.7%), Hungary (-0.6%) and Poland (-0.6%). In absolute terms, the largest losses are in the United Kingdom with about EUR61.3bn in private consumption relative to baseline, Germany EUR44bn, France EUR28.8bn, Spain EUR23.3bn and Italy EUR18.9bn.

Figure 9: Cumulative inflation rise (a) and private consumption decline (b) (2027–2030) from a one-off 2027 flood shock, calibrated to the 2015–2024 average maximum depth, relative to a no-flood baseline



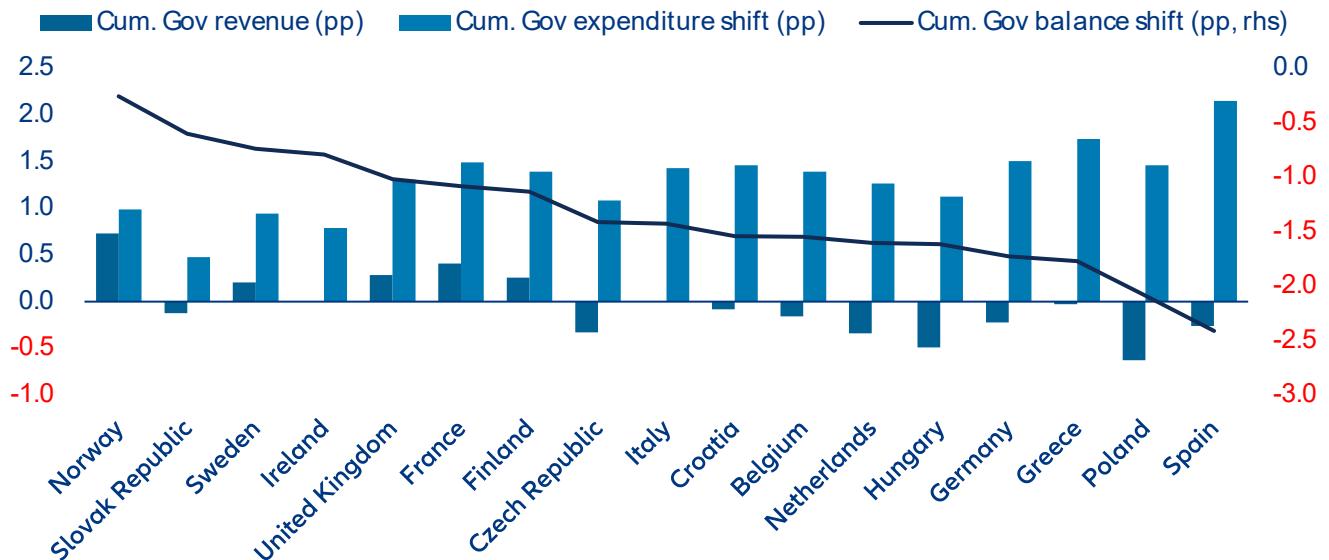
Sources: Allianz Research

Floods place sustained pressure on public finances, leaving governments with less fiscal space (Figure 10).

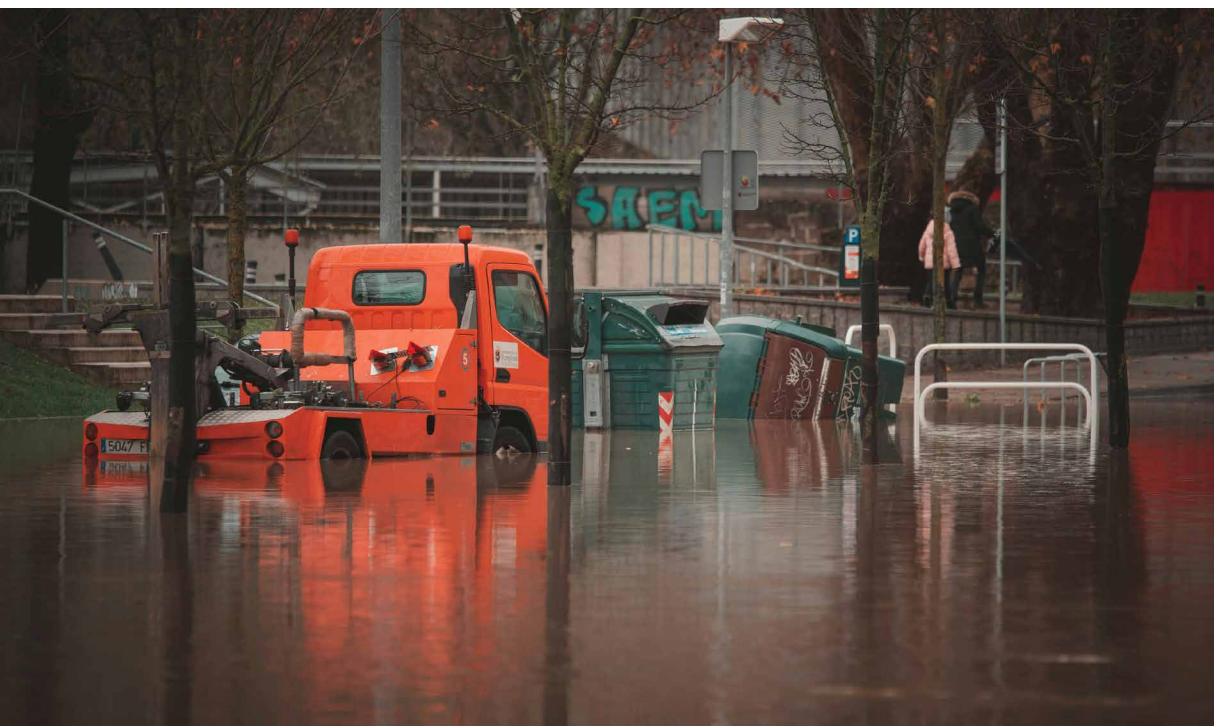
The government balance deteriorates in every country in the stress test, from a relatively contained -0.3 percentage points in Norway to -2.4pp at the upper end. The largest balance shifts are recorded in Spain (-2.4pp), Poland (-2.1pp), Greece (-1.8pp), Germany (-1.7pp), Hungary (-1.6pp) and the Netherlands (-1.6pp). The deterioration is mostly expenditure-driven, reflecting emergency response, reconstruction support and the public share of recovery costs. Spain records the largest expenditure shift, at about +2.2pp, followed by Greece (+2.2pp), Germany (+1.7pp), France (+1.5pp), Poland (+1.7pp), Germany (+1.5pp), France (+1.5pp), Poland

(+1.5pp) and Croatia (+1.5pp). The revenue channel is more mixed. It is positive in some countries, including Norway, France, the United Kingdom, Finland and Sweden, but negative in others, especially Poland (-0.6pp), Hungary (-0.5pp), the Netherlands (-0.3pp), Czechia (-0.3pp), Spain (-0.3pp) and Germany (-0.2pp). This mixed pattern is plausible: reconstruction and price effects can support some nominal tax bases, while weaker real activity and consumption reduce others. What is common across all countries is the pressure on the fiscal balance. Flood risk therefore creates a public finance problem in addition to a household income and capital accumulation problem.

Figure 10: Cumulative government balance (2027–2030) from a one-off 2027 flood shock, calibrated to the 2015–2024 average maximum depth, relative to a no-flood baseline



Sources: Allianz Research



Adaptation to flood-proof our economies

Floods are already Europe's costliest natural hazard, relying most heavily on public capital. Flood adaptation requires the most public capital of all climate hazards: of the 61 adaptation measures in our taxonomy, flood protection accounts for 65% of all primarily public measures, reflecting the dominance of large-scale infrastructure, retention and spatial planning in managing flood risk.⁵ The largest losses, the widest protection gap and the heaviest reliance on public investment together make closing the flood-adaptation gap not only an insurance, but a fiscal priority.

Concretely, flood adaptation is a layered risk-reduction strategy rather than a single type of investment. It starts with spatial planning: avoiding new development in floodplains, restoring natural retention areas and giving rivers more room to absorb peak flows. It then requires blue-green and grey infrastructure, including retention basins, dikes, levees, flood walls, sustainable drainage systems, porous pavements, green roofs, stormwater management, wastewater and sewage upgrades and pumping equipment. In already exposed areas, adaptation must move down to the asset level through flood-adapted construction, elevated foundations, protected utilities, backflow prevention, water-resistant

materials and the relocation of vulnerable equipment above expected flood levels. Finally, these physical measures need to be complemented by risk mapping, early-warning systems, evacuation planning and insurance incentives that reward actual risk reduction. The point is not to choose between nature-based solutions, infrastructure and property-level protection, but to combine them according to the type of flood risk: fluvial flooding requires river-basin management and retention, while pluvial flooding depends more on local drainage, urban design and building-level resilience.

One of the most positive developments since 2021 is the rapid improvement in flood risk information. Traditionally, flood management focused on river floodplains, mapping areas at risk from overflowing rivers and identifying exposed assets. While these remain important, the 2021 European flood event demonstrated that destructive floods are often compound events, combining fluvial flooding, when a river bursts its banks, with pluvial flooding, flash floods caused by intense rainfall and surface water floods from overwhelmed drainage systems. These flood types differ fundamentally. Fluvial flooding typically allows somewhat longer warning times and can often

⁵ From invisible to investible: An investment taxonomy for climate adaptation | Allianz

be managed through dikes, retention areas, and coordinated water management. The German Insurance Association (GDV) reports that less than 8% of addresses in Germany lie in the river floodplains⁶. Pluvial flooding, by contrast, can occur outside mapped floodplains, develops rapidly, and depends heavily on local factors such as urban drainage, topography, and land use.

Fragmented flood management leads to structural challenges for governance, but high-resolution flood mapping can help. In Germany, for example, river flood management is largely organized at the state and basin level, while pluvial flooding is primarily a municipal responsibility. This creates a structural challenge: municipalities play a critical role in managing risk but often lack the resources, data, or capacity to evaluate and implement effective measures at scale. Tools such as the surface water flood maps in North Rhine-Westphalia highlight areas at risk from pluvial flooding, including estimates of potential water depth and velocity⁷. These maps provide an essential bridge between regional-level hazard assessment and local decision-making. In practice, this makes flood risk operational rather than abstract: municipalities can use these maps to prioritise sewer and drainage upgrades, identify locations where temporary pumping may be needed and prepare emergency access routes. They also improve public risk communication by showing which homes, schools, care facilities and critical services are most exposed, allowing evacuation planning and adaptation spending to be targeted more precisely.

Artificial intelligence is accelerating the ability to evaluate adaptation options. AI-supported approaches allow rapid scenario testing, enabling stakeholders to simulate the impact of measures such as retention basins, drainage upgrades, or structural protections before committing to costly engineering studies. While they do not replace detailed local studies, they enable municipalities, planners, and citizens to better understand and discuss risk. The value of maps and models extends beyond identifying risk, to shaping behavior, such as guiding land use, informing building standards and supporting emergency planning.

Germany after the 2021 floods as a case study for European analysis: stronger frameworks, but

persistent implementation challenges. The 2021 floods made the implementation gap especially visible, and the five years since then provide a useful test of how far policy frameworks, risk mapping and local adaptation planning have actually moved from recognition to execution.

Germany has taken some important steps since the 2021 floods. One immediate response was the strengthening of early warning systems, including the introduction of alerts via Cell Broadcast (DE-Alert) to rapidly reach citizens in a potentially affected area via mobile phone during emergencies. Germany also adopted the Federal Climate Adaptation Act⁸, which entered into force in 2024. This creates a binding framework for adaptation policy across federal, state, and municipal levels. It requires systematic risk analysis, coordinated planning, and measurable targets for adaptation progress. Concrete implementation is still at an early stage, as the federal framework only entered into force in mid-2024. But some measures are already visible at local level, such as improved flood risk communication, even if large-scale infrastructure projects remain slow to deliver. The updated German Climate Adaptation Strategy⁹ complements this by establishing measurable targets for climate adaptation, including goals for infrastructure, land use, water management, spatial planning, business and civil protection.

There have also been improvements in flood mapping and risk communication, but delivery at scale remains slow. In response to the 2021 flood along the Ahr River, the German state of Rhineland-Palatinate revised the flood hazard maps a few months after the event, which incorporated improved hydrological understanding to better represent extreme runoff dynamics and flood extents¹⁰. A full revision of all floodplains in Rhineland-Palatinate was completed in February 2026¹¹.

Despite progress, delivery at scale remains slow.

Germany's federal system spreads planning and disaster-response responsibilities across multiple levels of government and aid organisations, creating coordination challenges, and large-scale projects such as retention basins face long planning and approval timelines. The National Flood Protection Programme

⁶ „ZÜRS Geo“ - Zonierungssystem für Überschwemmungsrisiko und Einschätzung von Umweltrisiken

⁷ LANUK Klima

⁸ BMUKN: The Federal Climate Adaptation Act

⁹ BMUKN: German Strategy for Adaptation to Climate Change

¹⁰ Überschwemmungsgebiet . Struktur- und Genehmigungsdirektion Nord

¹¹ Hochwassergefahren- und -risikokarten . Hochwasserrisikomanagement in Rheinland-Pfalz

(NHWSP) captures both the ambition and the lag: it bundles around 109 priority, supra-regional measures worth an estimated EUR6–7bn - co-financed 60% by the federal government and 40% by the Länder - targeting close to 1 billion m³ of new retention and roughly 40,000 ha of reconnected floodplain,¹² and federal modelling confirms these measures would lower the peak levels of future major floods by 10–50 cm over long river stretches¹³. Yet more than a decade after the 2013 floods, only a little over EUR500m had actually flowed through the dedicated „Preventive Flood Protection“ framework, with many NHWSP measures still in planning or early construction. The same lag appears locally: in the Ahr valley, communities and the district of Ahrweiler published their technical study of catchment-wide flood protection only in 2025 - four years after the disaster - with concrete measures still to be designed and built. This highlights a broader European challenge: adaptation frameworks have improved markedly, but delivery on the ground lags well behind.

The most effective flood adaptation measure — floodplain protection — is also the most politically challenging. Floodplains are natural water storage zones. When they are developed, flood risk intensifies. Giving rivers more room to flood and avoiding new development in floodplains are therefore key to risk reduction. However, this principle often conflicts with local economic and political realities. Municipalities depend on population and economic growth, which lead to development pressures—particularly in urban and waterfront areas. Additionally, floodplains are attractive to development because they are flat and require minimal grading. Another challenge is posed by rebuilding in affected locations after a major flood, either because it is permitted or required by authorities, or because homeowners seek to restore the communities they lost and remain emotionally attached to. Yet unless new construction incorporates resilience measures, these properties remain vulnerable to future floods.

Development in flood-prone areas requires strict building and planning requirements. Examples such as Hamburg’s HafenCity demonstrate that building in exposed areas can be compatible with risk management, if designed appropriately. This includes elevated structures, floodable public spaces, protected utilities, and clear evacuation and emergency strategies. The key is not simply whether to build, but how—and

under what conditions. If exposure cannot be avoided, it must at least be made resilient. In this way, development and nature-based flood solutions can be combined rather than treated as mutually exclusive.

Large-scale structural measures remain an important component of flood risk management, but should be used selectively. Levees, flood walls, and retention basins can significantly reduce the frequency and severity of moderate flood events. However, they also have limitations. Extreme floods can exceed design levels and overtop defenses. When this occurs, the impact can be even more severe, particularly if communities behind defenses have become less aware of the underlying risk or have accumulated additional values due to new construction. Structural protection is most appropriate where high exposures already exist, and relocation and individual property level retrofitting is not feasible or economically viable. In particular, it should not be used to justify new development as this would lead to increasing loss potential in catastrophic flood events. Importantly, it must be complemented by residual risk awareness, emergency planning, and building-level resilience.

While infrastructure plays a crucial role, properties need to be adapted at an individual building level. Especially in areas facing pluvial flooding and residual risk, flood-adapted building should become standard for all new construction. Already in 1995 the LAWA (Bund/Länder-Arbeitsgemeinschaft Wasser, Germany Working Group on Water Issues) concluded that adapted construction and individual prevention at the property level is the most cost effective and sustainable risk reduction method. Resilient renovation includes:

- Raising critical systems (electricity, heating)
- Choosing more resilient heating systems (e.g. replacing oil systems that cause contamination and avoiding wood pellets that expand when wet and can damage structures)
- Sealing entry points
- Installing backflow prevention
- Using water-resistant materials for walls, insulation and floors.
- Designing spaces for easier recovery

Tools such as the Hochwasserpass, a building specific flood risk certificate, help translate technical risk information into actionable guidance for property owners. They increase awareness, support

¹² 10 Jahre Nationales Hochwasserschutzprogramm (NHWSP)

¹³ Untersuchungen zur Ermittlung der Wirkungen von präventiven Hochwasserschutzmaßnahmen im Rahmen des Nationalen Hochwasserschutzprogramms

preparedness, and empower homeowners to take preventive action. Adapted construction also includes:

- Avoiding living spaces in basements
- Elevated foundations that allow for waterflow below or at the design flood level
- Waterproof material for walls, insulation and floors (at least up to the design flood level)
- Low value occupancies in ground floor below the design flood level.

These measures do not fully eliminate risk, but they significantly reduce the damage and recovery time. This is particularly important given the long disruption associated with flood events.

The economic case for targeted flood adaptation is strong and could significantly reduce economic damage. Without action, river-flood losses in the EU and UK are projected to rise more than six-fold to nearly EUR48–50bn a year under 3°C of warming by 2100, with close to 500,000 people exposed each year.¹⁴ Coastal flooding compounds the picture: absent adaptation, EU+UK annual coastal-flood damage could climb from about EUR 1.4bn today to nearly EUR240bn by 2100, of which roughly 95% could be avoided by pairing moderate mitigation with raised defenses.¹⁵ Using flood-risk modelling and cost-benefit analysis, the European Commission Joint Research Council (JRC) finds that reducing flood peaks with retention (detention) areas is the most economically attractive strategy: it can cut projected 2100 river-flood losses from about EUR50bn to EUR 8bn a year – an ~84% fall in expected losses, roughly back to today’s risk level – while returning about EUR4 for every EUR1 invested over 2020–2100.¹⁶ Strengthening dykes delivers a somewhat lower risk reduction (~70%) for comparable annual outlay, and building-level “damage-reduction” measures (the property-level retrofits discussed above) are found to be cost-efficient in most EU countries. These returns sit well within the wider evidence base: across all hazards, the Global Commission on Adaptation puts benefit-cost ratios for resilience investment at 2:1 to 10:1, and estimates that USD1.8trn invested globally across five adaptation priorities in 2020–2030 could generate USD7.1trn in net benefits.¹⁷

Adaptation is therefore preventive fiscal and

insurance policy. As insurance premiums reflect expected losses, public investment in flood defences and resilient construction is also an investment in insurance affordability and macro-financial stability. Despite these climate-induced trends, insurers have expanded flood coverage in many countries over the past 25 years. The protection gap for residential buildings in Germany shrank from 80% in 2003 to 43% in 2024, even as the building stock grew by three million structures. While the gap remains significant, it is not caused by a lack of insurance availability. Instead, it reflects limited awareness of flood risk among property owners and affordability challenges for those exposed to higher levels of risk.

A balanced approach to flood insurance: risk-based pricing and risk reduction. Coverage absorbs losses for individual homeowners after floods, but it also signals the presence of risk. Pricing insurance based on risk is therefore essential, as it provides incentives for adaptation and discourages development in areas highly prone to flood events. Risk-based pricing is an important element of fairness within the insurance pool, as it helps ensure that premiums more closely reflect the level of risk each policyholder contributes, thereby limiting cross-subsidization between lower- and higher-risk properties. However, without complementary measures, it can create affordability challenges for vulnerable households. Effective approaches combine:

- Risk-based pricing
- Ability for customers to choose limits and deductibles
- Support for risk reduction
- Incentives for resilient behavior
- Public-private partnerships with a focus on risk reduction

Public-private partnerships (PPPs) have long been used in Europe to address flood risk where private insurance markets alone cannot provide affordable and comprehensive coverage. Spain’s Consorcio de Compensación de Seguros and France’s Caisse Centrale de Réassurance (CCR) both operate on a risk-transfer mechanism in which risk is pooled nationally and backed by the state. In Spain, the Consorcio provides compensation for extraordinary risks—including floods—through a mandatory surcharge on insurance policies. France’s CCR works through a Nat Cat scheme, which combines private insurers with a state backed reinsurer.

¹⁴ JRC Publications - Adapting to rising river flood risk in the EU under climate change

¹⁵ JRC Publications - Adapting to rising coastal flood risk in the EU under climate change

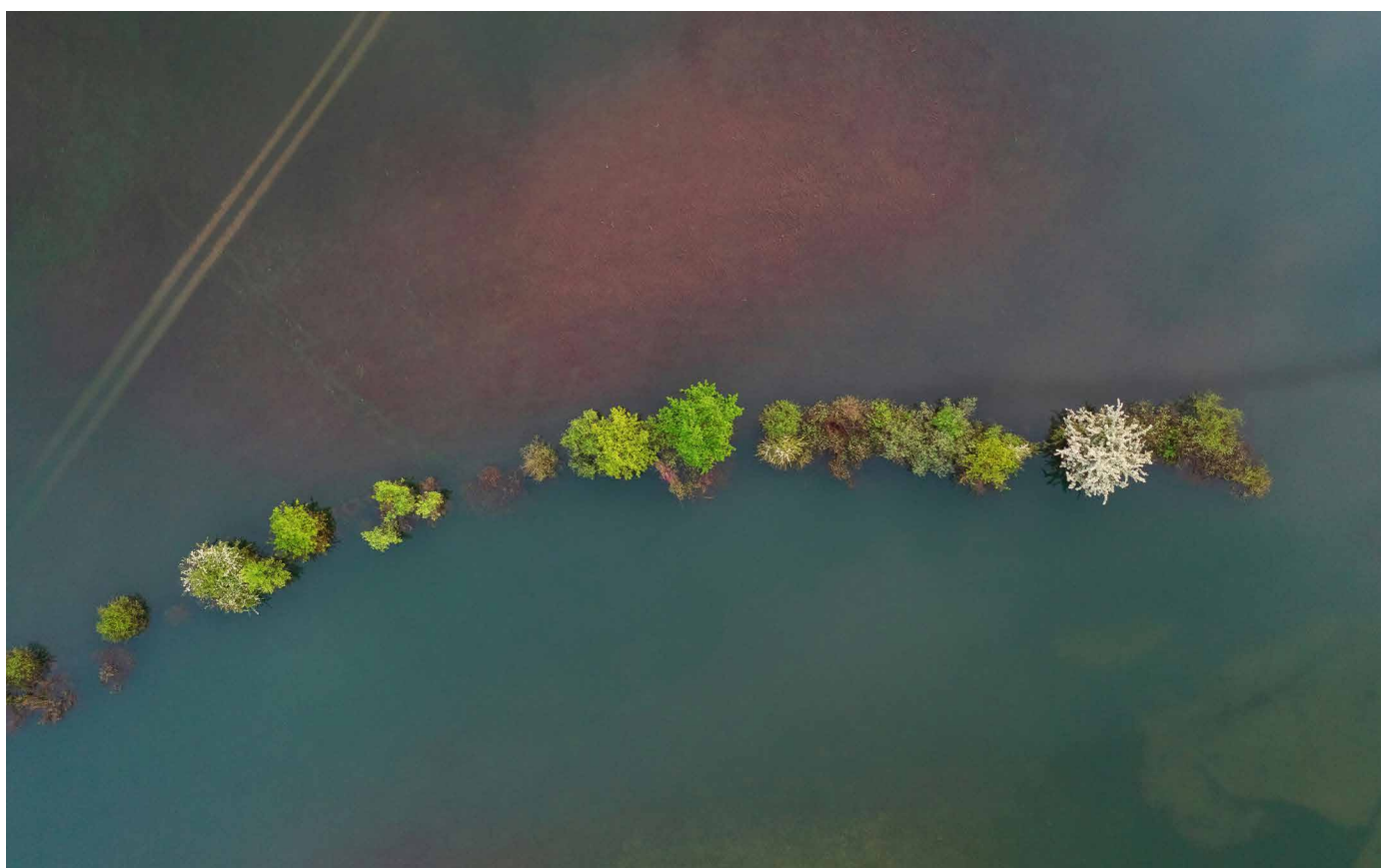
¹⁶ Cost-effective adaptation strategies to rising river flood risk in Europe | Nature Climate Change.

¹⁷ Adapt now: a global call for leadership on climate resilience - Global Center on Adaptation

The premiums increased significantly in 2025, after years of increasing risks and little focus on mitigation. Encouragingly, the Barnier Fund, which had been introduced dedicated to funding prevention, has recently returned to the political agenda. In the past it was chronically underfinanced and underused. Meanwhile the UK's Flood Re operates as a temporary public-private reinsurance scheme that enables insurers to transfer the flood risk of eligible household policies to a central pool, improving insurance affordability while supporting a gradual transition toward risk-reflective pricing by 2039. Through its *Build Back Better* program, it also incentivizes property-level resilience measures to reduce future flood losses.

Ireland and Germany are exploring PPP approaches to resolve existing protection gaps and growing affordability pressure. In Ireland, a government-led working group is developing a long-term strategy to address access to

flood insurance. The group has acknowledged that the problem “cannot be solved by insurance alone” and requires a public–private partnership approach. In Germany, the GDV is calling for a comprehensive natural hazard strategy that would cover extreme risks by combining universal insurance coverage (with an opt-out solution), binding adaptation measures and state participation. For both countries, these exploratory frameworks highlight an important implication for resilience investment: PPPs are most effective when paired with risk reduction. If insurance pricing reflects underlying risk, then public funding and policy should prioritize measures such as floodplain restoration, resilient construction, and infrastructure upgrades that reduce expected losses over time. In this way, PPPs can move beyond simply redistributing risk toward actively lowering it, thereby supporting long-term affordability and reducing future economic losses.



Appendix

A1 – The impact of floods on capital formation

Floods bear on capital through two channels that pull a measured investment series in opposite directions.

They destroy or damage part of the existing physical stock (buildings, factories and equipment, transport and utility infrastructure, agricultural land and crops) and they raise the uncertainty under which firms commit to new projects. Both of these depress investment in the immediate aftermath. Post-flood reconstruction is recorded as gross investment, so a flood can leave a positive imprint on the same aggregate once rebuilding gets under way. The net effect on a flow measure of investment is therefore an empirical question, and the sign and timing are what the regressions below are designed to recover.

The wider literature frames why the answer matters. Hsiang and Jina (2014)¹⁸ find that the national-income losses from tropical cyclones are persistent and do not recover within two decades, pointing to a capital channel that compounds rather than heals. At the city scale, Kocornik-Mina et al. (2020)¹⁹ reach a more sanguine conclusion for floods specifically, documenting rapid recovery of local economic activity and little permanent relocation. The companion analysis in this paper places investment at the center of the heat transmission, in line with Casey et al. (2024)²⁰, who argue that standard climate-economy models understate damages to capital-intensive activity. The exercise here asks the parallel question for floods: whether European gross fixed capital formation carries a detectable, datable response to flood events.

The analysis uses a balanced country-year panel of 36 European countries²¹ over 2010–2024 (540 observations), with no missing values in the outcome. The dependent variable is the log of gross fixed capital formation for the total economy (the AMECO series labelled OIGT), measured in volume terms at 2020 constant prices. Because the series is real and expressed on a national-currency base, country fixed effects absorb level differences in currency base and price level without contaminating the estimates with inflation.

Flood exposure is constructed from Copernicus Emergency Management Service (CEMS) activations aggregated to the country-year level²². Two treatment forms are used. The first is a binary indicator equal to one when a country records any flood activation in a given year; this is the case in 47 of the 540 country-years, spread across 17 countries. The second, which is the severity measure carried into the headline specification, is the average across affected NUTS-2 regions of each region's maximum inundation depth in centimeters, entered as $\log(1 + \text{depth})$. This measure is a regional average of regional maxima rather than the single deepest point recorded nationally; among treated country-years it averages roughly 125 cm.

Specification

The baseline is a two-way fixed effects (TWFE) model with country and year fixed effects, estimated in levels of log OIGT (Eq. 1). Identification rests on the meteorological exogeneity of flood timing conditional on the fixed effects: year effects absorb common shocks (the pandemic, euro-area financial conditions), country effects absorb time-invariant exposure and structure, and the flood term picks up within-country, within-year deviations.

¹⁸ [The Causal Effect of Environmental Catastrophe on Long-Run Economic Growth: Evidence From 6,700 Cyclones | NBER](#)

¹⁹ [Flooded Cities - American Economic Association](#)

²⁰ [Understanding climate damages: Consumption versus investment - ScienceDirect](#)

²¹ Main results consider only 17 European countries, where flood events were recorded for the period 2015 – 2024.

²² We use flood depth data from ClimRad, a tool developed by Allianz Research and EarthianAI. CEMS Rapid Mapping is activated on request by authorised users (national civil protection authorities, the ERCC and others), so activations reflect both the severity of an event and the decision to request mapping. This means that coverage is not fully uniform across countries or constant over time, as activation practice and user uptake have evolved since the service became operational in 2012. We therefore treat the activation count as an indicator of significant, officially recognised flood events rather than a complete census of all flooding. We rely on the severity measure (inundation depth) rather than activation frequency alone for the headline specification.

$$\ln(\text{OIGT}_{it}) = \beta \cdot \text{Flood}_{it} + \alpha_i + \lambda_t + \varepsilon_{it} \quad (\text{Eq. 1})$$

Because flood damage propagates into investment with a delay the preferred specification is a distributed-lag extension that enters the flood term contemporaneously and at four annual lags (Eq. 2). A Wald test on the sum of the lag coefficients summarizing the cumulative multi-year response. Standard errors are clustered at the country level throughout (36 clusters in the full panel).

$$\ln(\text{OIGT}_{it}) = \sum_{k=0}^4 \beta_k \cdot \text{Flood}_{i,t-k} + \alpha_i + \lambda_t + \varepsilon_{it} \quad (\text{Eq. 2})$$

Here Flood denotes the log average maximum depth, $\log(1 + \text{depth})$. The contemporaneous-only model (Eq. 1) is reported for reference; the distributed-lag model (Eq. 2) is the one from which the headline estimate is drawn. Because the number of clusters is modest, the inference on the headline coefficient is cross-checked with a restricted wild cluster bootstrap (Rademacher weights, B = 1999).

Results

Table A1 reports the distributed-lag estimates for the depth specification, where the effect of a single flood is allowed to play out over the event year and the four subsequent years. The contemporaneous coefficient is small but significant at the 10% level. The estimated effect grows over the following two years and is largest at the second lag, where a one-log-point increase in average maximum inundation depth is associated with a reduction in capital formation of about 1.1% (-0.0108 , $p < 0.05$). This is the only lag individually significant at the 5% level and the coefficient the paper carries forward. By the fourth year after the event the sign turns positive, though far from significant, which is what one would expect if reconstruction investment re-enters the capital formation measure and partially offsets the earlier shortfall. The depth regressor is zero in non-flood years, so moving from no flood to an average-severity event ($\log(1 + 125) \approx 4.8$) implies a cumulative shortfall on the order of 5% of capital formation two years out.

Table A1. Distributed-lag TWFE estimates: log gross fixed capital formation (OIGT) regressed on flood-depth severity in the event year and the following four years (lags 0–4).

Log avg-max depth	
Flood depth (t)	-0.0077* (0.0042)
Flood depth (t-1)	-0.0088* (0.0047)
Flood depth (t-2)	-0.0108** (0.0045)
Flood depth (t-3)	-0.0067 (0.0050)
Flood depth (t-4)	+0.0042 (0.0095)
Sum of coefficients	-0.0298
Fixed effects	Country + Year
SE cluster (clusters)	Country (36)
Observations	396

Notes: Dependent variable is log OIGT (AMECO gross fixed capital formation, total economy, 2020 constant prices). The treatment is the log average maximum depth, $\log(1 + \text{depth})$, entered at t to t-4. The common-sample size (N = 396) reflects the loss of early years to the lag window. Standard errors clustered by country in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Allianz Research

A2 –The impact of floods on household disposable income

Floods reach household income through three channels that operate at different speeds. The most immediate is direct asset destruction and damage which erodes both wealth and the income those assets generate. The second is labor-market disruption, as flooded workplaces cost days of work and, in the worst-hit areas, push some workers out of employment entirely. The third is business interruption, as damaged premises, markets and transport links suppress local activity and the wages and profits that flow from it. None of these is instantaneous in annual data: most damage is incurred and absorbed over the months following an event, so the income response is expected to build over one to two years rather than only in the year of the flood itself.

The analysis uses a balanced panel of 72 NUTS-1 regions across 15 European countries over 2013–2023 (792 region-year observations). The dependent variable is net disposable household income per inhabitant (ESA 2010, item B6N), drawn from the Eurostat regional accounts (nama_10r_2hhinc). Because B6N is published in current euros, the series is deflated to real 2013 euros using a cumulative price index built by chaining annual HICP headline inflation to a 2013 = 100 base.

Specification

The baseline model is a two-way fixed effects (TWFE) model with NUTS-1 region and year fixed effects (Eq. 1). Region effects absorb time-invariant structure (geography, industrial mix); year effects absorb common shocks, such as ECB policy and pandemic transfers. Conditional on these, flood timing is treated as quasi-random, since extreme-precipitation events are driven by meteorology rather than by income trajectories.

$$\ln(Y_{it}) = \beta \cdot \text{Flood}_{it} + \alpha_i + \lambda_t + \varepsilon_{it} \quad (\text{Eq. 3})$$

Because damage is absorbed over the months after an event, the retained specification is a distributed-lag version of Eq. 3 that enters log maximum depth at t , $t - 1$ and $t - 2$, tracing the dynamic profile rather than compressing it into a single contemporaneous coefficient. This is the same depth-in-lags form carried forward in the capital analysis, which keeps the severity measure and the dynamic structure consistent across outcomes. The identifying assumption is that, conditional on the fixed effects, the timing of floods is unrelated to prior income trajectories; a pre-trends check in growth rates supports this, with pre-flood coefficients indistinguishable from zero (Wald $p = 0.285$).

Results

The retained estimates are the distributed-lag depth specification, reported in Table A2. The contemporaneous effect is small and insignificant (–0.33% per log-point of maximum depth, $p = 0.111$), in line with the annual-data timing, then becomes significant at the first and second lags (–0.30%, $p = 0.031$; –0.37%, $p = 0.027$), summing to a three-year cumulative effect of about –1.0% per log-point of maximum depth. Because both sides of the regression are in logs, the coefficients are elasticities of real income with respect to flood depth: a flood twice as deep is associated with a cumulative real-income loss roughly 0.7% larger over three years (–0.0100 $\times \ln 2$).

Table A2. Retained specification : distributed-lag log maximum depth, real household income per inhabitant

Log max depth	Coefficient	p-value
Flood depth (t)	-0.0033	0.111
Flood depth (t-1)	-0.0030**	0.031
Flood depth (t-2)	-0.0037**	0.027

Notes: NUTS-1 + year TWFE; log maximum depth entered at t, t-1, t-2. Coefficients are elasticities of real income with respect to maximum depth. SE clustered at country (15 clusters). N = 648. * p < 0.10, ** p < 0.05, *** p < 0.01.

Source: Allianz Research

A3 – The construction and interpretation of flood-depth data

Flood depth gives the macro shock its physical scale. The variable is not a count of flood events and it is not a regional-average water level. It is constructed from satellite-mapped inundation footprints, linked to official event records, and then aggregated to the country-year or regional panel used in the regressions. This matters because a depth of 125 cm should be read as the characteristic depth of flooded footprints within an event-region, not as the depth of water across the whole country or region.

The underlying event dataset combines several layers of information (Table A3). The primary source is the JRC satellite-derived flood-depth map catalogue for Europe, which provides 20 m raster maps for 2015-2024 with pixel values expressed in centimetres. These maps are derived from systematic flood detection in the Sentinel-1 radar archive, with water depth reconstructed from the mapped flood extent and terrain elevation. Permanent and seasonal water bodies are flagged separately and excluded from the depth statistics. Copernicus EMS Rapid Mapping products are used to attach named event information, area of interest, flood extent and sensing-date metadata. EU-Hydro, geoBoundaries and Natural Earth are then used to attribute the footprint to the nearest river, the relevant administrative unit and the country.

The JRC maps are spatio-temporal clusters rather than named disaster events. Each relevant map was therefore matched to a Copernicus EMS Rapid Mapping activation by comparing the map date and the first-level administrative units intersected by the flood raster with activation records. Candidate matches were then checked visually in QGIS by overlaying the satellite flood footprint with the CEMS delineation products. This step avoids treating anonymous raster clusters as events before they are anchored to a documented flood episode.

For each matched map, the raster is processed into contiguous flooded footprints. Each footprint receives GIS attributes: planimetric area, country, administrative unit, nearest EU-Hydro river reach and zonal water-depth statistics. The matched CEMS vector product is then joined to the footprint, adding event-level information such as activation code, area of interest, flood type when available, sensing source and acquisition date. The final record therefore corresponds to a flood footprint within one event-region, rather than to an entire national event.

Table A3. Flood-depth dataset: source inputs and role in the construction

Input	Role in the construction
JRC satellite-derived flood-depth maps for Europe	Primary source of flood extents and water depths; 20 m rasters, with depth values expressed in centimetres.
Copernicus EMS Rapid Mapping vector products	Event delineations, areas of interest, maximum flood extent where available, sensing source and event metadata.
EU-Hydro river network database	Attribution of flooded footprints to the nearest river reach.
geoBoundaries and Natural Earth administrative boundaries	Country and first-level administrative-unit attribution, plus screening of flood locations.
QGIS manual verification	Visual overlay of satellite footprints and CEMS delineation products before the consolidated event table is retained.

Source: Allianz Research based on JRC, Copernicus EMS Rapid Mapping, EU-Hydro, geoBoundaries and Natural Earth

The depth statistics are deliberately conservative. First, each contiguous flood footprint is summarized by the median of its 20 m pixels. This reduces sensitivity to isolated extreme pixels near channels, in topographic lows or in areas affected by elevation-model artefacts. Second, the event-region median, mean and maximum are calculated across these footprint medians. The reported maximum is therefore the characteristic depth of the most severely inundated footprint, not the single deepest pixel in the event. This distinction is important because the deepest pixel can be much larger than the economically relevant flooded-area statistic.

The statistics describe flooded land only. A mean or maximum depth of 125 cm does not mean that the wider NUTS region, or the country, was under 1.25 metres of water. It means that, within the inundated footprints identified for that event-region, the representative footprint depth was around 1.25 metres. Areas outside the mapped footprint are dry by construction and do not enter the depth statistic.

The statistics are also not exposure-weighted. A footprint in a sparsely populated field and a footprint in a dense urban area receive the same weight when the event-region depth distribution is formed. Exposure is captured separately through the affected-population attribute and through the macroeconomic panel structure. The depth variable should therefore be interpreted as hazard intensity, not as damage, exposed capital or population-weighted risk.

A photograph showing a group of diverse hands of various skin tones stacked on top of each other, resting on a thick, textured tree branch. The background is a lush green forest with sunlight filtering through the leaves. The text "Our team" is overlaid in the center, with "Our" in white and "team" in orange.

Our team

Chief Investment Officer
& Chief Economist
Allianz Investment Management SE



Ludovic Subran
ludovic.subran@allianz.com

Head of Economic Research
Allianz Trade



Ana Boata
ana.boata@allianz-trade.com

Head of Macroeconomic & Capital
Markets Research
Allianz Investment Management SE



Bjoern Griesbach
bjoern.griesbach@allianz.com

Head of Outreach
Allianz Investment Management SE



Arne Holzhausen
arne.holzhausen@allianz.com

Head of Corporate Research
Allianz Trade



Ano Kuhanathan
ano.kuhanathan@allianz-trade.com

Head of Thematic & Policy Research
Allianz Investment Management SE



Katharina Utermoehl
katharina.utermaehl@allianz.com

Macroeconomic Research



Lluis Dalmau Taules
Economist for Africa & Middle East
lluis.dalmau@allianz-trade.com



Maxime Darmet Cucchiari
Senior Economist for UK, US & France
maxime.darmet@allianz-trade.com



Jasmin Gröschl
Senior Economist for Europe
jasmin.groeschl@allianz.com



Françoise Huang
Senior Economist for Asia Pacific
francoise.huang@allianz-trade.com



Maddalena Martini
Senior Economist for Southern Europe & Benelux
maddalena.martini@allianz.com

Outreach



Luca Moneta
Senior Economist for Emerging Markets
luca.moneta@allianz-trade.com



Giovanni Scarpato
Economist for Central & Eastern Europe
giovanni.scarpato@allianz.com



Heike Baehr
Content Manager
heike.baehr@allianz.com



Melissa Eddy
Content Manager
melissa.eddy@allianz.com



Maria Thomas
Content Manager and Editor
maria.thomas@allianz-trade.com

Corporate Research



Guillaume Dejean
Senior Sector Advisor
guillaume.dejean@allianz-trade.com



Maria Latorre
Sector Advisor, B2B
maria.latorre@allianz-trade.com



Maxime Lemerle
Lead Advisor, Insolvency Research
maxime.lemerle@allianz-trade.com



Pierre Lebard
Public Affairs Officer
pierre.lebard@allianz-trade.com

Thematic and Policy Research



Michaela Grimm
Senior Economist,
Demography & Social Protection
michaela.grimm@allianz.com



Patrick Hoffmann
Economist, ESG & AI
patrick.hoffmann@allianz.com



Simon Krause
Economist, ESG & Insurance
simon.krause@allianz.com



Hazem Krichene
Senior Economist, Climate
hazem.krichene@allianz.com



Kathrin Stoffel
Economist, Insurance & Wealth
kathrin.stoffel@allianz.com



Markus Zimmer
Senior Economist, ESG
markus.zimmer@allianz.com

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Director of Publications
Ludovic Subran, Chief Investment Officer & Chief Economist
Allianz Research
Phone +49 89 3800 7859

Allianz Group Economic Research
https://www.allianz.com/en/economic_research
<http://www.allianz-trade.com/economic-research>
Königinstraße 28 | 80802 Munich | Germany
allianz.research@allianz.com

 @allianz

 allianz

Allianz Trade Economic Research
<http://www.allianz-trade.com/economic-research>
1 Place des Saisons | 92048 Paris-La-Défense Cedex | France

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